Klamath Falls National Fish Hatchery

Annual Report for Fiscal Year 2025

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U.S. Fish and Wildlife Service

Klamath Falls National Fish Hatchery

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Table of Contents

Report Section	Page
Definitions/Acronyms	3
Executive Summary	4
Introduction	6
Fish Culture Operations Overview	8
Wild East Side Springs Lost River Sucker Spawning	13
Captive Shortnose Sucker Spawning	17
Wild Larval Sucker Collections and Early Rearing	21
Pond Growout	25
Net Pen Growout	38
Fish Distribution	43
Field Monitoring for Hatchery Fish	46
Partnerships/Collaborative Studies	51
Facility Construction	54
Future Work	59
Appendices	64

Definitions and Acronyms

The following definitions are/or acronyms being added to address questions from some of the external partners who reviewed this report and gave feedback in the past. These are not dictionary or policy definitions but rather laymen explanations for clarity in the report.

Definitions

Beneficial Occupancy – Where the government starts to use newly constructed infrastructure before the contractor has completed the entire construction project and turned everything over for use to close out the contract.

Production Fish – These are the fish raised to a target size of 200 mm total length, generally being 18-24 months old.

Production Goal – This is the target goal of the number of fish the hatchery would like to produce, raise, and/or release in a given year, based on staffing and space limitations.

Acronyms

BOR – Bureau of Reclamation

CJS – Cormack-Jolly-Seber Model

CY – Collection Year

ESS – East Side Springs

IRP – Influent Retention Pond

KBAO – Klamath Basin Area Office

KFNFH – Klamath Falls National Fish Hatchery

KLS – Klamath Large Scale Sucker

LRS – Lost River Sucker

SNS – Shortnose Sucker

UKL – Upper Klamath Lake

USBR – United States Bureau of Reclamation

USFWS – US Fish and Wildlife Service

USGS – United States Geological Survey

Executive Summary

The 2025 fiscal year (October 1, 2024 to September 30, 2025, hereafter FY25) brought a lot of unique challenges, but with the dedication of the staff supporting this project, we also experienced some very significant achievements and milestones at the Klamath Falls National Fish Hatchery ("KFNFH" or "Hatchery"). Two unique challenges that stand out were the unprecedented predation of otters and mergansers in the winter months, due in large part to limitations with our infrastructure during construction, which decimated our production and broodstock inventory, as well as the halt to all the of the BIL funded construction phases, which has crippled the development and completion of our needed infrastructure. But on the positive side, we had a great larval collection year, with 70,049 fry collected, and of these, approximately 65,847 were ponded for the growing season. In addition, the staff also spawned and hatched an estimated 62,022 larval Lost River Suckers on station from gametes collected from the East Side Springs population in Upper Klamath Lake ("UKL") and ponded approximately 53,400 for the growing season. And the most significant accomplishment for FY25, was that we were successful in spawning our captive Shortnose Sucker ("SNS") and Lost River Sucker ("LRS") and producing 44,589 viable SNS larvae and 5,591 viable LRS larvae. From these, 23,781 SNS fry were ponded for the growing season, from 22 distinct family groups that reflect well the genetic diversity of the wild population.

To help support the dwindling populations of Klamath Basin Suckers, the staff reared and repatriated 2,206 production fish through the fiscal year. There were also 417 fish reared and transferred for acoustic telemetry research in UKL. This brings the total number of production sized fish repatriated and/or transferred to 2,623 overall for FY25. In addition, 23,719 fry were transferred to the Klamath Tribes Hatchery to aid in their production efforts and a total of 112 Klamath suckers were salvaged and repatriated from the electrofishing efforts conducted by USBR staff over the winter in the local irrigation canals. This makes for a grand total of 26,454 fish released during FY25. Overall production numbers are down in FY25 since the stockings in the fall of 2024 consisted of the collection year ("CY") 2023 fish, which had record low collection and ponding numbers of wild fish. And unfortunately, our production in FY26 will also be lower, due in large part to the inventory losses of nearly 23,655 fingerlings from the 2024 lots which succumbed to the merganser predation mentioned above; otherwise, we likely would have had a record stocking in the fall of 2025, with the potential of over 20,000 production sized fish to be stocked. It is hoped that during FY26, we will carry a suitable inventory of fingerlings to see a record stocking in FY27, or the fall of 2026. However, we will not see steady increases in the amount and size of production sized suckers being produced from the facility on an annual basis until facility construction resumes and is completed.

To help the Klamath Basin move towards sucker recovery, the KFNFH staff continued to work hard to improve operational processes by conducting applied research, which focused on fish spawning and incubation techniques, and continued to expand monitoring efforts of hatchery released fish. For example, the pilot work for PIT tag monitoring in areas that had clusters from previous radio telemetry monitoring work within our office, which started in FY24, yielded better results in FY25 by deploying more submersible antennas, as well as litz antennas, in new areas, and with such staff were able to detect 508 unique PIT tagged fish, 223 of which were either reared from the Hatchery (194) or rehabilitated and repatriated from salvage operations (29), which is nearly half (43.9%) of the fish encountered having come through the hatchery at some point prior to repatriation into Upper Klamath Lake. This monitoring work has revealed, from direct detections, that we still have fish at large and surviving in the lake from each of the year classes stocked from the collection years 2017-2023, with the

oldest fish being over eight years old and spending over five years in the lake since initial stocking. This shows that the hatchery program is working, at least on a small scale, to supplement the natural populations. In addition, staff have been working with a Fisheries Statistician to develop a model to assess long term survival estimates and current population estimates, which will help us make better management decisions. These efforts will continue into the upcoming fiscal year, with expanded efforts of both passive and active tracking, so that we can further refine our rearing and stocking methods and get more Klamath suckers on the local landscape, as well as to better understand their habitat usage and persistence in the lake system post stocking.

And, while most of the current phases of construction were on pause for most of FY25, due to the freeze on BIL funding, the first construction contract awarded, Phase 1 to Morello Construction, is nearing completion, with most of the major tasks already done. This work has provided 11 new ponds, one as a water supply head pond and 10 ponds for fish rearing and production, six being 0.25 acre ponds, and four being 0.125 acre ponds. This now gives us 2.0 acres of newly constructed pond space, which is about one quarter of the planned facility infrastructure. We are also continuing to utilize 22 0.03 acre ponds in the P-series, which have been in use since at least 2016, but they are showing their age and need replacing soon. And there have been several new modifications awarded to Morello Construction, under the Phase 1 contract to make a partially built facility more functional in the interim, and these tasks should be completed in the spring or summer of 2026. Overall, we are grateful to have the space and infrastructure that we do have and look forward eagerly to getting more of the hatchery completed in the upcoming years. This will help us speed up the process of recovering these critically imperiled fish stocks.

Although temporary space constraints and inventory losses from predation will lead to reduced stocking numbers in FY26, or the fall of 2025, the future of the hatchery's production capabilities is still very bright and will only improve as new pond space comes online. Already staff are planning to implement study designs in FY26 that are geared towards improving our captive rearing and spawning techniques, further expanding our population monitoring of hatchery stocks in the wild, refining our documentation of standard operating procedures, and better compiling and reporting on historical data from the program's inception. The staff at both the hatchery and the Klamath Falls Fish and Wildlife Office (KFFWO) will continue to work closely with and learn from our vital partners within the Klamath Basin as we collectively work towards fishery recovery goals.

With all of that said, please give your attention to the following annual report for more information on the activities briefly described above during fiscal year 2025 at the Klamath Falls National Fish Hatchery.

Introduction

The Shortnose Sucker (*Chasmistes brevirostris*) and the Lost River Sucker (*Deltistes luxatus*) were listed by the US Fish and Wildlife Service (USFWS) as endangered in 1988. These species are long-lived freshwater fish endemic to select lakes and rivers in the upper Klamath Basin of southern Oregon and northern California, with lifespans of 30+ and 50+ years respectively. The Revised Recovery Plan for the Lost River Sucker (*Deltistes luxatus*, LRS) and Shortnose Sucker (*Chasmistes brevirostris*, SNS) called for the development of a controlled propagation program to prevent extinction. This was initially achieved in 2016 through a cooperative partnership with a local landowner of a fish rearing facility to use the existing geothermal water source, ponds, and infrastructure to rear wild caught sucker larvae and was then known as the Sucker Assisted Rearing Program (SARP). Due to the early success of the program at rearing Klamath suckers in ponds, the USFWS signed a 30-year lease with the landowner and designated the site as the Klamath Falls National Fish Hatchery ("KFNFH" or "Hatchery") in early 2022.

The Hatchery construction was tentatively planned for completion in 2026/2027, to expand the production capacity and infrastructure. However, during much of FY25, the BIL funded hatchery construction has been under a stop work order, as funding has been frozen and leadership up to the Department level has not given any status updates on whether or when the funding will be released to complete the awarded projects that have already started. Should construction efforts resume in FY2026, it will likely delay completion of hatchery construction beyond 2027/2028. However, upon completion, the Hatchery will include a total of 33 production ponds, totaling approximately 8.5 acres of rearing space, a new influent retention pond, an effluent retention pond, a hatchery and administrative building, a maintenance shop, chemical and feed storage buildings, a primary well, and security, telecom, and supervisory control and data acquisition (SCADA) control systems. This new infrastructure will allow for the annual production of approximately 60,000 18-24-month-old suckers to help support recovery goals and stabilize and rebuild existing sucker populations within Upper Klamath Lake (UKL) and its tributaries. For a conceptual image of the new hatchery infrastructure, please see **Figure 1**.

The current production target for the facility is to raise 8-10,000 18-24-month-old fish annually to an average size of 200 mm total length and in FY25, the KFNFH released or transferred 2,623 production fish, averaging approximately 223 mm total length. So far, approximately 77,243 production fish have been reared and released from the program since the first stockings in 2018. In addition, since 2022, the program has also released suckers into off-channel rearing locations or transferred fingerlings that were excess fish stocks (beyond our rearing capacity) as well as fry produced from experimental, wild-spawning activities on the lake and river side to other facilities. In FY25, fry transfers totaled 23,719 sucker fry. At the close of FY25, there have been approximately 20,269 fingerlings and 63,317 fry released or transferred to date from the program. There is also the ongoing development of an older refugia population that can act as a redundant captive broodstock should the need arise, with approximately 1,119 fish currently on station that range from the 2017-2024 collection years. In addition, suckers salvaged throughout the year from Klamath Project Operations are rehabilitated, tagged, and repatriated, with 112 total in FY25 and 7,336 salvaged suckers repatriated through the program to date. For more information on the collections and releases of suckers from the program's inception until the end of FY25, please see **Table 1**.

Figure 1 – Conceptual site plan for the new construction of the Klamath Falls National Fish Hatchery, which has a tentative completion date of 2026/2027.

Table 1 – Collections and releases from the Klamath Falls National Fish Hatchery since its inception in 2016, by federal fiscal years, running from October 1 through the following September 30. All total lengths (TL) reported are averages based on length-weight conversions from cumulative total weights of each type, production, fingerling, fry, and standard lengths [SL] and total lengths [TL] of salvage fish are averages from measurements for each year.

Fiscal Year	Larvae Collected	SARP Release	TL (mm)	Fingerling Release	TL (mm)	Fry Release	TL (mm)	Salvage Release	SL or TL (mm)
2016	4,134	-	-	-	-	-	-	-	-
2017	8,730	-	-	-	-	-	-	-	-
2018	9,544	2,355	147	-	-	-	-	784	102 [SL]
2019	24,426	4,497	189	-	-	-	-	1,586	103 [SL]
2020	40,603	11,774	223	-	-	-	-	1,928	94 [SL]
2021	106,710	13,394	208	-	-	-	-	1,689	143 [SL]
2022	51,929	12,768	193	4,655	118	19,993	10	-	-
2023	6,036	17,399	194	15,604	89	14,973	10	436	129 [TL]
2024	30,142	12,433	221	-	-	4,632	10	801	126 [TL]
2025	70,049	2,623	223	-	-	23,719	10	112	144 [TL]
Total	**352,303**	**77,243**		**20,259**		**63,317**		**7,336**	

Fish Culture Operations Overview

The Klamath Falls National Fish Hatchery (KFNFH) uses a unique water supply sourced from a geothermal well, permitted to supply up to 399 gallons per minute (GPM) of water at approximately 88^{o}C. To make this supply suitable for fish culture operations, it is currently retained in three serial flow through head ponds, to supply water to the P-series ponds on lease parcel two, and one larger influent retention pond (IRP), to supply water to the Tier-1 series of ponds on lease parcel one, whereby it is allowed to cool to ambient temperatures for use. It can also be used directly in ponds, in small quantities in relation to the pond volume, for exchange purposes year-round, or in larger quantities to artificially warm the ponds in the spring, fall, and winter months to alter or extend the growing season conditions for the fish.

During FY25, the outdoor facility included 22 0.03-acre ponds in the P-series and six 0.25-acre ponds in the C-series and four 0.125-acre ponds in the B-Series, both of which are in the new Tier-1 bank of ponds. While construction specification deficiencies were noted in nearly all ten Tier-1 ponds, during limited use in FY24, these deficiencies were repaired during the spring of 2025 and were fully available for use during the subsequent growing season. The 22 P-series ponds will eventually be demolished for Phases 3 and 5 construction activities to progress, which will not likely restart until FY27 at the earliest. For more information on the pond layout during FY25, please see **Figures 2 and 3.**

The original indoor intensive rearing facility, constructed in 2018 to replace a much smaller greenhouse, consisted of a 30'x100' greenhouse, which includes five 1300-gallon and three 500-gallon green circular tanks for processing hatchery fish, 13 150-gallon dark blue rectangular tanks, 12 90-gallon green circular tanks, and eight 60-gallon light blue circular tanks for larval sucker rearing, three 175-gallon light blue circulars for salvage rehabilitation, and four Research Racks, recirculating aquaculture systems (RAS), each with a 180-gallon light blue rectangular sump which were used for egg incubation and/or wet lab space for research. The greenhouse and all associated infrastructure will eventually be demolished for Phase 5 construction activities to progress, which will not likely restart until FY27 at the earliest. For more information on the tank layout in the greenhouse in FY25, please see **Figure 4.**

A newer addition to the indoor intensive rearing facilities, constructed in 2021 originally as a storage garage, but outfitted with plumbing and tanks in 2022, consisted of a 24'x40' pre-engineered metal building, which includes 18 90-gallon and one 1300-gallon green circular tanks, and four Research Racks, recirculating aquaculture systems (RAS), each with a 180-gallon light blue rectangular sump which were used for a conditioned water supply for egg incubation and/or wet lab space for research. This metal building is called the Salvage Building, as it was originally developed to hold wild caught fish in isolation or quarantine space, away from the general hatchery population. Because this building was considered when designing the overall hatchery buildout, it will not be demolished during Phase 5 construction activities, but will eventually receive upgrades to the plumbing supply and drainage capacity when these activities ensue. For more information on the tank layout in the Salvage Building in FY25, please see **Figure 5**.

The primary production cycle begins each year with the collection of wild larval suckers drifting down the Williamson River from early May through June, and occasionally extending into early July. Larval suckers are collected, either with stationary drift nets in the middle of the river channel during the night and early morning dark hours and/or by active pursuit with dip nets in the marginal vegetation during early to noon daylight hours. The fish are transported back to the Hatchery and are tempered to the hatchery water supply, enumerated into isolation rearing tanks, and placed on a four-day prophylactic treatment

protocol for removal of ectoparasites, namely a four-hour static bath of salt (0.5%) and an hour static bath of formalin (50ppm) prior to flushing the tank each day. Thereafter, mortality is monitored daily, and any ongoing fish health concerns are treated immediately to ensure that the wild caught fish do not pose a threat to established hatchery stocks before incorporation into the general population.

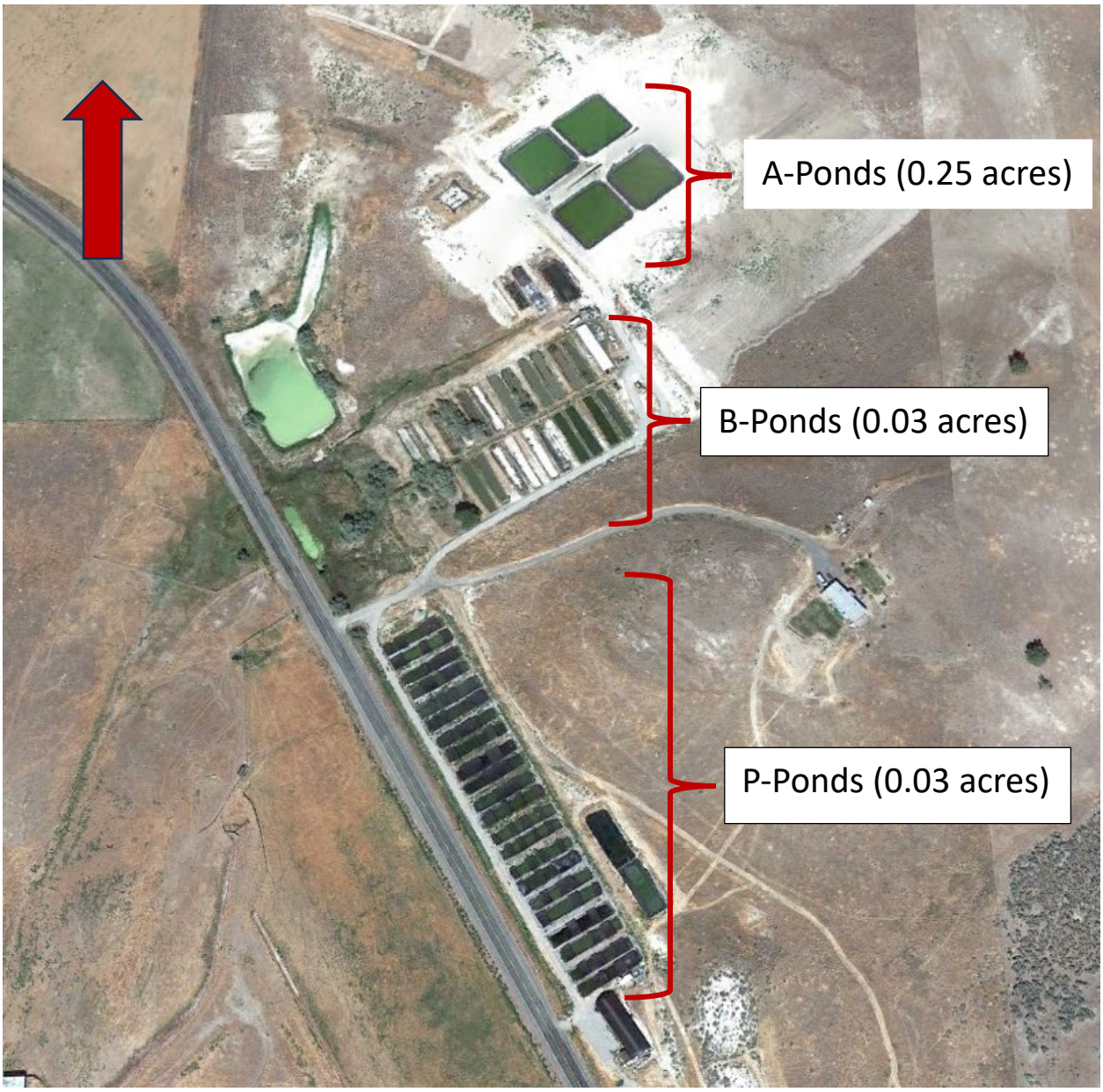

Figure 2 – An aerial view of the layout of ponds at the Klamath Falls National Fish Hatchery. During fiscal year 2023, the B-Pond Series were demolished for Phase 2 construction activities to commence. In fiscal year 2024, the A-Pond Series were demolished for Phase 3 construction activities to commence. During fiscal year 2025, the P-Pond series were the only ponds in this image that were available for broodstock management and production purposes.

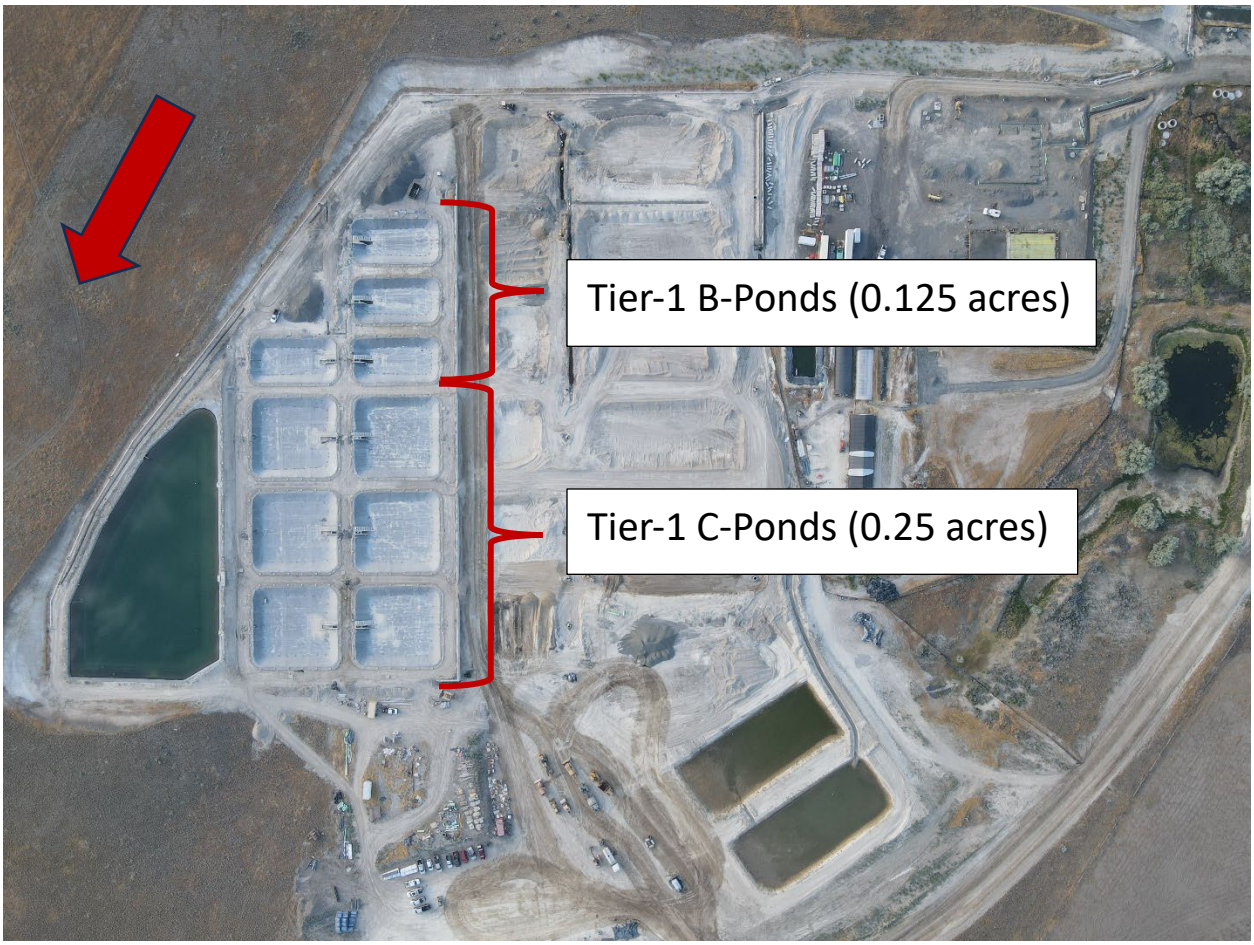

Figure 3 – An aerial view of the layout of the new Tier-1 ponds at the Klamath Falls National Fish Hatchery. This pond series includes six 0.25-acre C-Ponds (C1-C6) and four 0.125-acre B-Ponds (B1-B4). During fiscal year 2025, the Tier-1 pond series saw significant work accomplished to repair the initial construction efforts.

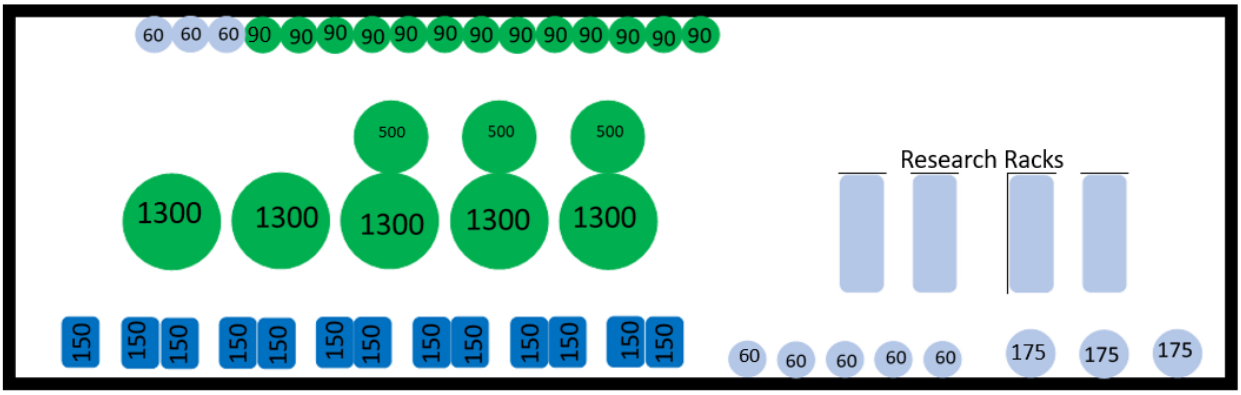

Figure 4 –Layout of tanks in the greenhouse (30'x100') at the Klamath Falls National Fish Hatchery in FY2025. Numbers indicate tank capacity in gallons.

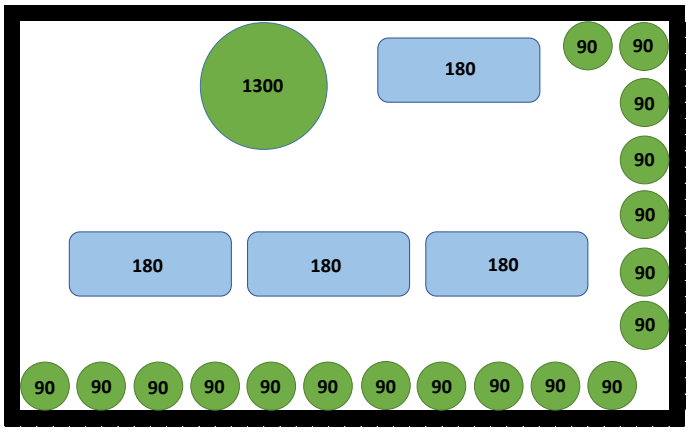

Figure 5 –Layout of tanks in the Salvage Building (24'x40') at the Klamath Falls National Fish Hatchery in FY2025. Numbers indicate tank capacity in gallons.

Once the larval suckers are placed in the quarantine tanks, they are fed live decapsulated and freshly hatched Artemia nauplii three to four times daily, usually at 7AM, 10AM, 1PM, and 4PM daily, and are later supplemented with commercially available dry feeds, such as Otohime A1 diets in a fine 75-150µm mash. All indoor culture units are cleaned each morning, monitored daily for adequate water quality, and fish mortalities are enumerated and removed from systems daily to monitor fish health and disease. After prophylactic treatments are completed and any observable mortality has subsided to normal levels, usually under 0.1-1.0% of the culture unit inventory, the larval fish are generally stocked directly into fertilized ponds with high zooplankton blooms for natural forage. Alternatively, the larval fish can also be feed trained during early rearing in tanks for up to four to six weeks prior to stocking in ponds, depending upon available pond space, manpower, and methodology needed. However, the former method, that of stocking fry directly to fertilized ponds after 5-6 days on station, seems to yield comparable survival, between 65-75%, and better overall growth and condition by the end of the first growing season, compared to long term intensive rearing techniques.

Once fish are ponded, they are generally harvested, split, and restocked at least three times during the 18-24 month growing cycle, usually every 6-8 months, to check inventory, growth, and/or to tag them. During that time, water quality is monitored daily by staff in each active culture unit to ensure that the fish have adequate dissolved oxygen (DO) concentrations to optimize growing conditions; this included ensuring morning DO measurements are above 5.0 mg/L. If the DO concentration is less than 5.0 mg/L, during morning DO measurements, fish are temporarily taken off feed for the day or given a reduced ration, and if DO is below 4.0 mg/L, aerators are immediately deployed. The fish are fed commercially available diets in progressively larger sizes as they grow, at rates that begin at 15-30% body weight (BW) daily and then are reduced progressively down to about 3-5% BW daily by the end of the first growing season. At this point the juvenile suckers have generally reached 80-100mm total length by November or December of their first year. During the cooler winter months, feed rates are usually reduced to about 1-2% BW, and reduced in frequency to one to three times weekly, depending upon water temperature. During the second growing season, which generally starts in April or May, feed rates range between 2-3% BW daily as the fish get older and their metabolism and growth slows. The fish are grown to the average target size of approximately 200mm total length by that second fall and then they are strategically

stocked and repatriated into various parts of Upper Klamath Lake and its tributaries. Prior to stocking out an entire year class, 300-500 fish total are randomly taken from multiple ponds to set aside as a refugia broodstock and incorporated into that captive population. After these refugia broodfish get larger and require thinning, a subset of suckers are randomly selected for repatriation as larger holdovers to stock at a larger size and these fish have served as annual large fish stockings or have been used by partners to do mesocosm or radio and acoustic telemetry studies.

One final component to the fish culture operations is seasonally rehabilitating salvaged suckers that were entrained in the forebay of the A-Canal headworks or unscreened diversions and canals of the Klamath Project. The US Bureau of Reclamation (BOR) staff, at the Klamath Basin Area Office (KBAO), salvages suckers from these locations and fish are transported to the USFWS hatchery. Upon their arrival, USFWS places suckers in temporary isolation, treats them using chemical therapeutants, scans each fish for a PIT tag, PIT tags them if untagged, and measures and records their length and weight. These wild, salvaged suckers are generally repatriated back to Upper Klamath Lake by hatchery staff. Some years result in large numbers of suckers being rehabilitated and repatriated and some years have none. This cooperative effort is important since these wild fish are some of the few suckers aged 0-2 years that have survived during their first few years of life. It may be that these fish have a greater likelihood of surviving into adulthood and rehabilitating them may represent an important sucker conservation measure.

Wild East Side Springs Lost River Sucker Spawning

Wild Lost River Sucker populations spawn in two different locations with strong site fidelity in the Upper Klamath Basin, so much so that they are considered two separate populations in the recovery plan due to the difference in reproductive behavior. The first and largest population, with approximately 2,401 LRS and SNS (Krause, J.R., 2025, 33) spawning in the Williamson and/or Sprague Rivers, near the town of Chiloquin, Oregon, during April and May. The other smaller population, with approximately 286 (Krause, J.R., 2025, 33) individuals, spawn on the east side of Upper Klamath Lake, on a series of springs commonly referred to as the East Side Springs, usually earlier in the spring than the river spawning population. US Geological Service (USGS) staff conduct annual monitoring of the East Side Springs using PIT antennas at Cinder Flats, Sucker Springs, Ouxy Springs, Barkley Springs, Hagelstein Park outlet, and Silver Building Springs to observe spawning use of these locations. During peak spawning activity, USGS staff set trammel nets along the shoreline to collect ripe Lost Rivers and deliver them to Hagelstein park via boat for USFWS staff and Ambodat staff to collect and fertilize gametes. This is the third consecutive year for experimental incubation trials to produce viable offspring for hatchery broodstock population and for production activities.

Spawning activities ran from April 2, 2025 to April 23, 2025 with a total of seven days staff collected gametes. A total of 46 family groups were made with female to male crosses ranging from 1:2 to 1:6 ratio. 67 suckers were handled by USFWS staff, 23 females and 44 males, collecting eggs from 18 females, and were boated back to their origin spring by USGS. To reduce stress on the fish, eggs and milt were collected once and then shared between the USFWS staff and Ambodat staff. After gamete collection, unfertilized eggs from a female were split into a designated number of bowls based on the cross, and each bowl of eggs was fertilized by one male's milt, silted, and rinsed. Thereafter, the eggs were transported to the KFNFH, disinfected with iodine, counted, and placed in separate jars per female for incubation and hatching. Out of the 67 LRS suckers handled by USFWS staff only one was detected via PIT antennas on the spawning grounds. Please see **Table 2** for more information on the adult LRS handled during these spawning events.

In total 122,293 eggs were collected, plus another 40 mL of eggs were flash frozen for thiamine analysis to SUNY Brockport, Brockport, NY 14420. See **Appendix 1** for thiamine egg results. The method of counting egg/mL changed from volumetric displacement method to the water displacement method to improve accuracy on 4/9/2025. The water displacement method was also used to calculate hatch rate as well as hand counting if the number of larvae was minimal. 58,140 fry were hatched from our first fry inventory and overall hatch rate was 47.5% (**Table 3**). Once larvae hatch, they are placed into a 7 qt. bin until they swim up and then are counted into the rack troughs. As sucker fry were developing and being moved around, which included some census counts, a secondary fry inventory showed approximately 62,022 fry hatched, or a 50.7% hatch rate. Once swim up occurred and the larvae had consumed their yolk sac, hatchery water was slowly added to the rack system over the following week. This year decapsulated, hatched artemia was fed 3-4 times daily, and small pinches of Otohime A1 was fed to the troughs as early as 10 days post hatch, which was much earlier than the normal 14-21+ days post hatch that was used in the past, without ill effect. Approximately 3,400 fry were randomly selected from the 46 family groups, 200 fry from each female, and stocked into 2 hatchery ponds to be used for broodstock from the ESS2025 lot. The approximately 50,000 remaining fry were stocked into 2 Tier-1 ponds in May 2025. Some of the excess milt from some males were used in some cryopreservation trials to assess post thaw motility and fertility rates using this milt. This year a protocol was written to preserve milt up for 7

days, using milt extender, because on the spawning day April 23, 2025, only females were captured by USGS, the first time in the history of USFWS collecting gametes.

The hatch rates of ESS LRS larvae during FY25 incubation efforts declined when compared to 70+% hatch rates experienced in past years. Efforts are made every year to refine the augmentation of water quality parameters, such as adjusting hardness and other vital mineral contents after stripping the water via reverse osmosis filtration system, and adjusting chiller settings to better stabilize incubation conditions for hatching and early rearing, but more work is needed to consistently improve hatch rates, fry development, as well as improve overall survival through early rearing. Experimenting and improving incubation, hatching, and feeding methods, as well as adjusting systems configurations and operations will continue to be ongoing priorities each year to determine the best management practices for sucker propagation at KFNFH.

Table 2 – The following lists summary information for the Lost River Suckers spawned at the East Side Springs of Upper Klamath Lake in April 2025, including the date of spawning, PIT tag and fin clip identifications, sex, descriptions of gamete use, fork length, spring location of collection, and the number of family groups created from each female. Fork length and spring location information was obtained from USGS. For thiamine samples collected, approximately 4 milliliters of green eggs were frozen from the female sampled before fertilization.

Date	PIT Tag	Fin Clip	Sex	Gamate Use	Fork Length	Spring Location	Family Groups
4/2/2025	0078DADDC5	3960-097	M	Spawned	560	Cinder Flats	
4/2/2025	0078DADDF2	3960-098	M	Not Spawned	518	Cinder Flats	
4/2/2025	0078DADDBB	3960-099	M	Spawned	575	Offshore Cinder Flats	
4/2/2025	0078DADDB0	3960-100	M	Not Spawned	501	Offshore Sucker Springs	
4/2/2025	003D4854AF	3960-101	M	Spawned	572	Cinder Flats	
4/2/2025	0078DADDE5	3960-102	M	Spawned	517	Cinder Flats	
4/2/2025	0078DADDB2	3960-103	F	Spawned	501	Cinder Flats	4
4/7/2025	0078DADC20	3960-104	M	Spawned	546	Cinder Flats	
4/7/2025	0078DADC04	3960-105	M	Spawned	528	Cinder Flats	
4/7/2025	1C2DB141F7	3960-106	M	Spawned	700	Offshore Sucker Springs	
4/7/2025	00774288EC	3960-107	M	Spawned	643	Cinder Flats	
4/7/2025	0078DADC40	3960-108	M	Spawned	520	Cinder Flats	
4/7/2025	0078DADC4D	3960-109	M	Spawned	600	Cinder Flats	
4/7/2025	257C67E299	3960-110	F	Spawned	729	Cinder Flats	2
4/7/2025	003D8B814E	3960-112	F	Spawned	650	Cinder Flats	2
4/7/2025	0078DADC3F	3960-113	F	Spawned	530	Cinder Flats	2
4/9/2025	0078DADC4F	3960-115	M	Spawned	540	Offshore Cinder Flats	
4/9/2025	0078DADC2E	3960-116	M	Spawned	525	Cinder Flats	
4/9/2025	003C08ECCA	3960-117	M	Spawned	664	Cinder Flats	
4/9/2025	0078DADC61	3960-118	M	Spawned	503	Cinder Flats	
4/9/2025	1C2D1DCF84	3960-119	F	Spawned/Thiamine	738	Cinder Flats	5
4/9/2025	0078DADC2D	3960-120	M	Spawned	523	Sucker Springs	
4/11/2025	0078DADC5A	3960-121	M	Spawned	520	Offshore Cinder Flats	
4/11/2025	0078DADC39	3960-122	M	Spawned	507	Cinder Flats	
4/11/2025	0078DADC64	3960-123	M	Spawned	492	Offshore Cinder Flats	
4/11/2025	0078DADC62	3960-124	M	Spawned	517	Cinder Flats	
4/11/2025	0078DADC08	3960-125	F	Spawned	601	Cinder Flats	4
4/11/2025	0078DADC18	3960-126	M	Not Spawned	503	Offshore Cinder Flats	
4/11/2025	0078DADC49	3960-127	M	Not Spawned	525	Cinder Flats	
4/14/2025	0078DADC09	3960-128	M	Spawned	532	Sucker Springs	
4/14/2025	003BBFCD43	3960-129	M	Spawned	628	Sucker Springs	
4/14/2025	0078DADC0E	3960-130	M	Spawned	524	Sucker Springs	
4/14/2025	003D8B814D	3960-131	M	Spawned	577	Sucker Springs	
4/14/2025	0078DADC26	3960-132	M	Spawned	522	Sucker Springs	
4/14/2025	003C088864	3960-133	F	Spawned	760	Cinder Flats	1
4/14/2025	003BF7C0D4	3960-134	F	Spawned	793	Silver Building Springs	3
4/14/2025	003BC1127A	3960-135	M	Spawned	623	Sucker Springs	
4/14/2025	003C088780	3960-136	M	Spawned	740	Offshore Silver Bldg	
4/14/2025	003D6D42E7	3960-137	M	Spawned	544	Silver Building Springs	
4/14/2025	1C2D80EFCC	3960-138	F	Spawned	751	Silver Building Springs	2
4/14/2025	0078DADC54	3960-139	F	Spawned/Thiamine	547	Sucker Springs	2
4/14/2025	0078DADC24	3960-140	F	Spawned	532	Sucker Springs	1
4/14/2025	0078DADC53	3960-142	F	Spawned	593	Silver Building Springs	2
4/16/2025	0078DADC5D	3960-143	M	Spawned	529	Sucker Springs	
4/16/2025	257C67E2D1	3960-144	M	Spawned	640	Sucker Springs	
4/16/2025	5510	3960-193	M	Not Spawned	690	Sucker Springs and Cinder Flats	
4/16/2025	DC44	3960-194	M	Not Spawned	540	Sucker Springs	
4/16/2025	0078DADC26	3960-195	M	Spawned	523	Sucker Springs	
4/16/2025	0078DADC33	3960-196	M	Spawned	507	Cinder Flats	
4/16/2025	DC51	3960-197	M	Spawned	504	Offshore Sucker Springs	
4/16/2025	0078DADC58	3960-198	M	Spawned	501	Cinder Flats	
4/16/2025	BC6F	3960-199	F	Spawned/Thiamine	747	Offshore Sucker Springs	6
4/18/2025	0078DADDCE	3960-200	M	Spawned	481	Offshore Cinder Flats	
4/18/2025	0078DADDD8	3960-201	M	Spawned	508	Silver Building Springs	
4/18/2025	003BC11270	3960-202	M	Spawned	665	Cinder Flats	
4/18/2025	257C67C79D	3960-203	M	Spawned	691	Cinder Flats	
4/18/2025	0078DADDC4	3960-204	M	Spawned	680	Cinder Flats	
4/18/2025	0078DADDDC	3960-205	M	Not Spawned	524	Offshore Sucker Springs	
4/18/2025	0078DADDB5	3960-206	F	Spawned/Thiamine	560	Sucker Springs	2
4/18/2025	0078DADDCF	3960-207	F	Spawned/Thiamine	552	Offshore Silver Bldg	2
4/18/2025	0078DADDD0	3960-208	F	Not Spawned	728	Sucker Springs	
4/18/2025	0078DADC56	3960-209	F	Spawned/Thiamine	562	Sucker Springs	2
4/18/2025	003C08E8FE	3960-210	F	Spawned/Thiamine	762	Cinder Flats	2
4/18/2025	0078DADDD5	3960-211	F	Spawned/Thiamine	571	Offshore Sucker Springs	2
4/18/2025	003BF7C0F1	3960-211	M	Spawned	716	Offshore Sucker Springs	
4/23/2025	0078DADDA8	NA	F	Not Spawned/Thiamine	747	Sucker Springs	
4/23/2025	003C08EC75	NA	F	Not Spawned/Thiamine	710	Offshore Cinder Flats	
4/23/2025	003D77CD71	NA	F	Not Spawned/No Eggs	NA	NA	
4/23/2025	8812	NA	F	Not Spawned/No Eggs	738	Sucker Springs	

Table 3 – The following summary information lists family group crosses and egg collection volumes and incubation hatch rates for the Lost River Suckers spawned at the East Side Springs of Upper Klamath Lake in April 2025. In total 46 family groups were collected in fiscal year 2025. This table includes the date which females were spawned with each male, the total volume of eggs collected, number of eggs per mL per female, total number of eggs, 1st fry count and hatch rate, and 2nd fry count and hatch rate. The 1st fry count was done by a water displacement method and the 2nd fry count was hand counted. Please note that the total number of eggs reported are estimated based on an extrapolated count and volume; thus, hatch rate is an estimate, and therefore, can be greater than 100% at times. The last 4 digits of the PIT tag ID that were recorded on 4/16/2025 for the female and one male used cannot be verified with our records or USGS data, indicating that a transcription error may have occurred during data collection.

Date	Female	Male	Male	Male	Male	Male	Male	Family Groups	Egg (mL)	# Eggs/mL	Total Eggs	1st Fry Count	Hatch (%)	2nd Fry Count	Hatch (%)
4/2/2025	0078DADDB2	0078DADDC5	0078DADDBB	003D4854AF	0078DADDE5	-	-	4	66	49.33	3256	280	8.6	97	3.0
4/7/2025	257C67E299	0078DADC20	0078DADC04	-	-	-	-	2	62	44	2728	210	7.7	158	5.8
4/7/2025	003D8B814E	1C2DB141F7	00774288EC	-	-	-	-	2	185	48.33	8941	6890	77.1	5629	63.0
4/7/2025	0078DADC3F	0078DADC40	0078DADC4D	-	-	-	-	2	65	53.66	3488	788	22.6	713	20.4
	Escaped						-							3672	
4/9/2025	1C2D1DCF84	0078DADC4F	0078DADC2E	003C08ECCA	0078DADC61	0078DADC2D	-	5	55.7	88.23	4914	1422	28.9	1987	40.4
4/11/2025	0078DADC08	0078DADC5A	0078DADC39	0078DADC64	0078DADC62	-	-	4	45.4	122.94	5581	3100	55.5	1596	28.6
4/14/2025	003C088864	003BBFCD43	-	-	-	-	-	1	13	83.89	1091	267	24.5	495	45.4
4/14/2025	003BF7C0D4	0078DADC09	003BBFCD43	0078DADC0E	-	-	-	3	180	75.45	13581	8471	62.4	7762	57.2
4/14/2025	1C2D80EFCC	003D8B814D	0078DADC26	-	-	-	-	2	267	77.62	20725	13825	66.7	15202	73.4
4/14/2025	0078DADC54	003BC1127A	003C088780	-	-	-	-	2	64	122.31	7828	3000	38.3	4155	53.1
4/14/2025	0078DADC24	003D6D42E7	-	-	-	-	-	1	4.7	129.17	607	258	42.5	672	110.7
4/14/2025	0078DADC53	003BC1127A	003D6D42E7	-	-	-	-	2	9	62.5	563	560	99.6	815	144.9
4/16/2025	BC6F	0078DADC5D	257C67E2D1	0078DADC26	0078DADC58	DC51	0078DADC33	6	116.7	100	11670	3768	32.3	3768	32.3
4/18/2025	0078DADDB5	0078DADDCE	0078DADDD8	-	-	-	-	2	55	136.36	7500	3714	49.5	3714	49.5
4/18/2025	0078DADDCF	0078DADDD8	003BC11270	-	-	-	-	2	65	100	6500	3790	58.3	3790	58.3
4/18/2025	0078DADC56	257C67C79D	0078DADDC4	-	-	-	-	2	47	107.14	5036	2072	41.1	2072	41.1
4/18/2025	003C08E8FE	0078DADDCE	0078DADDC4	-	-	-	-	2	109	96.77	10548	4596	43.6	4596	43.6
4/18/2025	0078DADDD5	003BF7C0F1	003BC11270	-	-	-	-	2	98	78.95	7737	1129	14.6	1129	14.6
								46			**122293**	**58140**	**47.5**	**62022**	**50.7**

Captive Sucker Spawning

While the primary goal of the hatchery is to raise wild caught larval suckers past the recruitment bottleneck size prior to stocking, current population size and structure of wild adults indicate that there could be a time in the near future whereby too few SNS and LRS are present to spawn in the Williamson River to allow for effective collections of larval fish to raise at the hatchery. Part of the goals for the hatchery is to establish and provide viable broodstock populations that can serve both as a refuge population in the event of catastrophic losses in the wild, but also for use in propagating fry for future stockings, especially if wild larvae cannot be obtained each year for production purposes. Starting with wild larval fish from the 2017 collection year, and each year since, both SNS and LRS have been randomly selected and incorporated into a broodstock population, recognizing that these long lived fish take years to reach sexual maturity; generally wild SNS reach sexual maturity between the ages of 4-6 years old and wild LRS between the ages of 7-9 years old; observations at the hatchery reveal that captively raised fish of both species tend to reach sexual maturity at a younger age than wild fish, especially SNS males which may begin producing milt at ages 2 or 3 years of age.

Since the spring of 2021, hatchery staff, with assistance from fish reproductive specialists at the regional level, have tried various methods to get SNS and LRS to ovulate in the hatchery environment and then to effectively incubate viable fertilized gametes. In that first year, we were able to get several SNS females to ovulate with injections of HCG, human chorionic gonadotrophin, and we obtained plenty of milt from males so that we could fertilize gametes. However, we were unable to hatch any viable fry due to limitations in our water supply for incubation purposes. During the next few years, we have found that excessive handling and stress can cause both SNS and LRS to reabsorb their eggs and shut down for the season. However, it is necessary to catheterize females to check egg diameter and growth to time the onset of hormone injections, all of which requires handling. And it is no surprise that we have found that the fish are not well synchronized to spawn at the same time but likely mature within at least a three to four week window, just like the fish in the wild. Over the course of several years, we have also identified limitations with our geothermal water supply that preclude proper gamete development and good hatching rates, requiring treatment and stripping through a reverse osmosis process and then reconstituting the mineral contents thereafter.

Beginning in March 2025, staff harvested broodstock ponds and strategically split the older, mature suckers into two 1/8 acre ponds. Suckers were initially separated by size, >375mm total length for LRS and >300mm total length for SNS and SNS/KLS hybrids. The smaller fish were placed into different ponds. The goal was to consistently monitor one pond for egg diameter and track water temperature, while the other pond wasn't touched until water temperature was close to optimal spawning temperature units, signs of ovulation, and egg ripeness of the monitored fish. Temperatures were monitored closely in late April and early May when pond temperatures reach 14-16^{o}C consistently (**Figure 6**). One pond of mature adults were harvested and assessed for egg ripeness on April 1, 2025. Females who felt swollen were catheterized to collect eggs and measure diameter. Both mature ponds were again harvested on April 21, 2025 and 101 suckers were injected with the first dose of hormones, human chorionic gonadotrophin (HCG) or Ovaprim, and again on April 22, 2025, the following morning. HCG was given at a rate of 100 µl/kg injected intramuscularly in the dorsal area to ripen for three days. The female SNS are given two doses of HCG every 24 hours, totaling a rate of 100 µl/kg injected intramuscularly in the dorsal area, before being placed into a separate "female" tank, or pond net pen. Staff checked for gravidity and ovulation rates of the females and milt volumes of the males on April 25, 2025 when the ovulating

females were spawned. Ages of fish spawned were recorded to determine when hatchery held suckers are reaching sexual maturity. Overall SNS ages ranged from 4 to 8 years old, and LRS ranged from 6-8 years old. For more information, see **Table 6** for the number of each species and sex spawned in FY25.

On April 25 and on May 12, 2025, a total of 12 female SNS were crossed with 1-2 males each depending upon milt volumes and observed motility. A total of 22 family groups collected overall from the wild captive SNS broodstock. In addition, on April 25, 2025, a total of 4 female Williamson River (WR) LRS were crossed with 2 males each, for a total of 8 family groups collected overall from the captive LRS broodstock. Average volume of eggs was 34.5 mL, with one female giving 95 mL from the SNS, and an average of 23.8 mL from the LRS. The milt volume average from 22 males was 1.9 mL for SNS, and 2.5 mL for LRS. Motility of extended milt was observed and assessed using a microscope, where a percent motility was assigned to each male. Average motility of the SNS males was 93% while the LRS males averaged 100%. From these efforts, a total of 83,452 eggs were collected from the shortnose broodstock; 36,387 larvae hatched for a hatch rate of 43.6% on the first fry inventory, and a second inventory later yielded 44,589 larvae for a final hatch rate of 53.4% (**Table 5**). In addition, a total of 25,366 eggs were collected from the WR LRS broodstock; 3,433 larvae hatched, for a hatch rate of 13.5% on the first fry inventory, and a second inventory later yielded 5,591 larvae, for a total hatch rate of 22.0% (**Table 6**).

Early rearing of these captively produced larvae was challenging, as the fry were much smaller than the wild fry collected in the Williamson River, which are generally 1-2 weeks old, post hatch. The San Francisco strain artemia nauplii that we normally feed daily were too large for these fry to eat, and some fish started to die from starvation. Out of necessity, staff started feeding Otohime A1 diet shortly after yolk sac absorption to stop this mortality. The SNS fry readily responded to the feed and started growing, whereas there were nearly total losses of the LRS fry within several weeks post hatch. Approximately 23,781 larval SNS were successfully reared for several weeks, prior to stocking into a fertilized pond for future production purposes. During the 2026 spawning season, we plan on feeding newly hatched fry live rotifer cultures, as well as Otohime A1 diet once the fish absorb their yolk sac, before transitioning fish to artemia nauplii. Please see **Table 5** and **Table 6** for more information on the fish that were spawned and the egg inventories and hatch rates for each female used.

Figure 6 – Daily water temperature of a broodstock pond (B1) from March 20, 2025 to May 31,2025. Water temperature was collected with a HOBO Tidbit temperature logger. Temperature was monitored one time per hour and the Tidbit was placed ~1 foot from bottom of the pond near the kettle. A linear trendline is included as well.

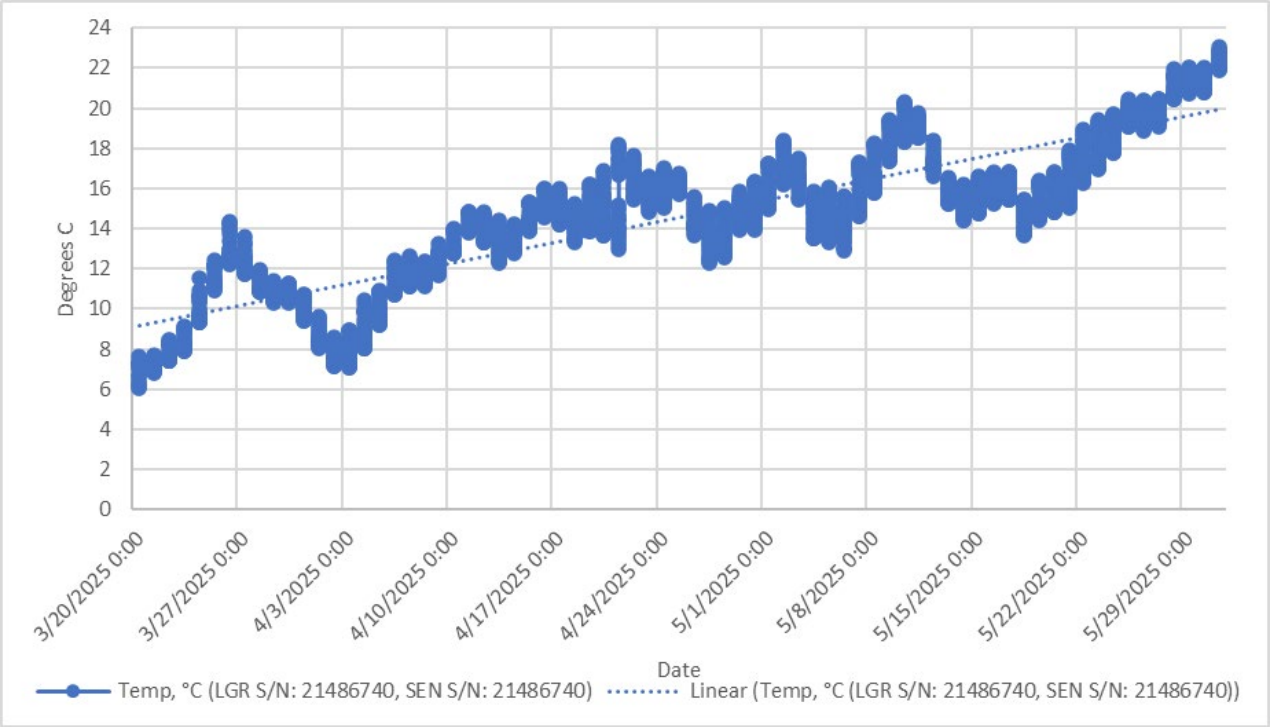

Table 4 – Number of broodstock capture year class per species and sex spawned at KFNFH in spring 2025.

Capture Year (CY)	SNS		LRS	
	F	M	F	M
2017	4	5	2	1
2018	3	6	0	1
2019	2	8	2	1
2020	1	2	0	0
2021	0	3	0	0
N/A	0	0	0	2

Table 5 – Shortnose broodstock spawned at KFNFH in the spring 2025. Summary of shortnose broodstock egg incubation hatch rates spawned from KFNFH in spring 2025. In total 22 family groups were collected. In the table is the date which females were spawned with each male, the total volume of eggs collected, number of eggs per mL per female, total number of eggs, 1st fry count and hatch rate, and 2nd fry count and hatch rate. The 1st hatch count was done by a water displacement method and the 2nd hatch count was hand counted. Please note that the total number of eggs reported are estimated based on an extrapolated count and volume; thus, hatch rate is an estimate, and therefore, can be greater than 100% at times.

Date	Female	Male	Male	Family Groups	Eggs (ml)	#Eggs/ml	Total Eggs	1st Fry Count	Hatch (%)	2nd Fry Count	Hatch (%)
4/25/2025	003C0884B0	003D7C5EB8	003C08BA73	2	170	115.38	19615	6881	35.1	10165	51.8
4/25/2025	003C087DD8	003C08CADD	003C08BA87	2	260	83.33	21667	10428	48.1	12620	58.2
4/25/2025	003C08CB44	003D77D415		1	5.8	115.38	669	625	93.4	644	96.2
4/25/2025	003D78B150	003D77D45B	003D7C6D3E	2	35	166.67	5833	1719	29.5	2157	37.0
4/25/2025	003C08C757	003C08C4C7		1	12	88.42	1061	265	25.0	251	23.7
4/25/2025	003D78D247	003D77DA49	003C088060	2	64	113.33	7253	1921	26.5	2533	34.9
4/25/2025	003D78A5BB	003C0884F6	003D77E270	2	44	81.92	3605	2055	57.0	2141	59.4
4/25/2025	003D77E2A0	003C08BCA7	003D78A835	2	NA	NA	2161	2161	100.0	1954	90.4
4/25/2025	003C08C52B	003C087DE1	003C08CAF4	2	110	71.43	7857	2052	26.1	2485	31.6
4/25/2025	003C08CADE	003D78CB7A	003C087E1B	2	90	78.95	7106	4396	61.9	5858	82.4
5/12/2025	003D7C6F7D	003C088515	003D77E228	2	55	92.11	5241	3833	73.1	3730	71.2
5/12/2025	003C08C87C	003D77E1DE	003E14A9CF	2	10.1	119.33	1384	51	3.7	51	3.7
				22			**83452**	**36387**	**43.6**	**44589**	**53.4**

Table 6 – Lost River broodstock spawned at KFNFH in the spring of 2025. Summary of LRS broodstock egg incubation hatch rates spawned from KFNFH in spring 2025. In total 8 family groups were collected. In the table is the date which females were spawned with each male, the total volume of eggs collected, number of eggs per mL per female, total number of eggs, 1st fry count and hatch rate, and 2nd fry count and hatch rate. The 1st hatch count was done by a water displacement method and the 2nd hatch count was hand counted. Please note that the total number of eggs reported are estimated based on an extrapolated count and volume; thus, hatch rate is an estimate, and therefore, can be greater than 100% at times.

Date	Female	Male	Male	Family Groups	Eggs (ml)	#Eggs/ml	Total Eggs	1st Fry Count	Hatch (%)	2nd Fry Count	Hatch (%)
4/25/2025	003D7C7238	003C08CA5F	003E43CF8E	2	11.6	186.67	2165	51	2.4	0	0.0
4/25/2025	003C08C52D	003D77CF8E	003D7C857F	2	36	147.27	5302	21	0.4	9	0.2
4/25/2025	003C087FD3	003C08CA5F	003D7CA171	2	130	115.00	14950	3361	22.5	5582	37.3
4/25/2025	003D77D187	003C08CB2F	003D7CA171	2	28	105.33	2949	0	0.0	0	0.0
				8			**25366**	**3433**	**13.5**	**5591**	**22.0**

Wild Larval Sucker Collections and Early Rearing

During the Spring of 2025, the Hatchery staff, along with assistance from the KFFWO staff and BOR staff, collected a total of 70,049 larval suckers from the lower Williamson River, near the town of Chiloquin, Oregon. Staff spent a total of 21 collections days on the river between the dates of May 8th and June 12th. The species composition of this catch was presumably a mix predominantly of LRS and SNS, with some potential for limited portion of Klamath Largescale Sucker (KLS). Collection methods predominately used a passive method of deploying drift nets in the river channel overnight this year, as well as using active methods like long handled dip nets to actively collect larval suckers from a small flat-bottomed boat and/or by wading through marginal emergent vegetation during the early daylight hours. For more information on the daily collection trends, as well as overall comparisons of recent larval collection years, please see **Table 7** and **Figures 7,**and **8**.

All collected sucker fry were transported back to the Hatchery daily in 45 quart, insulated RTIC coolers, tempered and hand counted into 90-gallon flow through circular rearing tanks, with a flow through 0.3% salt treatment upon stocking. Prophylactic treatments comprising of a three hour static bath 0.5% salt treatment immediately followed by a one hour static bath 50ppm formalin treatment were administered four consecutive days thereafter. Fry were fed freshly hatched decapsulated Artemia nauplii three to four times daily until they are stocked in a fertilized fry pond. Tanks are siphoned and inspected daily or every other day to collect mortalities and to clean the tanks of debris and uneaten food. If fry are reared for more than 14-21-days, they are supplemented with dry commercial fish diets three to four times daily until stocked into a fertilized fry pond. Experimentally in FY25, fry that were in tanks longer than four days post collection were fed Otohime A in small amounts two to three times a day, to see if they would experience any ill effects. Newly collected fry were able to safely consume the Otohime A diet without staff observing any gut impaction, as had been observed in past years, especially when fry were fed undecapsulated artemia nauplii, which was probably the primary cause of the gut impaction. Fry were ponded within 5 to 19 days post collection.

The overall goal during early rearing the past few years has been to minimize the time that newly collected wild fry stay in indoor tanks before being ponded. Fish health related mortalities from Costia infections during the intensive rearing period are a known issue when fry stay in tanks for longer periods of time. Fortunately, disease didn't seem to be much of an issue in FY25 during early rearing indoors. There is also an unknown cause of mortality in tanks around 45 to 60 days post collection, which may be due to a lack of an unknown nutritional need, as they transition to bottom feeding behavior. And with fry being so small, necessitating the use of 400µm mesh screening to retain them in the culture unit, tank overflows happen on occasion, leading to unknown losses. To help prevent or limit this possibility in FY25, bubble curtains were installed for the first time around the exterior tank standpipes to deter the screen from clogging and to limit unobserved mortalities due to tank overflows at night. These wild fish meet the best growing conditions in fertilized ponds and hence grow larger and have better condition by the end of the growing season, with similar or slightly better returns on survival, for much less work and attention compared to being intensively reared for longer periods of time. Stocking fry into fertilized ponds immediately after collections and initial treatments are complete will become the standard method in future years. Total mortality observed from all the tanks was 3,652 fry, which was approximately 5.1% of the entire stock, with unobserved mortality, primarily from several noted tank overflow events, at approximately 0.9% of the entire stock, which was much lower than in years past. Overall, a total of 65,847 fry were stocked in ponds in May and June 2025, for an overall tank survival

during early rearing at 94.0%. For more information on specific survival and mortality in each culture unit used during the early rearing season, please see **Table 8.**

In 2022, staff began sampling approximately 1% of the daily total collection for larval drift genetic analysis, to determine the species composition of larvae floating down the Williamson River in the spring. This allows the hatchery staff to potentially determine if one species is performing better than another during the spawning period and whether current pond culture practices are better than another to favor certain species, since the species proportion could change on station by the end of the production cycle from what was initially represented in the overall catch. Both LRS, SNS, and KLS are all collected and ponded together during the first 6-18 months of rearing, since we cannot identify them upon collection and during the first growing season. From the data collected so far, it appears that the number of and proportion of LRS present in the larval drift has been going down in the past few years. In 2022, the proportion of LRS was higher than SNS, but in 2024 and 2025, were only around 10-13%. Whereas KLS seem to be staying consistent at about 10-11% from year to year. (**Figure 9**) Please note that larval genetics from the 2023 cohort were only collected for three days and were not included in the **Figure 9** bar chart, as the drift was negligible at only 6,038 fry collected and staff concluded it wasn't worth sacrificing 1% of the fry because all surviving larvae were needed for production and broodstock purposes. In short, from the 3 days of samples run in 2023, 88% were shortnose and 4% were Lost Rivers. This genetic data seems to corroborate what has been observed the past few years with adult population monitoring of LRS exhibiting sharp declines in the wild.

Table 7 – Historical catches of larval suckers from the lower Williamson River, Chiloquin, Oregon. These are used for production and broodstock at the Klamath Falls National Fish Hatchery. (*) Note that 2023 was the first year when collections were limited in the number of fry available during the collection season.

Year (Spring)	# Collected	Primary Collection Method
2016	4,134	Drift Net (Passive Method)
2017	8,730	Drift Net (Passive Method)
2018	9,544	Drift Net (Passive Method)
2019	24,426	Drift Net (Passive Method)
2020	40,603	Drift Net (Passive Method)
2021	106,710	Drift Net (Passive Method) / Dip Net (Active Method)
2022	51,929	Dip Net (Active Method)
2023	6,036*	Dip Net (Active Method)
2024	30,150	Drift Net (Passive Method) / Dip Net (Active Method)
2025	70,049	Drift Net (Passive Method)/ Dip Net (Active Method)
Total Collected to Date	**352,303**	

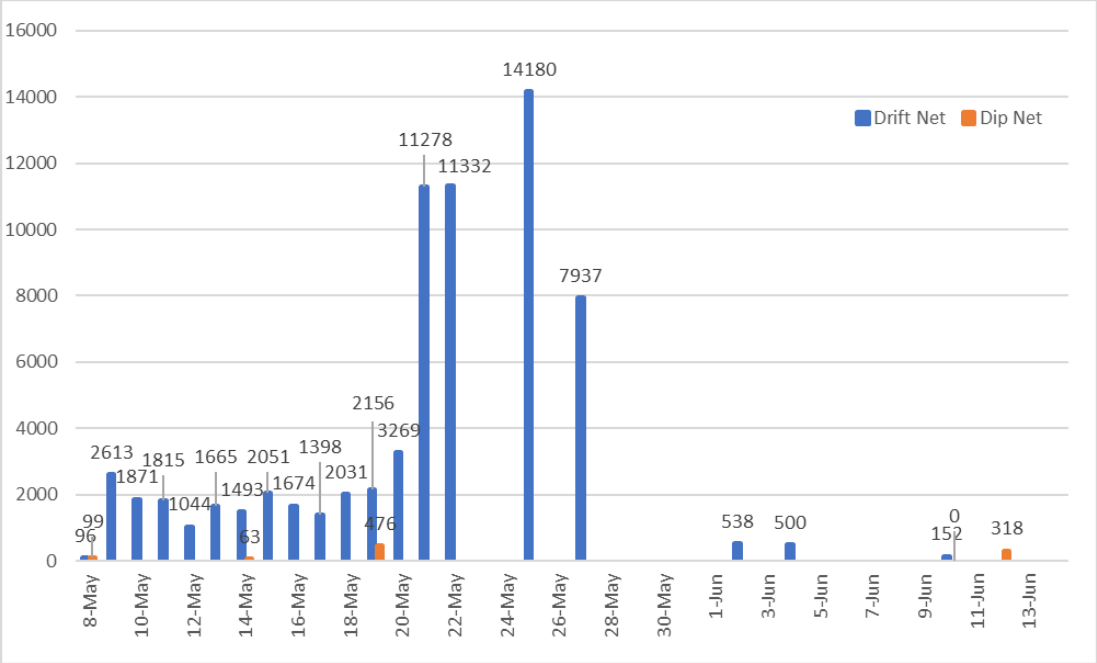

Figure 7 – Number of larval suckers captured per day in the drift nets versus dip netting on the lower Williamson River, near Chiloquin, OR during the spring 2025 for the Klamath Falls National Fish Hatchery. Total number of larval suckers collected and placed in tanks was 70,049.

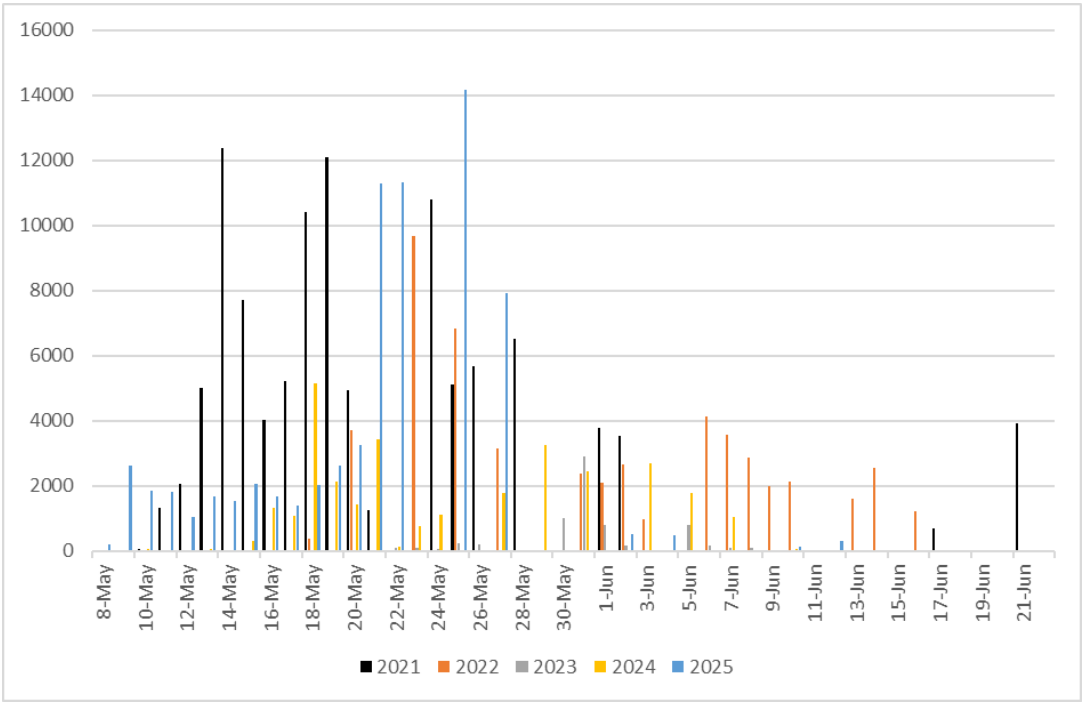

Figure 8 – Number of total daily larval sucker collections from years 2021 to 2025. Collections occur in the spring via dip netting or drift nets set on Modoc bridge on the Williamson River, Chiloquin, OR.

Table 8 – Wild larval sucker collection date, what tank (culture unit) they were placed into, number placed in tank, number of removed mortality, number that was stocked to ponds, survival percent, and mortality percent. Unobserved mortality is calculated by the difference in observed mortality vs number counted to the ponds. Overall, unobserved mortality was calculated as less than previous years due to only counting larvae into tanks and not out to ponds, to reduce staff time and handling stress.

Date	Culture Unit	Collected	# Mortality	# Stocked	Tank Survival (%)	Observed Mortality (%)	Unobserved Mortality (%)
5/8/2025 & 5/9/2025	C1 (Salvage Building)	2808	150	2622	94.6	5.4	0.0
5/10/2025	C2 (Salvage Building)	1871	116	1736	93.7	6.3	0.0
5/11/2025	C3 (Salvage Building)	1815	85	1712	95.3	4.7	0.0
5/12/2025 & 5/13/2025	C4 (Salvage Building)	2709	185	2497	93.1	6.9	0.0
5/14/2025 & 5/15/2025	C5 (Salvage Building)	3607	213	3348	94.0	6.0	0.0
5/16/2025 & 5/17/2025	C6 (Salvage Building)	3072	201	2839	93.4	6.6	0.0
5/18/2025 & 5/19/2025	C8 (Salvage Building)	3531	204	3327	94.2	5.8	0.0
5/19/2025 & 5/20/2025	C9 (Salvage Building)	4401	145	4221	96.7	3.3	0.0
5/21/2025	Restock C1/C2/C4 (Salvage Building)	11278	938	10340	91.7	8.3	0.0
5/22/2025	Restock C3/C5/C6 (Salvage Building)	11332	571	10413	92.1	5.1	2.8
5/25/2025	Restock C7/C8/C9 (Salvage Building)	14180	429	13785	81.8	2.5	15.6
5/27/2025	C10/C11 (Salvage Building)	7937	280	7632	96.5	3.5	0.0
6/2/2025 & 6/4/2025	C15 (Salvage Building)	1038	42	958	94.1	4.1	1.8
6/10/2025	Restock C1 (Salvage Building)	152	0	141	100.0	0.0	0.0
6/12/2025	Restock C2 (Salvage Building)	318	3	276	88.2	1.0	10.9
		70049	**3562**	**65847**	**94.0**	**5.1**	**0.9**

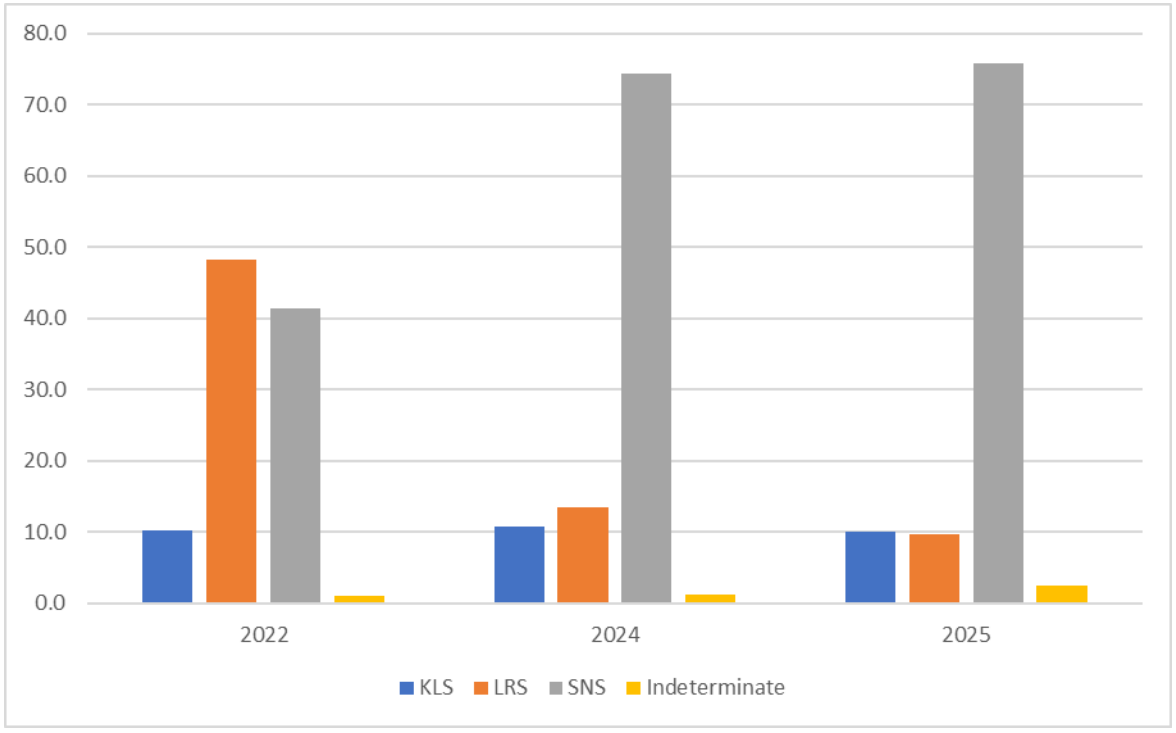

Figure 9 – The species composition percentages from the larval suckers collected from the Williamson River, in Chiloquin, Oregon. Approximately 1% of the total daily drift collected was sampled and analyzed. The data from 2023 is not included as the drift was negligible and not representative, as very few samples were collected due to historically low catches.

Pond Growout

Predation

During FY25, we had some major setbacks in production due to predation on both our old ponds, as well as with some of the newly constructed ponds that were in use before all the infrastructure was completed. The first round of predation was observed when something appeared to be eating fish on the margins of the P-ponds, leaving fish heads, scales, and gonads in small piles in December 2024. This was occurring primarily with our broodstock ponds, the largest and most valuable fish on station. Upon the first major snowfall, this predation intensified and we were able to identify river otter tracks and slides in the snow around the ponds. During this time, foot and body traps were purchased and set along the ponds and in the wetlands towards the north part of the property, and a skilled trapper also attempted to trap the otters as well, but quit after several unsuccessful weeks. Towards the end of December, we did end up getting one otter with the body traps we set in the culvert under the road, that drains all the facility water, but the predation persisted. In addition, an electric fence barrier was installed but this failed to keep the otters out, given large snow events that made the electric fence useless for weeks on end. In an effort to get these fish further away from the road and wetlands, whereby we saw many signs of otter tracks and slides, all of the broodstock fish were harvested in early January and moved to ponds C5 and B3 in Tier 1 pond series for winter storage, although these were still needing warranty repairs to pond grades and liner adjustments the following spring. Since these ponds are further away from where the otters were frequenting, we did not see any signs of otters thereafter on the broodstock ponds.

During the three months while the broodstock fish were in the P-series ponds, from mid-October 2024 to early January 2025, approximately 164 fish were lost, which was a 9.6% loss. This is significantly higher than normal mortality experienced in an entire year across all broodstock lots. The hardest hit broodstock were in pond P2, which had only 74% survival overall for that short period of rearing. Fortunately, most of the smaller fish in the P-series ponds did not seem to be targeted by the otters, although their access as vectors to the ponds did lead to more fish health issues in subsequent months, which normally are not seen during the winter and early spring months. For instance, cases of cold water bacterial infections and fin rot were observed in several P-ponds, especially pond P10, which also had abnormally lower survival at just under 70% survival, which could also have been potential consumption losses from otters. In addition, a handful of juvenile fish started exhibiting abnormal behavior, such as swimming at the surface along the water's edge during the day, which is very uncommon in winter and early spring. Fungal infections were observed on moribund fish, mainly on fins and some heavy colonization on the dorsum of the body and these were proactively removed from the populations. Fish Health specialists were contacted and some sampling was performed, but nothing determinate was identified.

Another and heavier predation event occurred in late February, when we had a small window when our pond ice melted and when all the other water bodies in the area were still frozen. During a two day window, over the weekend, many mergansers converged on the Tier 1 ponds, none of which had bird netting installed yet by the contractor building these ponds, and these birds ravaged our fingerling inventory, as well as the broodstock ponds. While efforts to deter the birds were made, they did not scare easily and kept coming back after staff had completed their shifts. The following Monday, staff purchased several thousands of dollars of pull rope and immediately installed this on the ponds in a grid like pattern, as a temporary deterrent, which seemed to work thereafter, as it broke the flight patterns of

the birds from getting on and off the water. In the subsequent days, staff installed old bird netting from demolished ponds, piecing them together to cover these larger areas, and no further bird predation was noted thereafter, although many mortalities were collected in the subsequent weeks from delayed mortality.

While it was unclear at the time, just how badly the inventory of fish on station had been affected by the merganser predation, we determined that it would be best to harvest all fingerling and broodstock ponds in the Tier-1 series immediately in early March to assess the damage and treat any stressed or injured fish. The results of this inventory work was shocking, especially when we harvested the fingerlings ponds, to find that most ponds were nearly empty of fish. Fingerling ponds in the Tier-1 series, namely ponds C3 and C6 (CY2024) lost 9,434 fish or had a survival of 9%, and ponds B1, B2, and B4 (ESS2024) took an overall loss of 14,221 fish with only 21% survival. Wild broodstock (CY17-CY23), namely ponds C5 and B3, had an overall loss of 666 fish, or 44% of the inventory, with only 56% survival. When broodstock ponds were harvested and during the fish processing it was noted that most fish were over 250mm total length, meaning that most fish lost were from the CY23 and CY22 lots. Once PIT tags were analyzed it was found that only two fish from the CY2023 lot survived, one shortnose and one Lost River. The CY2022 lot survival averaged 36% for both species. During the handling of broodstock, staff observed larger fish with heavy bird strikes from the mergansers, some with missing lips, cuts, scale loss, and abrasions. (**Figure 10**) Broodstock were stocked back into ponds, and again later processed in April 2025 for spawning efforts, and another 23 fish were missing from the inventory, for an overall survival of 97% between March and April. Predation was still observed throughout FY25, mostly from herons working the ponds or walking on the bird netting, however tracks from racoons, cats, and/or otters were occasionally seen on the perimeter pond lining through the 2025 growing season. Efforts were periodically made to tighten the roping as it degraded in the sun to keep the netting off the water. For more information on pond survival throughout the fiscal year, please see the section *Survival*, as well as **Tables 10, 11,** and **12** below.

Figure 10 – Photos of wild broodstock suckers with wounds from the merganser predation event in February 2025, at the Klamath Falls National Fish Hatchery.

Water Quality

Pond water quality was monitored each morning between 600 am and 730 am using a handheld YSI meter to measure temperature and dissolved oxygen concentrations. Parameters are ideally taken about one foot off the bottom of the pond. In instances where stratification was observed two readings were taken at the bottom and top of the water column. These parameter measurements determine daily feed rations for each pond. During such daily fish husbandry activities, ponds were also checked for fish mortality and dead fish were removed to reduce the spread of disease on station. All of such data is recorded into an online Microsoft spreadsheet, using staff work phones, and is available in real-time to partners, such as the USFWS California Nevada Fish Health Unit, so that their staff can monitor our water quality and mortality records as needed, especially if abnormal trends or problems are observed and reported. Close monitoring and observation are the first lines of defense to effectively culture Klamath Suckers and allow the staff to make daily management decisions and adjustments to help care for and protect these captive stocks.

Aside from the basic morning pond checks, there are many other things that require regular monitoring and managed adjustments to meet production goals. Because of the geothermal water source, some production ponds are augmented by adding small flows of hot water in the spring and fall to increase growth potential. These flows are adjusted daily to meet the targeted morning temperature threshold. Staff also monitor the ponds' planktonic blooms, and when these start to diminish, ponds are fertilized with alfalfa pellets, at a rate of 100-200 lbs. per acre, and with super phosphorous, at a rate of 8-10 lbs. per acre. Also, when ponds are on the opposite end of the spectrum, with heavy, dark blooms, and the ponds are supersaturated, staff will flush the ponds with fresh water from the head ponds in an attempt to lower the dissolved oxygen and pH levels. In addition, pH readings are collected and recorded monthly during the warmer seasons with a HANNA pen for monitoring purposes. Also, total ammonia-nitrogen is measured monthly using a LaMotte Colorimeter and unionized ammonia is calculated from that to ensure that blooms are functioning properly and we are not overfeeding our ponds. And during the spring months, temperature HOBOs are continuously monitoring temperature every hour in one or two broodstock ponds for timing the onset of captive spawning operations.

Dissolved oxygen levels in the ponds were overall optimal for the year, generally staying above 6 mg/L on average. This was most likely due to the milder summer temperatures we experienced. Dissolved oxygen stratification occurred in some of the Tier 1 ponds, during the warmer summer months, and this may have resulted in a minor acute mortality event in one pond, where 11 dead fish were collected from pond C6 when dissolved oxygen levels were measured below 2 mg/L on one morning; fortunately this pond had good survival overall for the season. To help mitigate and minimize pond stratification, flushing flows from the IRP was sent to these ponds. Compressed oxygen bottles and micro pore air stones were also added to the ponds to make a refugia area during the low dissolved oxygen events. While we currently lack sufficient electrical capacity to run aeration at every pond in the Tier 1 pond series, an electrical modification is underway so that in the future we will be adding pond aerators, as well as diaphragm air compressors. Having this equipment in place will allow us to continuously move water within each pond to improve nutrient cycling for the bloom and to provide emergency aeration capacity. Over the summer months, surface aerators were placed on timers in the wild broodstock ponds, B1 and C3, to run overnight and mitigate potential crashes in dissolved oxygen due to the heavier loading on these ponds. Aerators are also operated in ponds that measures below 4 mg/L dissolved oxygen, for

emergency use. For more information on the pond water temperature and dissolved oxygen levels throughout the fiscal year, please see **Figures 11** and **12**.

Water quality samples were also collected when high fish mortality was occurring in August and September of 2025 in the ESS2025 ponds, namely ponds C1 and C2. In addition, samples were taken from pond C6 and the IRP as controls during these sampling events. Some of the parameters sampled were for those that we do not have equipment to measure, including looking at heavy metal concentrations, the presence of pesticide residuals, and if harmful cyanobacteria were present in the ponds, producing biotoxins. Approximately $5,000 worth of samples were sent to the following laboratories: the Sprague River Water Quality Lab located in Chiloquin, Oregon (OR), the Neilson Research Corporation located in Medford, OR, Matrix Sciences located in Sherwood, OR, and Bend Genetics LLC located in Sacramento, CA. The Sprague lab completed analysis from ponds C1, C2 and C6 (control) and results concluded that the total phosphorus and total nitrogen in pond C1 were double the concentrations in ponds C2 and C6. The Neilson lab completed a top-35 analysis for ponds C1, C6, and the IRP and the main take away was that the aluminum levels were higher in pond C1 than in pond C6 and the IRP respectively. The Matrix Science lab assessed pesticide residuals in pond C1 versus pond C6 and IRP, but nothing abnormal was noted. The Bend Genetics lab assessed a water sample from pond C2 and found 3 types of cyanobacteria in the sample but no biotoxins present. For more information on the results from these various laboratories, please see **Appendices 2-5**.

Figure 11 – Daily average temperature (degree C) and dissolved oxygen (mg/L) in ponds at the Klamath Falls National Fish Hatchery is broken down quarterly for fiscal year 2025. Pond temperature and dissolved oxygen readings are taken between 600 am and 730 am every day.

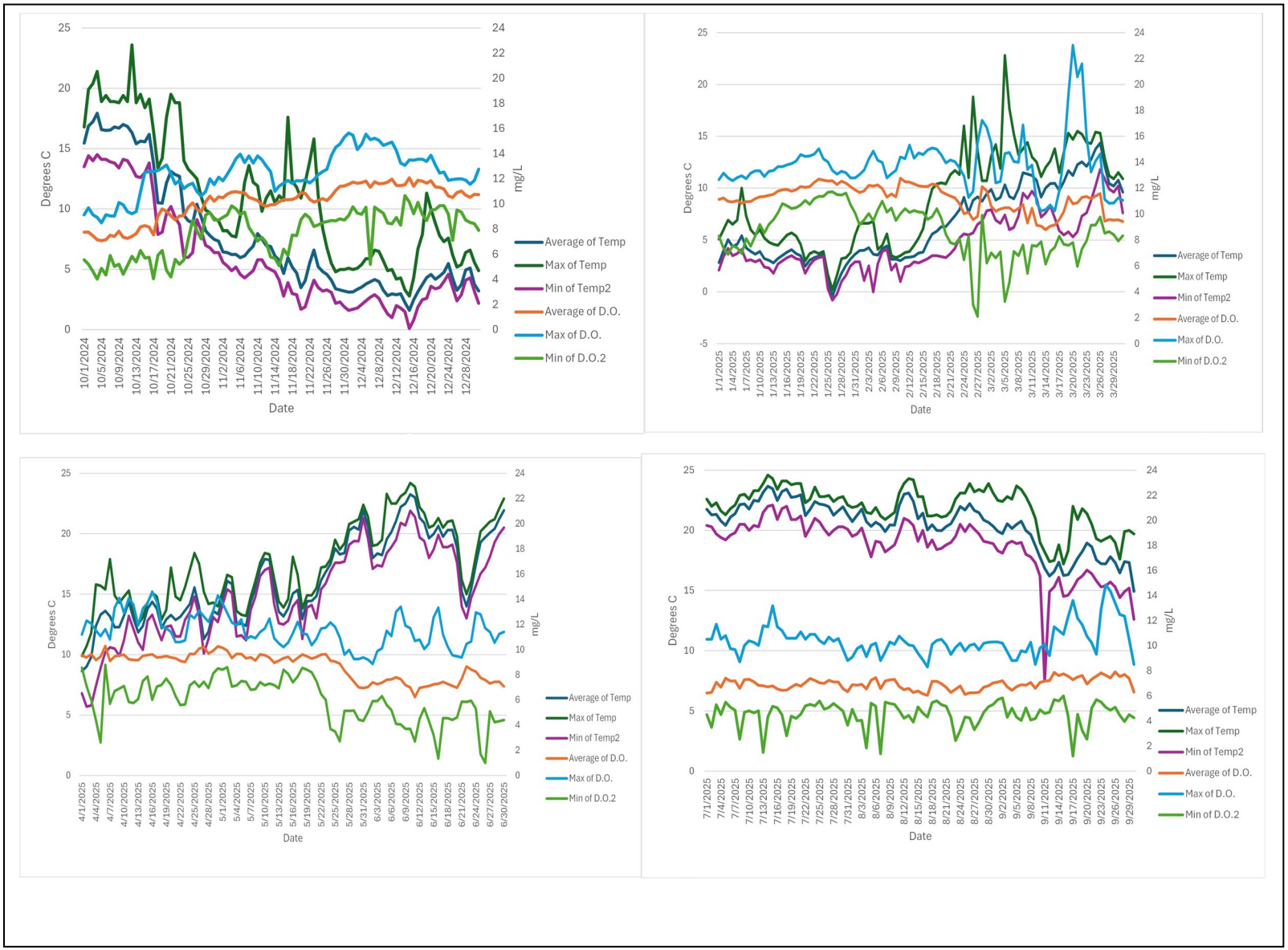

Figure 12 – Daily average, maximum, and minimum temperature (degree C) and dissolved oxygen (mg/L) in ponds at the Klamath Falls National Fish Hatchery is broken down quarterly for fiscal year 2025. Pond temperature and dissolved oxygen readings are taken between 600 am and 730 am every day.

Feed

Klamath suckers are reared mainly in ponds ranging from 0.03 acres up to 0.25 acres currently at the hatchery. During the warmer months, the ponds are managed to produce a planktonic bloom to encourage the production of macroinvertebrates including rotifers, cladocerans, copepods, and insect larvae, all of which suckers naturally forage on in the wild. Having the ability to forage naturally allows a hatchery reared fish to retain its wild characteristics and helps to reduce domestication. Plankton tows are taken periodically during the beginning of the larval ponding stage to ensure forage food is available. Fish growth is sped up as they graze on live feed in the ponds, while their primary diet consists of commercially available and nutritionally complete fish feed pellets, which include full vitamin packs, sufficient animal protein and fat concentrations, and other vital nutrients. In FY25 the hatchery used a variety of commercial diets to feed the various fish sizes on station. For more information on the feed used this past year, please see Table 9.

We experienced a change in the June Sucker diet manufacturer in the spring of 2025 as Skretting, the previous manufacturer, closed all plants in the US and stopped making this diet altogether. Fortunately, the Rangen feed company was able to start manufacturing the diet for the hatchery in a 3.2 mm pellet and a #2 crumble (1.0-1.4 mm broken pellet), with the latter being similar in size to the June Sucker diet we were getting from Skretting in a 1.0 extruded pellet. The new feed from Rangen met all of the same specifications to the formulation of the Skretting diet, as the recipe was designed by the USFWS Bozeman Fish Technology Center. The new June Sucker diet was analyzed quarterly by the nutrition lab at the USFWS Abernathy Fish Technology Center and was within the manufacturer's range for protein %, lipid %, ash %, and moisture % for this particular diet formulation. The peroxides, free fatty acids, and rancidity were also analyzed and were found to be within range. One note is that the free fatty acids were higher in the Rangen diet, at ~8%, than the Skretting diet, at ~5%, from the previous year's analysis, and staff observed the color of the Rangen diet to be darker in color, and oilier, than the Skretting diet.

This change in feed manufacturers may be one reason for the significant mortality we started observing at the end of July, through August, and into September, among the ESS2025 ponds, namely ponds C1 and C2. While we tried to unsuccessfully to chase down a plethora of causative agents, be they water quality or fish health related, the causative agent may simply have been related to the palatability of this "new" diet; the fish likely just stopped eating it. All of the fingerling ponds were switched over to grind of the Rangen June Sucker #2 crumble in early July when the Rangen Salmon starter #1 was used up, which we use as a high protein stopgap diet during early pond rearing, until we could get the new June Sucker diet from Rangen for use as a growout diet. This may be a plausible hypothesis because the wild CY2025 lot, ponds C4 and C5, which is ~80% SNS, as well as the captive 2025 SNS lot, pond C6 did not experience the same type and duration of mortality as the ESS2025 lot did; and none of the age 1+ LRS on station, wild WR or ESS, seemed to exhibit any noticeable deleterious effects from the diet switch. While the former had a gradually diminishing condition factor over time, very little growth, and high mortality rates since July, indicating that fish went off feed around that time, the latter all had steady growth, lower mortality rates, and good survival and returns into the fall of 2025. Growth rates calculated from monthly sampling with hoop nets confirm this trend. One other noteworthy difference was that the ESS2025 in ponds C1 and C2 were initially stocked at 100,000 per surface acre, which is higher than our normal densities stocked in the other lots, which ranged between 60-80,000 per surface acre.

Table 9 – Feed information for FY2025.

Developmental Stage	Feed Type
Larval suckers	Otohime A/A1 (intensive); Rangen Starter #0/ June #2 grinded to powder
Fingerlings (>30 mm TL)	Rangen starter #1/June #2 grind
Juvenile (>120 mm TL)	June #2 crumble (1 mm pellet)
Broodstock/Age 2+ (>200 mm TL)	June #2 crumble/ June 3.2 mm pellet

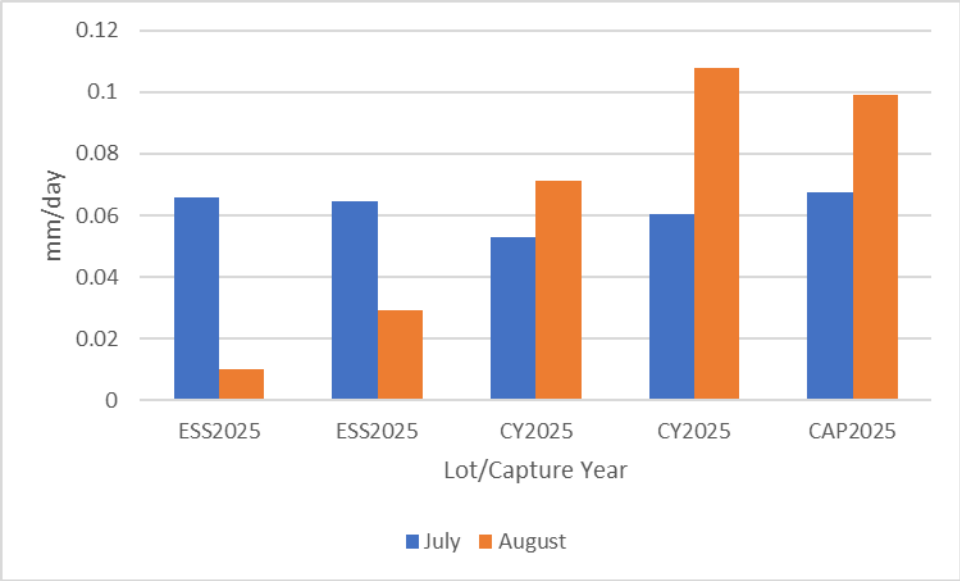

Figure 13 – Growth rate of the 2025 lots in July and August in the tier 1 ponds. ESS2025 are C1 and C2, CY2025 are C4 and C5, and CAP2025 are shortnose in C6. Samples taken once a month with hoop nets set overnight

Pond Survival

All ponds were harvested in the spring and fall to assess growth and survival and to be restocked into hatchery ponds or to stocked off station. Suckers were PIT tagged in fall 2024 at 6 months old, and these, along with suckers too small to tag were counted and batch weighed into a pond. Any fish that were previously tagged and stayed on station where only counted and batch weighed to gain survival and growth metrics. Broodstock were processed every time they were handled to collect individual total length (mm), weight (g), species morphological ID, and to collect a genetic fin clip from the caudal if needed. Broodstock survival ranged from 56% to 100%, with the average survival of 84%. This is below average for the hatchery overall, but the losses are mainly due to the unknown otter predation and merganser predation event in fall/winter 2024/2025. With the new deterrents, new cabling, and netting system, pond survival improved from the spring of 2025 to the fall of 2025, but the possibility of otter

predation will be something to account for into the future. For more information on pond survival and performance throughout each phase of FY25, please see **Tables 10**, **11**, and **12**.

The survival of the CY2024 and ESS2024 lots ranged from 24% to 74% from October 2024 to March 2024. In the summer of 2024, the CY2024 and ESS2024 lots had to be intensively reared through August, as there were construction delays on the Tier 1 ponds being completed and being ready for use. The mortality was high in these lots during that time, due to Costia infections and higher water temperatures in the greenhouse tanks. These production lots never seemed to recover and were stunted in growth when harvest came around in October and November 2025. No CY2024 or ESS2024 ponds were harvested by the end of the fiscal year 2025 to assess the juvenile growth and survival stage.

In September 2025, the Tier 1 ponds containing ESS2025 lots were harvested early due to daily mortality rates being very high. In addition, we started harvesting the CY2025 lot at the end of September to get ahead of the pending government shutdown and to jumpstart and get ahead on fall pond inventory operations, which normally start up in October of each year. The California Nevada Fish Health Center sampled suckers from each of these lots throughout July, August, and September as mortality spikes occurred. A breakdown of the diagnostic summaries for these sampling events can be found in **Appendix 6**. Given that no causative agents could be identified during the various sampling events, the Fish Health staff recommended harvesting the ESS2025 from the ponds into clean tanks even though the handling mortality would potentially be high. Fish were treated in tanks with clean flow through water, O2 injection, therapeutic treatments including 0.5% salt treatments and 50ppm formalin static baths, as well as constant feed availability via belt feeders with starter diet mixed with vegetable oil. The hatchery staff also performed a mini diet trial between 2 tanks containing ESS2025 fish to see whether the new June diet was palatable for them and we found that they consumed it when it was covered in vegetable oil. We had to cover the feed in oil as the suckers were emaciated and we needed to employ every tactic we had to encourage them to eat, including feeding thawed blood worms as well. However, it is still unclear whether the poor survival and performance of these two ESS2025 ponds was due to palatability issues with the new Rangen June Sucker diet or not. So moving into FY26, we will likely stop using the Rangen June Sucker diet altogether and start using a tried, tested, and good performing alternative diet, the Rangen EXTR-450 grower diet, which is commonly used for trout and sturgeon grow out. We found that both LRS and SNS had equally good performance to the Skretting June Sucker diet used in some past feed studies on station in FY22 and FY23.

When the ESS2025 fish were in greenhouse tanks in September, we experienced an abnormal mortality event, where 635 dead fish, all with wounds to the caudal peduncle, were found the first morning, one day post-harvest. It was noted that many backswimmers (Notonecta, or water boatman) had also been added to the tanks from the harvest. After staff received some stinging bites during the handling of these fish, and with a little research, it led staff to the conclusion that the backswimmers may have threatened the health of the fish in these ponds, given the inordinately high amount of backswimmers present in each tank and their abnormally aggressive nature this harvest season. Staff also observed first hand that many of the fish were being attacked by the backswimmers directly in the tanks, and the fish were jumping and flashing, trying to get these biting insects off of them. While backswimmers have always been present in the ponds and subsequentially in the tanks when fish are harvested and moved, we have never seen an outcome like this. Vegetable oil was applied to the coat the surface of the water at 1.5 mL/m2. The vegetable oil ultimately suffocates the backswimmers when they come to the surface for air, as it coats their plastron and makes gas exchange harder. Fish mortality in these tanks dropped the

following day to 121 and most of the backswimmers were dead. This abnormally high post-harvest mortality could also be due in part to the fact that these fish were already in a compromised or weakened state and in poor physical condition, perhaps making them more susceptible to attacks from backswimmers in the tanks. In the future we will be applying a higher application rate of 4.5 mL/m2 of vegetable oil prior to stocking ponds with larval fish, plus periodic maintenance applications at 1.5 mL/m2 as backswimmer infestations reappear.

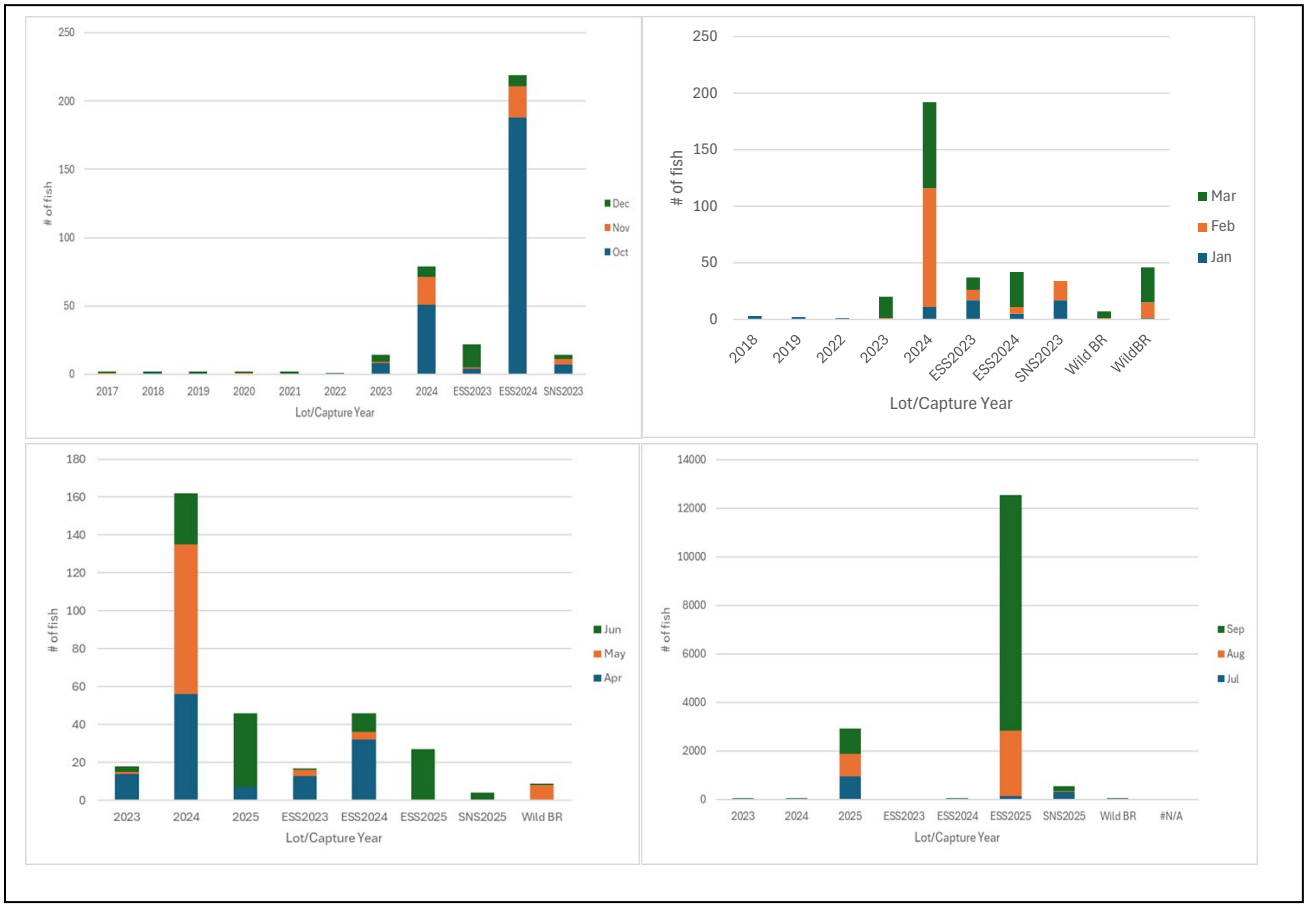

Figure 14 – Observed mortality broken down by fiscal year quarter than by lot/capture year present at Klamath Falls National Fish Hatchery. Wild broodstock (BR) is capture year 2017-2023.

Table 10 – This summary lists the Spring 2024 to Fall 2024 pond performance, including pond number, collection year, stocking date, number of suckers on stocking date, average weight per fish, total weight of all fish in each pond, average total length at stocking date (mm, measured or derived from a length weight conversion), date harvested, number of suckers on harvest date, growth, and survival during the production cycle.

Pond	Lot	Stocked Date	Start #	Start g/f	Start Wt	Start Wt	Start TL	Harvest Date	End # Total	End g/fish	End Wt	End WT	End TL	Days in Pond	Months in Pond	Growth (mm/day)	Harvest Morts	Survival
					(lbs)	(g)	(mm)				(lbs)	(g)	(mm)		(EL-SL)/# Days in Pond			
P0	2020	5/15/2024	127	237.01	66.3	30100	306	10/15/2024	122	349.31	93.87	42616	322	153	5.1	0.10	0	96
P1	2018	12/1/2023	86	477.87	90.52	41097	345	10/8/2024	73	552.97	88.91	40367	374	312	10.4	0.09	0	85
P2	2018	12/4/2023	86	426.07	80.71	36642	336	10/7/2024	76	478.68	80.13	36380	357	308	10.3	0.07	0	88
P3	2019	12/1/2023	91	426.51	85.49	38812	335	10/16/2024	85	504.14	94.39	42852	365	320	10.7	0.09	0	93
P4	2019	11/28/2023	91	458.22	91.85	41698	341	10/14/2024	69	528.59	80.34	36473	368	321	10.7	0.08	0	76
P5	2020	5/15/2024	132	237.01	68.91	31285	306	10/15/2024	124	689.34	188.28	85478	322	153	5.1	0.10	0	94
P6	2021	12/28/2023	266	175.03	102.55	46559	261	10/8/2024	219	275.21	132.75	60270	302	285	9.5	0.14	0	82
P7	2017	5/17/2024	53	673.55	78.63	35698	426	10/7/2024	49	739.00	79.76	36211	410	143	4.8	-0.11	0	92
P8	2017	5/17/2024	32	673.55	47.51	21568	426	10/7/2024	31	821.77	56.11	25475	417	143	4.8	-0.07	0	97
P9	2022	10/27/2023	401	89.90	79.40	36049	217	10/9/2024	367	181.32	146.58	66546	269	348	11.6	0.15	0	92
P10	2024	8/28/2024	1767	4.65	18.11	8222	81.2	10/23/2024	1495	9.66	31.81	14441	106	56	1.9	0.44	84	85
P11	2024	8/28/2024	1669	5.66	20.79	9440	86.6	10/27/2024	1276	11.88	33.38	15154	111	60	2.0	0.41	8	76
P12	2024	5/28/2024	3027	0.05	0.33	151	17.9	10/30/2024	1201	15.34	40.57	18420	121	155	5.2	0.67	4	40
P13	2024	8/28/2024	640	5.69	8.02	3640	86.8	10/30/2024	320	18.56	13.08	5940	129	63	2.1	0.67	5	50
P14	2024	8/28/2024	1600	8.21	28.94	13141	98.1	10/27/2024	1365	13.54	40.70	18478	117	60	2.0	0.32	26	85
P15	ESS2024	8/28/2024	1865	5.77	23.71	10763	87.2	11/4/2024	1974	10.52	45.75	20770	108	68	2.3	0.31	68	106
P16	ESS2023	8/27/2024	422	33.62	31.25	14188	157	10/23/2024	364	75.81	60.78	27596	202	57	1.9	0.79	3	86
P17	ESS2023	8/27/2024	486	38.71	41.44	18814	165	10/23/2024	433	66.55	63.47	28817	194	57	1.9	0.52	1	89
P18	2023	8/27/2024	467	47.24	48.60	22062	176	10/21/2024	310	66.17	45.19	20514	182	55	1.8	0.11	2	66
P19	2023	8/27/2024	457	44.48	44.78	20329	172	10/22/2024	325	53.60	38.37	17419	178	56	1.9	0.10	36	71
P20	2023	8/27/2024	492	49.46	53.60	24333	179	10/21/2024	453	58.19	58.06	26360	179	55	1.8	0.01	8	92
C1	2024	8/2/2024	9510	0.36	50.44	22902	35	11/6/2024	7666	9.12	154.08	69951	97	96	3.2	0.65	19	81
C2	2023	8/19/2024	1345	47.05	139.39	63282	176	11/4/2024	1262	84.79	235.69	107005	195	77	2.6	0.25	0	94
C4	ESS2024	8/2/2024	14250	0.51	16.01	7268	39	12/18/2024	6034	7.35	97.70	44357	95	138	4.6	0.41	24	42
C5	ESS2023	8/19/2024	908	34.82	69.63	31612	159	11/4/2024	875	102.86	198.24	90002	218	77	2.6	0.77	2	96
Wild Brood (17-22)			**1365**						**1215**									**89**
	2023		**2761**						**2350**									**85**
	2024		**18213**						**13323**									**73**
	ESS2023		**1816**						**1672**									**92**
	ESS2024		**16115**						**8008**									**50**
	Overall		**40270**						**26568**									**66**

Table 11 – This summary lists the Fall 2024 to Spring 2025 pond performance, including pond number, collection year, stocking date, number of suckers on stocking date, average weight per fish, total weight of all fish in each pond, average total length on stocking date (mm, measured or derived from a length weight conversion), date harvested, number of suckers on harvest date, growth, and survival for ponds during the production cycle. [1]Broodstock in p series ponds when the first otter predation was observed. [2]Broodstock moved to the tier 1 ponds before and after total # after the merganser event. [3]Initial and end total numbers post spawning and bird strike wounds fall out.

Pond	Lot	Stocked Date	Start #	Start g/f	Start Wt	Start Wt	Start TL	Harvest Date	End # Total	End g/fish	End Wt	End WT	End TL	Days in Pond	Months in Pond	Growth (mm/day)	Harvest Morts	Survival
					(lbs)	(g)	(mm)				(lbs)	(g)	(mm)			(EL-SL)/# Days in Pond		
P0	2023	10/23/2024	478	75.29	79.27	35989	190	1/14/2025	420	75.58	69.92	31743	205	83	2.8	0.18	0	88
P1	2022	10/10/2024	367	181.30	146.58	66546	269	1/13/2025	350	183.38	140.97	63999	276	95	3.2	0.07	1	95
P2	2021	10/9/2024	219	275.21	132.75	60270	302	1/14/2025	161	248.77	88.22	40052	306	97	3.2	0.04	0	74
P3	2020	10/16/2024	246	347.74	188.28	85478	322	1/13/2025	245	352.89	190.44	86458	327	89	3.0	0.06	0	100
P4	2019	10/16/2024	85	504.14	94.39	42852	365	1/7/2025	81	514.41	91.78	41667	371	83	2.8	0.07	0	95
P5	2019	10/16/2024	69	528.59	80.34	36473	368	1/7/2025	67	538.99	79.54	36112	376	83	2.8	0.10	0	97
P6	2018	10/9/2024	73	552.97	88.91	40367	374	1/6/2025	62	534.77	73.03	33156	375	89	3.0	0.01	0	85
P7	2018	10/9/2024	76	478.68	80.13	36380	358	1/7/2025	72	492.82	78.16	35483	365	90	3.0	0.08	0	95
P8	2017	10/8/2024	80	771.08	135.87	61686	413	1/6/2025	71	795.65	124.43	56491	429	90	3.0	0.18	0	89
P9	SNS2023	10/24/2024	1060	66.91	156.22	70923	183	3/3/2025	947	64.62	134.79	61195	186	130	4.3	0.02	3	89
P10	ESS2023	10/25/2024	500	71.02	78.06	35441	197	3/3/2025	349	71.95	55.31	25111	202	129	4.3	0.04	7	70
P11	ESS2024	11/5/2024	903	10.83	21.53	9776	107.6	2/26/2025	818	11.12	20.04	9099	109	113	3.8	0.01	5	91
P12	2024	11/8/2024	1296	3.70	10.57	4799	75.2	3/4/2025	1034	3.74	8.51	3862	75	116	3.9	0.00	37	80
P13	2024	11/8/2024	1358	3.85	11.51	5224	76.2	3/4/2025	1143	3.87	9.75	4425	76	116	3.9	0.00	64	84
B1	ESS2024	11/6/2024	909	10.96	21.94	9963	108	2/25/2025	61	7.74	1.04	472	96	111	3.7	-0.11	0	7
B2	ESS2024	12/20/2024	1815	4.19	16.75	7605	78	2/25/2025	123	3.87	1.05	476	76	67	2.2	-0.04	1	7
B3	WildBR	1/6/2025	467	333.79	343.35	155879	337	3/10/2025	299	438.11	288.54	130995	369	63	2.1	0.51	0	64
B4	ESS2024	12/23/2024	4112	8.76	79.38	36038	100	2/25/2025	1250	8.87	24.41	11083	101	64	2.1	0.01	3	30
B1	Wild BR	3/11/2025	241	515.28	273.53	124183	390	4/1/2025	241	515.27	272.39	123665	390	21	0.7	0.00	1	100
B2	Wild BR	3/11/2025	208	514.49	235.71	107014	390	4/21/2025	195	572.89	246.07	111714	403	41	1.4	0.32	0	94
B3	Wild BR	4/1/2025	240	515.27	272.39	123665	390	4/21/2025	240	515.27	272.39	123665	390	20	0.7	0.00	0	100
B4	Wild BR	3/11/2025	411	267.75	242.39	110044	313	4/22/2025	402	288.04	253.78	115216	321	42	1.4	0.19	2	98
C3	2024	10/27/2024	5401	12.01	142.90	64876	111	3/4/2025	124	6.54	1.79	811	91	128	4.3	-0.16	9	2
C5	WildBR	1/7/2025	1059	258.08	602.00	273307	310	3/11/2025	561	374.76	463.09	210243	351	63	2.1	0.65	0	53
C6	2024	11/8/2024	4959	12.04	131.50	59701	106	3/3/2025	802	7.61	13.45	6106	96	115	3.8	-0.09	3	16
Wild Brood (17-23) 10/24-1/25[1]			**1693**						**1529**									**90**
Wild Brood (17-23) 1/25-3/25[2]			**1526**						**860**									**56**
Wild Brood (17-23) 3/25-4-25[3]			**860**						**837**									**97**
	2023		**1060**						**947**									**89**
	2024		**13014**						**3103**									**24**
	ESS2023		**500**						**349**									**70**
	ESS2024		**7739**						**2252**									**29**
	Overall		**24006**						**9877**									**41**

Table 12 – This summary lists the Spring 2025 to Fall 2025 pond performance, including pond number, collection year, stocking date, number of suckers on stocking date, average weight per fish, total weight of all fish in each pond, average total length on stocking date (mm, measured or derived from a length weight conversion), date harvested, number of suckers on harvest date, growth, and survival for ponds during the production cycle.

Pond	Lot	Stocked Date	Start #	Start g/f	Start Wt	Start Wt	Start TL	Harvest Date	End # Total	End g/fish	End Wt	End WT	End TL	Days in Pond	Months in Pond	Growth (mm/day)	Harvest Morts	Survival
					(lbs)	(g)	(mm)				(lbs)	(g)	(mm)			(EL-SL)/# Days in Pond		
P0	ESS2024	2/27/2025	656	7.55	10.9141	4955	95											0
P1	ESS2024	2/27/2025	713	9.22	14.48018	6574	102											0
P2	ESS2024	2/28/2025	874	10.89	20.95815	9515	108											0
P3	ESS2023	3/9/2025	342	71.95	54.19824	24606	202											0
P4	2024	3/6/2025	581	4.02	5.147577	2337	77											0
P5	2024	3/6/2025	498	3.66	4.017621	1824	75											0
P6	2024	3/4/2025	500	7.83	8.623348	3915	92											0
P7	2024	3/4/2025	414	7.05	6.431718	2920	93											0
P8	2024	3/6/2025	501	3.83	4.22467	1918	76											0
P9	2024	3/7/2025	496	3.59	3.922907	1781	75											0
P10	2023	3/9/2025	454	63.13	63.12996	28661	184											0
P11	2023	3/11/2025	490	66.00	71.23128	32339	187	9/25/2025	456	146.28	145.63	66118	238	198	6.60	0.26	4	93
P12	ESS2025	4/28/2025	1725	0.01	0.038106	17	11											0
P13	ESS2025	4/28/2025	1728	0.01	0.038106	17	11											0
P14	2025	6/12/2025	635	0.03	0.04207	19	15											0
P15	2025	6/12/2025	500	0.03	0.03304	15	15											0
P16	2025	6/12/2025	641	0.03	0.042291	19	15											0
P18	2025	6/12/2025	641	0.03	0.042291	19	15											0
P19	2025	6/12/2025	500	0.03	0.03304	15	15											0
P20	2025	6/12/2025	667	0.03	0.044053	20	15											0
B1	Wild BR	4/27/2025	195	584.44	251.0249	113965	407											0
B2	2025	5/24/2025	7548	0.01	0.165639	75	11											0
B3	2025	5/21/2025	7923	0.01	0.174449	79	11	9/29/2025	4971	6.10	66.78	30318	89	131	4.37	0.60	111	63
B4	2025	5/20/2025	7531	0.01	0.165881	75	11	9/30/2025	5641	8.82	109.53	49725	100	133	4.43	0.67	39	75
C1	ESS2025	5/2/2025	25530	0.01	0.562335	255	11	9/11/2025	12697	4.42	123.47	56057	80	132	4.40	0.52	1144	50
C2	ESS2025	4/28/2025	25009	0.01	0.550881	250	11	9/10/2025	12131	3.58	95.58	43393	74	135	4.50	0.47	1429	49
C3	Wild BR	4/22/2025	640	416.38	586.9736	266486	363											0
C4	2025	6/9/2025	19995	0.02	0.880837	400	13											0
C5	2025	6/6/2025	19266	0.03	1.273084	578	15											0
C6	SNS2025	6/6/2025	23781	0.04	2.095154	951	17											0
Wild Brood (17-23)			**835**						**835**									**100**
	2023		**944**						**910**									**96**
	2024		**2990**						**2990**									**100**
	2025		**65847**						**61005**									**93**
	ESS2023		**342**						**342**									**100**
	ESS2024		**2243**						**2243**									**100**
	ESS2025		**53992**						**28281**									**52**
	CAP2025		**23781**						**23781**									**100**
	Overall		**150974**						**120387**									**80**

Net Pen Growout

Background

In 2019, the USFWS initiated the Net Pen Project in Upper Klamath Lake (UKL). Each spring, wild-caught (as larvae), hatchery-raised and PIT-tagged (age 1+), Lost River and Shortnose Suckers were introduced into two adjacent net pens for extended grow out prior to repatriating to UKL. The objective was to provide an in-situ rearing location, away from predation, to increase growth beyond what is experienced in the hatchery and to provide extended acclimation to wild conditions before release. Due to limited hatchery pond space, net pens allow for some suckers to grow and acclimate, and captive stocks that remain at the hatchery can be held at lower densities. Typical stocking of the net pens occurs in late spring to early summer depending on lake conditions and ice off for access and harvest occurs in the fall of that same year, before ice forms on the lakes. A net pen structure has two separate pens, with each pen having 1,500 sq/ft of surface area with ¼" net mesh.

After very poor survival was observed in 2019, modifications in 2020 raised the bird predation netting 5ft above the water by installing an elevated railing, as suspected avian predators may have infiltrated the netting in 2019. These modifications were not completed until the summer and reduced the 2020 season to a fall growout from September 16th to November 12th. In the summer of 2020, another net pen system was developed for installation in Gerber Reservoir, as hypothesized that poor water quality might be limiting sucker survival in UKL. Increased monitoring of the net pens started in 2020 with deployment of two Biomark 5ft wagon wheel PIT tag antennas for fish survival, and using the nearby (300ft) USGS RPT water quality monitoring station to check dissolved oxygen concentrations with more frequent checks of the system of once every week to two weeks. The UKL net pen at Rattlesnake Point and the newly developed Gerber net pens were deployed and stocked in the spring of 2021. The Gerber net pen required an early harvest in July of 2021, due to low surface elevations and limited water depth for the fish, but growth and survival were high in Gerber net pens, with survival between 83-97.3% and length gains ranging from about 30-45 mm during this short duration. The UKL net pens also required an emergency harvest in August of 2021, due to long term anoxic conditions forming and resulting in significant losses prior to harvest, with 100% mortality in one pen and only 40.4% survival in the other pen and with length gains of approximately 47-60mm during this short duration.

With the problems experienced, the UKL net pen was moved to a new location in Agency Lake in 2022, at the entrance of Four Mile Canal with the two antennas still being deployed for fish survival and using a Yellow Springs Instruments (YSI) EXO1 water quality sonde, to better monitor water quality parameters throughout the growing season. This new location was selected because water quality data collected in the fall of 2021 indicated better water quality than in UKL. The Gerber net pens were not deployed in 2022, as the lake elevation was too low to operate. As the 2022 net pen season closed, we observed the highest survival on record, ~48%, from the UKL net pens at Four Mile Canal, although the fish were in poor condition, hypothesized due to competition and crowding from a large biomass of fathead minnows and blue chubs in the nets with the suckers. Subsequent net pen operations at Four Mile Canal in 2023 included biomass removal in July and adding rope/wood structure to increase food availability but no difference was seen. In 2024 the UKL net pen was used to accomplish two different studies, first being able to hold adult SNS thru the spawning season to check sexual ripeness and the other as a holding area for a USGS/USFWS joint acoustic project in fall of 2024, no production fish were stocked in

2024.For more information on historical UKL/Agency Lake net pen performance, survival, and growth, please see **Table 13**, for location of deployment see **Figure 15**

The Geber net pen was not stocked in 2022 due to low water levels continuing in the reservoir and the fear of another early harvest potentially needed in the summer months. In 2023 the Gerber net pen was stocked in the spring and we experienced a full growing season with acceptable lake levels. One issue in 2023 was low survival in one pen compared to the other, with survival approximately 96.5% for the west pen versus the 45.2% survival in the east pen. During the fall harvest, holes were found in the east pen with the low survival which most likely resulted from fish escapement. During the 2023 deployment, the net pen captured thirty-three small suckers, with an average total length of 103mm, identified as SNS/KLS most likely originating from natural recruitment in Gerber reservoir as no fish were stocked that small inside the net pen, and those fish were tagged and stocked into Gerber Reservoir. During 2024, the Gerber net pen was stocked as usual, with holes repaired, but due to escapement issues and permitting concerns, only fish morphologically identified fish as SNS or KLS were stocked in the Gerber net pen for growout. For more information on historical Gerber net pen performance, survival, and growth, please see **Table 14**, and for deployment location compared to known Gerber spawning areas check **Figure 16.**

UKL and Gerber Net Pen 2025

Due to limited staff capacity and a lack of fish available at KFNFH, neither net pen operation was stocked nor operated in FY25. This was due in large part to the significant losses of fish from the merganser predation that occurred in February of 2025. Infrequent visits were made to check anchor deployment and line tension, as water levels changed, and to ensure no other issues arose during the 2025 season. A YSI sonde was placed in the same location as previous years at the Gerber net pen to continue water quality data collection and monitoring, as part of a long-term data set for the USFWS recovery team at this site. The winch barge that has been in UKL since the start of program was moved to Gerber Reservoir as those net pen anchors need adjustment and the staff will be moving the Gerber net pens to a deeper location for the 2026 deployment. In addition, staff will be moving the UKL/Agency Lake net pens to a new location in the main part of UKL at Mid North, as this will give the fish in the pens access to deeper water during the growing season. And plans are currently being made to use about 8,000 fingerling fish from the inventory at KFNFH to stock all net pens in the spring of 2026.

Table 13. Upper Klamath Lake sucker net pen survival thru each operation year since first deployed in 2019. Each operation year is divided by the different fish pens (UKLN, UKLS, UKLW, UKLE) and the year the fish were captured as larval fish from the Williamson River (CY). The average total length (TL) in millimeters (mm) stocked into the pen and harvested out of the pen that same year to show growth and numbers stocked into the pen and harvested out of the pen to show survival of each operation year. Operation year 2020 is an outlier year as fish were not stocked until fall and only held for two months, and no production fish were stocked in 2024 and 2025.

Operation Year	Fish CY	Pen ID	Stock TL (mm)	# Stocked	Harvest TL (mm)	# Harvested	Survival (%)
2019	2018	UKLN	160	534	0	0	**0.0**
2019	2017	UKLS	236	480	263	10	**2.1**
2020	2019	UKLN	224	220	233	180	**81.8**
2021	2019	UKLN	228	100	0	0	**0.0**
2021	2020	UKLN	134	900	0	0	**0.0**
2021	2019	UKLS	233	150	280	87	**58.0**
2021	2020	UKLS	150	1350	210	519	**38.4**
2022	2021	UKLW	96	1999	142	947	**47.4**
2022	2021	UKLE	105	1852	155	897	**48.4**
2023	2021	UKLW	170	1005	221	29	**2.9**
2023	2021	UKLE	171	1001	220	31	**3.1**
2024	experimental use only no production fish stocked						
2025	No Fish Available - Not Operational						
Overall				**9591**		**2700**	**28.2%**

Table 14. Gerber net pen sucker survival and growth data since the first use of the Gerber net pen in 2021, each operation year is divided by the different fish pens and by the Fish CY (Capture year from the Williamson River as larval fish). The average total length (TL) is in millimeters (mm) of fish stocked in and harvested out of the pen the same year, also shown is number of fish stocked and harvested in the same year for survival of that grow out year. In 2021 fish were pulled out of the net pen early (July) due to low lake levels, and GERE in 2023 show low survival however suckers likely escaped out of multiple holes that were found in net at harvest.

Operation Year	Fish CY	Pen ID	Stock TL (mm)	# Stocked	Harvest TL (mm)	# Harvested	Survival (%)
2021	2019	GERE	235	100	269	97	**97.0**
2021	2020	GERE	140	898	174	827	**92.1**
2021	2019	GERW	230	150	265	146	**97.3**
2021	2020	GERW	138	1331	173	1105	**83.0**
2022	Low Water - Not Operational						
2023	2021	GERE	170	1005	225	454	**45.2**
2023	2021	GERW	171	1036	214	1000	**96.5**
2024	2022	GERE	149	648	205	624	**96.3**
2024	2022	GERW	148	730	201	693	**94.9**
2025	No Fish Available - Not Operational						
Overall				**5898**		**4946**	**83.9%**

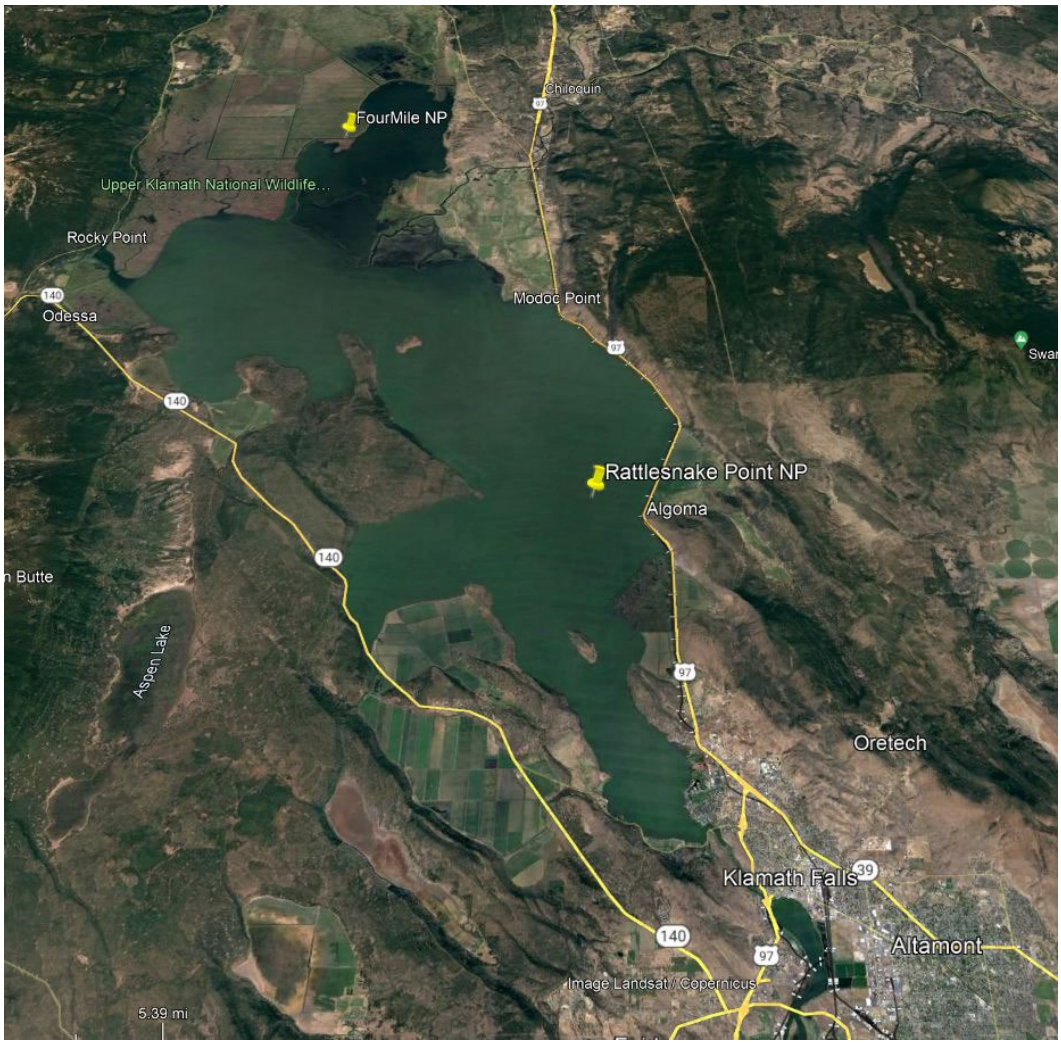

Figure 15 – Locations of the previous net pens location, Rattlesnake Point, in Upper Klamath Lake, and the current net pen location, Four Mile Canal, in Agency Lake, near Chiloquin, Oregon.

Figure 16 – Location of the net pens in Gerber Reservoir (see yellow pin in the middle of the lake, near Bonanza, Oregon.

Fish Distribution

Since the inception of the SARP program, suckers have been released in the spring of each year, when fish are 22-24 months old. Starting in 2020, there have also been fall releases of fish that grew faster and had gotten to the production target size of 200 mm total length by as early as 17-18 months old, with the smaller fish being graded out for growout during a second winter before release the following spring. However, during the fall of 2022, after ponding approximately 7,100 smaller fish, winter rearing space was exhausted, due to vacating the B-series ponds for demolition and construction of the new hatchery building, and we began releasing production fish regardless of size, without grading, in mid-November 2022. As pond space would also diminish in the fall of 2023, with the vacating of the A-series ponds for demolition for Phases 3 and 4 of construction to continue, all production fish were released without grading the fish. Hence, the average size of fish released declined during both years (**Table 1**).

In total, for FY25, dating from October 1, 2024 through September 30, 2025, approximately 2,206 production fish of ~200 mm total length or greater were released at four different locations. In addition, 417 larger production fish were transferred to The Klamath Tribes Hatchery (KTH), as part of a joint USFWS/USGS acoustic telemetry operation. An additional 23,719 fry, collected from the Williamson River, were also transferred to KTH to aid in meeting their production goals. In addition to production releases and transfers, 112 salvage fish were rehabilitated and released by KFNFH at Malone Springs three different times during the fiscal year. The total number of fish distributed for FY25 is 26,454, including production releases, transfers, and salvage fish combined. For more information on historic and current fish distributions, please see **Tables 1** and **15**.

Production sized fish were released in four locations, two of which were typical stocking locations that have been used for the last three years, Odessa Creek and Beatty Gap and the other two non-typical stocking locations were at the south end of Hanks Marsh where USFWS PIT antenna equipment was deployed and at mouth of Four-Mile Canal, the release of acoustic tagged USFWS/USGS study fish. The overall average total length of SARP released or transferred fish was about 223mm, which includes larger study fish for the acoustic telemetry study. If you exclude those larger 30-month old fish, and only look at those reared for ~18th months, the overall average total length was about 219mm. For the first time ever, the SARP production releases included LRS fish that were artificially spawned at the UKL East Side Springs, and were incubated, hatched, reared at KFNFH to target size, and released at Hanks Marsh and Odessa Springs, totaling 1,165 fish. This is the first time we have closed the whole loop on artificial production from the program. All other production fish were captured as larval fish from the Williamson River during the larval drift in May/June. Salvage fish released in FY25 came from BOR salvage operations of the A-Canal Forebay, C-Canal, and J-Canal. Those fish were brought to KFNFH by BOR staff, tempered into rearing tanks, prophylactically treated, scanned for pre-existing PIT tags, PIT tagged if untagged, and then processed to collect length and weight data before being released; generally fish were held no longer than one week at KFNFH prior to repatriation. In addition, all production fish released into the wild were PIT tagged, usually one year before release, at the end of their first growing season on station, for future monitoring and evaluation of hatchery reared fish in UKL.

The number of SARP production fish released was down considerably compared to FY20 thru FY24 as the average number released in those year was 13,553, with the range between 11,774-17,399 released annually (**Table 1**), as this is due to lack of capacity with ongoing demolition and construction at KFNFH. Construction of the new hatchery infrastructure has caused KFNFH to lose pond space starting in fall of

2022, to get new ponds built in the pre-existing location of old hatchery ponds. In FY23 and FY24, the KFNFH lost approximately 72.3% of its pond space for sucker rearing and broodstock management, leading to reduced capacity of being able to rear fish at safe densities through the warm summer months. Low numbers of fish at KFNFH at the start of FY25 due to this lack of capacity was exacerbated with high mortality of ponds in January thru April due to otters and avian predation (see the heading "Predation" under the "Pond Growout" section for more information). With the hatchery in construction mode, there was a lack of animal predation defenses for ponds. The low carry over inventory numbers of fish led to no stockings for UKL or Gerber net pens for growout this season. And predicting into FY26, it will be another low distribution year due to no increase in rearing capacity with hatchery construction, the limited stock of CY2024 fish on hand, and trying to meet a new goal of external partners of increasing the number of larger fish to keep at the hatchery for additional growout before release in future years.

A large part of the FY25 distribution for KFNFH were fish used for the USFWS/USGS acoustic telemetry project which focused on juvenile hatchery reared sucker movement and survival, which was a part of a much larger USGS acoustic study monitoring wild adult suckers in UKL. At the end of FY24, 602 fish were transferred to the Four-Mile Canal net pen to be surgically inserted with an acoustic tag, 406 came from the lower Klamath refuge ponds and 196 from KFNFH. All telemetry fish were originally SARP fish collected as Williamson River larval but those held at the refuge ponds at the start of FY24 were larger, as KFNFH lacked space to house them for future telemetry and large fish releases. Some of the fish from the refuge ponds were already counted as a distribution into the wild, as the refuge pond historically was a long term extended holding pond, and once they left KFNFH were counted as a distribution in the past. With the refuge pond changing as a short-term holding spot the 304 fish added in spring of 2024, these were subsequently counted as a transfer.

The fish harvested from the refuge pond during FY24, which were to be transferred to the net pen for telemetry had an atypical survival for those ponds of 65%, usually its higher, causing the need for more fish to meet the demands of the telemetry project study design, hence adding the additional 196 fish from KFNFH. Fish before and during acoustic tag surgery had an increased mortality while holding in the net pen, causing a low release number of acoustic tagged fish. There was a continuous YSI water quality sonde placed in the net pen checking dissolved oxygen and temperature, but nothing was noted as the cause of mortality. With the observed increase in mortality in the net pen, the California/Nevada Fish Health Center did a diagnostic check on October 16th, 2024, with high bacterial loads noted, most likely from the accumulation of dead and dying fish within the pen. With the numbers of mortality between tagged and untagged fish being pretty even, Fish Health noted that the mortality was most likely not from the tagging surgery, but from the surgical wound being open to the lake environment's high bacterial load, and that non-tagged fish were possibly infected while in the refuge ponds or from being held in the high bacteria load within the net pen. On November 5th 275 fish were released next to the net pen, which included 160 acoustic tagged fish. The USGS acoustic receivers near the release point detected all of the fish released shortly after release. A repeat of the telemetry study took place in fall of 2025 as 417 fish were transferred to The Klamath Tribes Hatchery, to be held for surgery and recovery, instead of using the UKL net pen, along with no holding time at Lower Klamath Refuge Ponds as another possible source of infection or mortality, to hopefully increase post-surgery and post stocking survival. These transferred fish will be counted as a distribution in FY 26 as they were released in December of 2025.

Table 15. Summary of all fish distributions for repatriation or transfer from the Klamath Falls National Fish Hatchery during the fiscal year of 2025, dating from October 1, 2024 through September 30, 2025.

Stock Date	Species	Lot	#	Weight (lb.)	Actual TL (mm)	#/lb.	Projected TL (inches)	Projected TL (mm)	From Ponds/Salvage Area	Stocking Location
10/24/2024	ESS LRS	2023	360	60.62	202	5.94	8.12	206	P16	Hanks Marsh Lakeside Boat Ramp
October Total	**ESS LRS**	**2023**	**360**	**60.62**	**202**	**5.94**	**8.12**	**206**		
11/5/2024	LRS	Mixed	46	16.28	265	2.82	10.41	264	4-Mile Canal Net Pen	4-Mile Canal Net Pen
	SNS	Mixed	230	114.25	226	2.01	11.08	281	4-Mile Canal Net Pen	4-Mile Canal Net Pen
11/6/2024	SNS	2023	765	141.94	194	5.39	7.98	203	C2	Beatty Gap
11/7/2024	ESS LRS	2023	805	183.49	218	4.39	8.99	228	C5	Odessa Springs Boat Ramp
11/8/2024	LRS	Salvage	4	0.21	139	19.12	5.50	140	Forebay	Malone Springs
	SNS	Salvage	43	4.98	171	8.63	6.82	173	Forebay	Malone Springs
	Unknown	Salvage	50	1.31	110	38.28	4.15	105	Forebay	Malone Springs
11/19/2024	Unknown	Salvage	10	0.43	121	23.26	4.90	125	C Canal	Malone Springs
Novmeber Total	**SNS**	**2023**	**765**	**141.94**		**5.39**	**7.98**	**203**		
	ESS LRS	**2023**	**805**	**183.49**		**4.39**	**8.99**	**228**		
	LRS	**Mixed**	**46**	**16.28**		**2.82**	**10.41**	**264**		
	SNS	**Mixed**	**230**	**114.25**		**2.01**	**11.08**	**281**		
	LRS	**Salvage**	**4**	**0.21**		**19.12**	**5.50**	**140**		
	SNS	**Salvage**	**43**	**4.98**		**8.63**	**6.82**	**173**		
	Unknown	**Salvage**	**60**	**1.74**		**34.48**	**4.30**	**109**		
12/23/2024	Unknown	Salvage	5	0.51	154	9.80	6.54	166	J Canal	Malone Springs
December Total	**Unknown**	**Salvage**	**5**	**0.51**		**9.80**	**6.54**	**166**		
5/21/2025	Unknown	2025	6,043	0.10	10	60430.00	0.36	9	Salvage Building Larval Tanks	Transfer to Klamath Tribes Hatchery
5/23/2025	Unknown	2025	17,676	0.29	10	60951.72	0.36	9	Salvage Building Larval Tanks	Transfer to Klamath Tribes Hatchery
May Total	**Unknown**	**2025**	**23,719**	**0.39**		**60817.95**	**0.36**	**9**		
9/25/2025	SNS	2023	417	134.80	238	3.09	9.60	244	P11	Transfer to Klamath Tribes Hatchery / Acoustic Telemetry
September Total	**SNS**	**2023**	**417**	**134.80**		**3.09**	**9.60**	**244**		
FY25 Spp. Total	**LRS**	**Total**	**50**	**16.49**		**3.03**	**10.17**	**258**		
	SNS	**Total**	**1455**	**395.97**		**3.67**	**9.07**	**230**		
	ESS LRS	**Total**	**1165**	**244.11**		**4.77**	**8.74**	**222**		
	Unknown	**Total**	**23,784**	**2.64**		**9022.19**	**0.67**	**17**		
Stocking Total	**Production**		**2206**	**516.58**		**4.27**	**8.62**	**219**		
	Salvage		**112**	**7.44**		**15.05**	**5.67**	**144**		
Transfer Total	**Fry**		**23719**	**0.39**		**60817.95**	**0.37**	**10**		
	Production		**417**	**134.80**		**3.09**	**9.60**	**244**		
Overall Total	**Production**		**2623**	**651.38**		**4.03**	**8.80**	**223**		
	Fry		**23719**	**0.39**		**60817.95**	**0.37**	**10**		
	Salvage		**112**	**7.44**		**15.05**	**5.67**	**144**		
	All Spp.		**26454**	**659.21**						

Field Monitoring for Hatchery Fish

Continuation of the FY24 monitoring efforts took place in FY25 with the purpose of increasing detections of hatchery reared fish in known juvenile sucker habitat. USGS has a considerable number of PIT tag monitoring arrays in UKL and its tributaries but the area to cover is massive with USFWS focusing on juvenile suckers and their habitat preference. This monitoring effort would give some clues to the habitat usage of juvenile fish seasonally throughout the spring, summer and fall months based on the patterns of PIT tag detections with equipment deployed. Consulting with fisheries modelers at the Abernathy Fish Technology Center and utilizing data from recent USFWS radio telemetry work, several locations were selected initially. In FY25, we continued to use most of the FY24 locations, namely Hanks Marsh, Odessa Creek, and Thomason Creek. However, we discontinued using the Williamson Oxbow location because of a lack of detections the previous year. Areas that were added for monitoring in FY25 included the entrance to Algoma Pond, Rattlesnake Point and Barnes/Agency reconnection with some exploration of North Hanks Marsh.

In the spring of 2025, six submersible 5' diameter Biomark Pit antennas were deployed at each location in Odessa Creek, Thomason Creek, Hanks Marsh, and Barnes/Agency, with slight alterations from the previous year's monitoring locations to maximize antenna efficiency. Three 80' litz land-based PIT antenna cables with IS1001 readers, charged with 24v batteries and solar panels were deployed, two in Hanks Marsh in April near submersible antennas to test differences in equipment and one Litz was placed in a new location based on the tip of Rattlesnake Point in May 2025. No significant differences were seen between equipment types, but the litz setup had more range with greater size and an increase in power. In June 2025, with the known and tested ability of the new litz antennas, the Hanks Marsh submersible antennas were moved to the north side of Hanks Marsh to investigate potential SARP fish usage. In addition, two submersibles were deployed in the pinch point under Highway 97 bridge into the Algoma backwater. Each antenna was checked monthly to flip fresh batteries on submersibles and to download the available data during each visit. Also slight adjustments to the antenna locations were made at each site upon retrieval due to decreasing lake levels.

There were minor issues with occasional battery strength or connection issues, but the vast majority of equipment and sites worked with no issues. Thomason Creek antennas had issues with antennas moving during spring months and had to be ground tethered to avoid movement, also buoys would go missing from an apparent chop from boating activity. Staff could usually find the antenna due to GPS waypoint and still find the antenna working with no issues, this is true except for two antennas early in the FY25 season. Two antenna buoys went missing in April of 2025 and with repeated search efforts with boat gaffs and hooks could not be found during the summer, water was deep, turbid and not near the waypoint. In October when pulling all equipment out of Thomason Creek, extended searches took place and staff found one of the two missing antennas. Staff will continue the search for the missing antenna during the spring and summer of 2026. All the antenna issues and battery failures are logged in the antenna metadata for data analysis and all of the antenna metadata and detections data was sent to USGS for upload into their database.

Focusing on SARP fish detections, the Hanks Marsh and Odessa Springs arrays had the majority of the detections from the initial deployment until the end of June; a similar pattern or trend was observed in FY24. A full field season at Thomason Creek showed an improvement in detections from the year before but it is difficult to handle equipment in the area. Of the three new areas for antennas, Rattlesnake Point

did well with 54 unique detections using only a single litz cable and no submersible antennas. As for the Algoma and Barnes/Agency arrays, only 13 total SARP fish combined were detected during the year. The exploration of submersible antennas in North Hanks Marsh resulted in only six unique tag detections. A total of 508 unique detections in field season 2025, with 44% of those being SARP or salvage fish that passed through the hatchery. In addition, a total of 285 unknown tag detections were also recorded, which could be adult suckers, Redband Rainbow Trout, or possibly experimentally released juvenile Chinook Salmon. For more information on the total number of unique fish detections, please see **Table 16**.

Of the 194 SARP fish detected with the USFWS monitoring effort, 86 (44.3%) of those were reared/long term acclimated for up to six months in one of three large net pen setups at Gerber Reservoir, Rattlesnake Point in Upper Klamath Lake, or Four-Mile Canal in Agency Lake. The majority of the fish detected, approximately 79.9%, were two to three years of age (CY23/CY22), with two of the fish detected being eight years of age (CY17), with the majority of the older fish detected at the Odessa Creek antennas. For more information on age and net pen acclimated number of fish detected in USFWS monitoring efforts, please see **Table 17.**

With the 2025 monitoring at the six locations and looking at all unique tag detections per day, a pattern was found with majority of the detections taking place in the spring with a significant drop off the end of June and then an uptick in the number of detections in the fall (**Figure 17).** When focused on just SARP juvenile fish that were detected in 2025, there is similar trend but without the uptick in detections in the fall (**Figure 18)**. Unknown tags are still being investigated to see what is being detected in these monitoring areas in the fall. When focused on SARP detections per site, the same trend is seen through all the sites, with Odessa Creek having more detections in the fall than all the others, as Odessa, Rattlesnake Point, and Hanks Marsh showing detections of SARP fish past August 5th (**Figure 19).**

Continued analysis of the monitoring data will occur in FY 26 with inquiring of unknown tags in USGS database, and diving into individual detection points at each monitoring location to see patterns of shallow or near land vs deep/ distance from land locations. Evaluation of sites will take place with continuation of most in FY 26 to evaluate trends and increase detections of SARP fish. Hanks Marsh, Odessa, and Rattlesnake Point will continue due to high detection numbers, and Barnes/Agency most likely with the high interest in the recent reconnection of the wetland and sucker use of the habitat. Unsure of Algoma due to low tag detection or Thomason with equipment issues being used in FY26 as more evaluation will be needed for those locations. Additional equipment was purchased in FY25 and are being prepared to deploy in FY26 to investigate other locations of possible SARP fish including Howards Bay, Ball Bay, Eagle Point, and Bear Island.

Table 16. Summary of all of the unique fish detections at six different sites where submersible antennas detected PIT tags in Upper Klamath Lake and its tributaries, using USFWS monitoring equipment, for the calendar year of 2025; this only included detections of SARP fish six months post stocking. The unique tags that are listed as unknown are either adult wild suckers or other species that are PIT tagged in the system.

Monitor Location	Deployment	Retrevial	Unique Tag	SARP tag	Salvage tag	Unknown Tag
Algoma	6/11/2025	10/24/2025	3	2	0	1
Barnes/Agency	4/24/2025	10/29/2025	18	11	3	4
Hanks Marsh	3/25/2025	10/27/2025	121	75	8	38
Odessa Creek	3/25/2025	10/30/2025	269	42	11	216
Rattlesnake Point	5/29/2025	10/27/2025	54	37	2	15
Thomason Creek	3/24/2025	10/30/2025	43	27	5	11
Calendar Year 2025 Total			**508**	**194**	**29**	**285**

Table 17. Summary of all of the unique SARP fish detected at six different sites where submersible antennas detected PIT tags in Upper Klamath Lake and its tributaries, using USFWS monitoring equipment, for the calendar year of 2025; this included detections of SARP fish six months post stocking. This table also shows the SARP fish detected that were reared in a USFWS net pen setup (Gerber, UKL Rattlesnake Point, or Agency Lake Four-Mile Canal), and the age of the fish detected; the "CY" represents the collection year of the fish collected as larvae and brought to the Klamath Falls National Fish Hatchery for rearing.

Monitor Location	Deployment	Retrevial	SARP tag	Netpen Fish	CY23	CY22	CY21	CY20	CY19	CY18	CY17
Algoma	06/11/25	10/24/25	2	2	0	2	0	0	0	0	0
Barnes/Agency	04/24/25	10/29/25	11	5	5	4	1	0	0	0	0
Hanks Marsh	03/25/25	10/27/25	75	34	35	36	2	1	0	1	0
Odessa Creek	03/25/25	10/30/25	42	16	12	6	4	15	1	2	2
Rattlesnake Point	05/29/25	10/27/25	37	19	14	21	1	1	0	0	0
Thomason Creek	03/24/25	10/30/25	27	10	10	10	2	2	1	2	0
Calendar Year 2025 Total			**194**	**86**	**76**	**79**	**10**	**19**	**2**	**5**	**2**

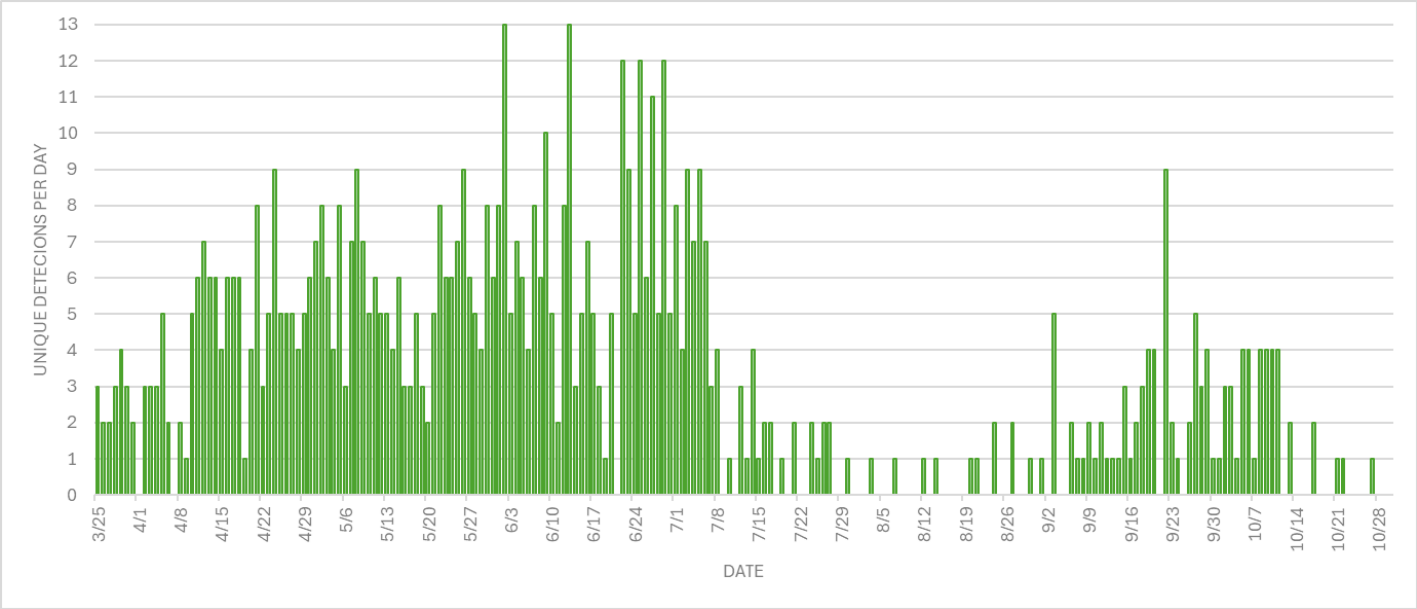

Figure 17 – All PIT tag detections, including SARP, salvage, and unknown fish in the calendar year 2025 from USFWS PIT tag monitoring efforts at six different locations in Upper Klamath Lake and its tributaries, with submersible antennas showing daily unique detections thru the field season. Most antennas were deployed by April of 2025, with all antennas deployed by June of 2025, and all antennas were retrieved at the end of October 2025.

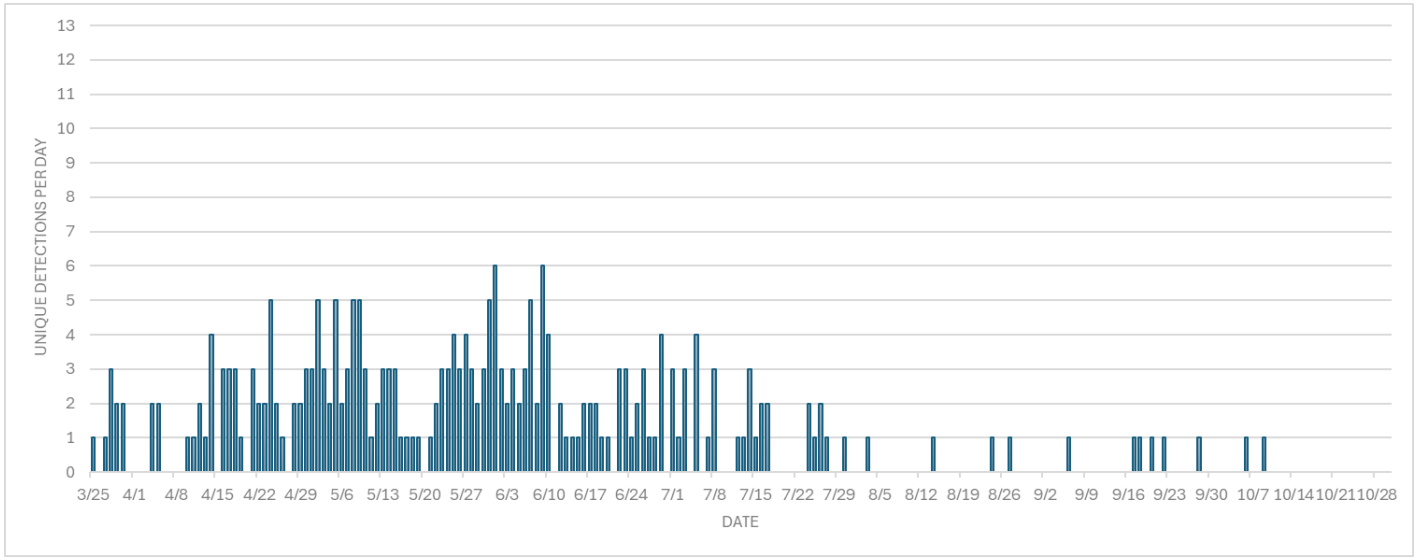

Figure 18 – Known SARP fish detections in the calendar year 2025 from USFWS PIT tag monitoring efforts at six different locations in Upper Klamath Lake and its tributaries, with submersible antennas showing daily unique detections thru the field season. Most antennas were deployed by April of 2025, with all antennas deployed by June of 2025, and all antennas were retrieved at the end of October 2025.

Figure 19 -- Known SARP fish detections in calendar year 2025 from USFWS PIT tag monitoring efforts, divided into the six areas of focus in Upper Klamath Lake and its tributaries. Horizontal dotted lines represent when equipment was deployed and retrieved, with no major issues or long data gaps found on equipment. Most antennas were deployed by April of 2025, with all antennas deployed by June of 2025, and all antennas were retrieved at the end of October 2025.

Partnerships/Collaborative Studies

USGS/USFWS Acoustic Sucker Juvenile Tag Study

The USGS started an acoustic tagging and monitoring study of adult suckers in UKL in 2024 with the deployment of over 70 acoustic telemetry receivers in a lake-wide array in UKL and its tributaries. This will provide high resolution movement and survival estimates of the fish in this large body of water. With juvenile sucker movement and survival from being largely uncertain, using this telemetry array to get more data on juvenile suckers was called for. With the difficulty of tracking untagged juvenile fish in the wild, these smaller fish would need to be provided by KFNFH. An older and larger SARP cohort was being held at the KFNFH with the purpose of being used for stocking larger fish from the program, and some were utilized for the USFWS radio telemetry program. These fish were moved to the Lower Klamath Refuge ponds in October of 2023 as pond space was contracting at KFNFH with construction. During the winter of 2023/2024 the USFWS radio telemetry program was ending its stocking of radio telemetry fish, making the fish at the Lower Klamath Refuge ponds available. With a proposal for acoustic telemetry study of juvenile suckers being approved, these fish would be used to assess LRS and SNS survival across different size classes, so these fish were targeted to gather information that will supplement the adult study.

In March of 2024, the refuge pond fish were harvested and inventoried, and a little over half the fish that survived were stocked in the Williamson River, as part of planned large fish release near spawning grounds. The other half of the fish were restocked into a refuge ponds, with another 304 added from KFNFH making a total of 647 fish that will be used for acoustic telemetry in fall of 2024. In September 2024, 405 suckers were harvested and transported from Lower Klamath Refuge Ponds to the USFWS net pen setup in Four-Mile Canal in Agency Lake, as a holding place for tag surgery and a four-to-six-week recovery. Due to the lower than average expected survival in the refuge ponds, an additional 196 suckers from KFNFH were stocked at the net pen at the same time to ensure sufficient numbers for the juvenile study. Shortly before surgery took place and after surgery, unexpected mortality was observed. Thereafter, upon further investigation, the continuous water quality sonde placed in the net pen showed no dissolved oxygen or temperature issues along with no signs of escapement from the net, nor any signs of predation on the fish. The California Nevada Fish Health Center visited the net pen setup on 10/16/2024 to collect samples to investigate the mortality issues at the net pen. With equal mortality on the control fish and the surgery fish, surgery was not likely the main cause of the mortality, as most fish were likely unhealthy prior to surgery with bacterial infections and coupled with being held in wild conditions in the lake and with the accumulation of sick and dead fish in the net pen, it likely created epizootic conditions. On November 5th of 2024, 275 fish were released from the Four-Mile net pen at the site of the net pen, 160 acoustic tagged fish were included in that release. Latest verbal update from USGS show all fish were detected in November 2024 and are getting data on at least a third of the fish thru the summer, Matthew Sholtis USGS giving presentation on these detections at annual Sucker Technical Symposium January 2026.

Of what was released in November of 2024, it was mostly SNS of two size classes, fish around 210 mm and 240 mm total length, which led to the desire for a repeat acoustic telemetry study the following year with focus on SNS of two size classes to investigate survival by size and stocking method. For the 2025 telemetry study fish will come directly from the KFNFH, with no holding period at the Lower Klamath Refuge ponds to avoid that variant in the study and delivered to the tagging crews at the Klamath Tribes

Hatchery. With the unexpected mortality from avian predators and otters at KFNFH during the winter of 2024/2025, there was an inventory limitation for this study resulting in no smaller size class being available for acoustic telemetry. In September 2025 417 SARP fish were transferred to The Klamath Tribes Hatchery for holding, surgery and recovery, with 349 of those getting an acoustic telemetry tag (Innovasea V7x2 or V7x4) implanted or underwent control surgery with an incision and suture but no tag. Decreased mortality and infection took place compared to the 2024 surgery, as there were 5 mortalities during the entire holding and recovery time at KTH, with 8 being re-ponded into Ambodat ponds, and 60 not having a tag placed but released as a production fish with just a PIT tag. Fish were stocked on December 4th 2025, with 203 fish stocked via boat; 85 study fish (acoustic tag or surgery control) and 16 non-study (PIT tag only) fish were released at Rattlesnake Point (42.4055849, -121.8720721) and 87 study fish and 15 non-study fish were released at Mid North (42.4387344, -122.0123557). The total number of fish released by truck was 201, 87 study fish with 15 non-study fish were released at Henzel Park boat ramp and 85 study fish with 14 non-study fish were released at Odessa Springs. Acoustic tags will be tracked using a ~30 acoustic receiver array throughout the winter and ~80 acoustic receiver array during summer months. The USGS is responsible for data collection, analysis and report writing for this study, with KFNFH and KFFWO staff assisting.

USFWS SF Bay Delta, Will Smith Sucker PIT Tag Model Building

With over 80,000 PIT tagged SARP fish from KFNFH released into UKL and its tributaries from the beginning of the program to 2025, analysis of the tag detection data that we have on hatchery reared fish and creating recommendations for the hatchery on improving survival of hatchery-reared SNS and LRS is needed. A statistical model is needed to look at year to year survival of hatchery reared fish and as the program grows, it will become a proper tool to evaluate the program. In April of 2025 a scope of work was created and approved with Will Smith, a Fisheries Statistician from the SF Bay Delta USFWS office, to create a Cormack-Jolly-Seber (CJS) model, run the model to ensure proper conditions, and create code to use and train KFFWO and KFNFH staff how to use for annual analysis of hatchery reared fish. With the creation of the scope of work, Will Smith was able to investigate data to evaluate the type of model and best use of the data available to help evaluate the program. There has been constant communication with Will Smith and revisions have been ongoing throughout FY25 with building a model that meets the needs of the hatchery program, to better inform management decisions. No reliable results have come from the model building yet, with a variety of parameters still being evaluated. A functional model will be fully developed, with code sharing, and virtual training on using the model by the close FY26 or the beginning of FY27.

USFWS Abernathy Fish Technology Center, Matt Smith Hatchery Genetic Management Plan

Since the start of the KFNFH and SARP, one of the goals for recovery was to maintain and preserve genetic diversity of the wild populations along with creating a captive broodstock program. With this goal, the KFNFH required a scientifically defensible genetic management plan (GMP) to conserve the genetic diversity of the naturally spawning populations while supporting overall recovery goals. Genetic management plans are best supported when they are justified by a comprehensive understanding of genetic patterns of the species. Completion of an ongoing genetic evaluation of the naturally spawning

populations, captive broodstock, and larval collections is recommended for the development and implementation of the GMP. The KFNFH collects and will continue to do fin clips of selected captive broodstock, one percent of larval drift, and wild adults that are used for spawning at the UKL Eastside Springs LRS population, this with other genetic material collected by USGS and BOR in their field work will be able to create defensible protocols and have a monitoring/evaluation plan to ensure future genetic diversity. The statement of work with Matt Smith, Geneticist at the Abernathy Fish Technology Center, started in January 2025 to create a hatchery genetic management plan for the KFNFH, with a first draft expected initially in the fall of 2025, but due to delays, the first draft is not expected until early 2026.

USFWS Abernathy Fish Technology Center, Benjamen Kennedy Sucker PIT Antenna Trawl Pilot

With more activity of monitoring and evaluation of hatchery reared suckers, there is an ongoing need to increase the number of SARP fish detected in UKL for modeling and representation. The vast majority of the fish released by KFNFH have a PIT tag implanted in them before release, making detections of the PIT tag key for evaluation. The USGS has several PIT antenna arrays around UKL and with good coverage of the entry to sucker spawning grounds in the Williamson River. UKL is a massive shallow lake and is difficult to get good coverage with passive antennas or with large trap nets, as noted with the low catches from the USGS juvenile cohort monitoring. In 2020, a trawl was tested in capturing juvenile suckers in UKL, with the trawl being loaned out from the Lodi USFWS office. In the limited trawling with a net that was not designed for UKL or the KFFWO boats used, the netting did collect a small number of juvenile suckers over the summer. In 2021, a former sucker recovery biologist ordered a trawl with the specifics of capturing juvenile suckers in UKL but due to a shift in staffing and priorities it was not used. With research into increasing PIT detections by consulting PIT antenna experts at Columbia Basin PIT Tag Information System (PTAGIS) and Abernathy Fish Technology Center, a pilot study with the UKL designed trawl and an unused Biomark raft antenna system with two IS1001 readers, a master controller, battery pack, and antennas to create a PIT antenna trawl. In June 2025 a statement of work was created and approved with Benjamen Kenndy, Fisheries Biologist at the Abernathy Fish Technology Center, to create a pilot PIT antenna trawl that can detect fish actively and mark their location in the lake. The bases of the trawl was to utilize the unused KFFWO gear, which needed to be transported to Benjamin in July of 2025 to retrofit into a PIT antenna trawl. The design, build, and ground testing was completed in August 2025 and staff were planning for a UKL water test run in October 2025. However, this effort was cancelled with the looming government shutdown, which required cancelling travel for all non-priority work. As of December 2025, staff plan to test the antenna trawl system in UKL during the Spring of 2026. This will enable staff to make needed modifications early in the season and to test operations again throughout the 2026 field season to evaluate functions as part of a pilot study.

Facility Construction

After several years of planning, environmental compliance, permitting, finalizing conceptual designs for various phases of construction, and developing bid packages, the first two phases of construction were awarded to contractors in fiscal year 2022 to do initial site grading, build 11 ponds, add underground infrastructure, and build a hatchery and maintenance building. Phases 3 and 4 of construction were awarded to contractors in fiscal year 2023 and will entail doing more site grading, building 21 additional ponds, adding more underground infrastructure, and drilling a new primary geothermal well. And during the beginning of fiscal year 2025, Phase 5 was awarded to renovate the infrastructure on the second lease parcel, to include major site grading, the installation of retaining walls, rerouting portions of the existing geothermal supply line from the old well, constructing four additional ponds, and leveling a pad for a future fish culture building to be located, with work tentatively planned to break ground in the spring of 2025. Shortly into the calendar year of 2026, a stop work order was issued to most BIL funded phases, including Phases 2, 3, and 5; although no such order was issued to the Phase 4 vendor. And prior to the end of fiscal year 2025 and the ensuing government shutdown, the Phase 6 contract was awarded, which centered on the installation of perimeter fencing to most of the lease area 1 portion of the site, starting in early fiscal year 2026. In addition to these contracts being awarded, numerous modifications have also been made and awarded to date, with nearly $36.3 million obligated so far from both Fisheries and Aquatic Conservation funding (1311) for hatchery operations and Bipartisan Infrastructure Law funding (BIL). It is also anticipated that at least one additional phase of construction may be needed to complete the hatchery infrastructure, perhaps with a performance period running from 2026 to 2028.

The new hatchery will consist of a one-acre head pond water supply influent retention pond (IRP), a bank of Tier-1 ponds (C1-C6 and B1-B4) consisting of six quarter-acre and four eighth-acre production ponds for fry and fingerling production, a lower bank of Tier-2 ponds (C9-C14 and D1-8), capable of serial reuse of water from Tier-1 ponds, consisting of six quarter-acre and eight half-acre production ponds for juvenile production, another lower bank of broodstock ponds (A1-A6) consisting of six twelfth-acre ponds, a hatchery and maintenance building, a half-acre effluent retention pond (D10), an additional bank of lease area 2 ponds (D11 and A7 and A8) consisting of one half-acre and two twelfth-acre isolation/production ponds, as well as an additional third-acre head pond water supply for lease area 2. In addition to this, the facility will have a hatchery and administrative building, a maintenance shop and storage garage, feed and chemical storage buildings, a wide array of underground utilities and water works running to all the infrastructure, including power to each pond kettle, cooled and geothermal water supply and a drain line from each pond, telecommunication and SCADA conduit for internet and remote monitoring, and fencing around the entire site. For more information on the different phases of construction, please see **Figure 15**.

Phase 1 Construction, which was 1311 funded and awarded to Morello Construction in the summer of 2022, is nearing completion, with tentative completion by the summer of 2026. The overall project is approximately 97% complete, however several recent modifications require the Phase 2 and Phase 4 contractors to compete their work in tandem, and this is delaying some final contract tasks and punch list items, due to the BIL fund freeze. Otherwise, all major site grading and below ground piping and conduit is complete and the above ground geothermal water supply piping from the well is also complete and operational. The IRP head pond has been completed. Concrete kettles have been poured and renovated in all the Tier-1 ponds and the pond linings were installed; these ponds underwent some

warranty renovations during fiscal year 2024, due to tolerance deviations on the kettle screen and stop log channels, and again in fiscal year 2025, due to them not meeting grading specifications, which became apparent during their limited use in late fiscal year 2024. Fortunately, these deficiencies have all been corrected to an acceptable degree. The large retaining wall separating Tier-1 and Tier-2 ponds is also complete and all of the levee roads around ponds have gravel and are set to finish grade. Electrical services were installed but recent design limitations have surfaced and have since required several modifications, which should be completed by the spring of 2026. Also, the installation of metal cabling and bird netting has been installed on all ten of the production ponds, and the cabling has been installed on the IRP head pond, however the bird netting initially installed on this pond was damaged during installation and will be replaced under warranty or through a modification to a stronger net mesh. The overall completion dates will be weather dependent and there are still several modifications to the contract that will be worked on into the spring or summer of 2026.

Phase 2 Construction, which was BIL funded and awarded to RJS Construction in the fall of 2022, was very slow out of the gate but was underway during fiscal year 2024. The original contract performance period was to complete all work by the fall of 2024, within 24 months for the performance period. However, it became apparent prior to the start of fiscal year 2025, that RJS would not complete things prior to the end of the 2024 calendar year, as they were nearly a year behind schedule on deliverables to the contract, especially in regard to constructing the hatchery building. Then, with the stop work order to this phase in early 2026, we have been told that the deliverables would be over one and half years out once BIL funds are unfrozen and construction is remobilized. Hence, a tentative completion date, as well as the fate of this project overall, is currently unknown.

To date, all major Phase 2 construction site grading to the utility elevations is done and the septic sanitation system is complete and signed off on by county inspectors. The maintenance building was erected and dried in; however no interior finish work has been started and there is no power pulled to conduit or panels in the building. The underground utilities are nearly installed for the hatchery building, and the concrete footers have been poured, but the building foundation slab has not been poured; in addition, the building super structure and the insulated roof panels were delivered and are onsite, ready to be erected on the future slab. Once construction resumes and the buildings are complete, then the whole parking lot needs to be brought to finish grade and paved with asphalt. There will also be the installation of a feed storage building and a chemical storage building once the parking lot is paved. But these construction tasks will not be completed unless BIL funds are unfrozen, and the stop work order on this phase is lifted.

Phase 3 Construction, which was BIL funded and awarded to Diversified Contractors in the summer of 2023, is well underway, and the initial contract completion date stretching into the summer or fall of 2026. However, as was the case with most BIL funded construction, it was put on a stop work order in early 2026, once those funds were frozen and remains that way. Hence, a tentative completion date, as well as the fate of this project overall, is currently unknown. Already, the basic forming of many of the fourteen Tier-2 ponds (C9-C14 and D1-8) is complete and they were going to start pouring concrete kettles prior to the stop work order. All remaining construction tasks will not be completed unless BIL funds are unfrozen, and the stop work order on this phase is lifted.

Phase 4 Construction, which was BIL funded and awarded to KLR Inc. in the summer of 2023, includes the drilling of a new primary geothermal well. Currently, the 6" test well was drilled to 400' and they have hit the correct aquifer on the first try, which was confirmed with electronic logging of data. Thereafter, they installed the 18" production well with a stainless-steel casing since the water is so corrosive. This work was all completed during fiscal year 2024 and no appreciable work has been completed since then on this phase of construction. While Phase 4 was not put on a stop work order in early 2025, because the work was deemed essential to continue and the contractor initially agreed to complete the work at their own risk, assuming that they would eventually be paid out by the end of the fiscal year. However, the contractor never showed up in fiscal year 2025 and missed the opportunity that all BIL funded contractors were paid through 9/30/2025 for any outstanding work charged. This negligence has greatly delayed the installation of this new well and we are currently waiting for this contractor to show up and complete the remaining work to date, with little to no communication from them as to what their construction schedule will be.

The remaining work on the Phase 4 contract includes pouring a small concrete pad around the well casing and then installing the pump and associated discharge to the water delivery system at the site. In addition, a covered structure will be constructed over the well to protect the equipment from the elements. We have already run power to the well site, as well as connection plumbing to the existing water delivery system, through a modification to the Phase 1 contract and just need KLR to finish their part to make everything operational. After over three years of waiting for approval from the State of Oregon's Water Resources, we finally got the necessary permitting to operate this new well, making it the primary point of diversion. The dilemma we currently face is that we now need to pump water from this new diversion point and monitor flows during the calendar year of 2026, to show that beneficial use has been made and documented, to justify using this diversion point into the future. We may need to seek an extension to the permit prior to the close of fiscal year 2026, if this new well is not operational by then. Lastly, while the current contract also had an option to drill a second well within the lease boundary, it does not appear that we will get permission to move forward on this, so this task will likely be deobligated from the contracts.

Phase 5 Construction, which was BIL funded, was awarded to Suulutaaq Inc. in November of 2024, early in fiscal year 2025. Shortly after the kickoff meeting and the notice to proceed was given in December of 2024, the contractor started doing materials submittals and were planning to break ground during the spring of 2025. However, prior to that, as was the case with most BIL funded construction, it was put on a stop work order in early 2026, once those funds were frozen and remains that way to date. Hence, a tentative completion date, as well as the fate of this project overall, is currently unknown. This phase will tentatively include the last of the pond construction on the second lease area to include site grading on this parcel, including the demolition of 14 P-ponds (P0-P13) and the greenhouse building, and the construction of three additional production ponds and rehabilitation of the existing head pond supply on this parcel. During this phase, a reroute of a portion of the geothermal water supply piping will also be included to match the new site elevations. The remaining construction tasks will not be completed unless BIL funds are unfrozen, and the stop work order on this phase is lifted.

Phase 6 Construction, which was 1311-funded and awarded to Aetna in August of 2025, just as the fiscal year was closing out. A kickoff meeting was scheduled for the fall and it happened in December 2025, after the fall government shutdown was over, when the notice to proceed was given. This phase will

install perimeter cyclone fencing around the first lease parcel, on the property line boundaries. It is scheduled to begin in the spring of 2026 and be completed by the summer.

Additional tasks on another future phase of construction may also include the installation of a SCADA system for the entire facility, as well as additional installations of security systems across the facility. There may also be the construction of an additional pole barn building to hold larger tanks for holding feral broodstock in isolation on lease parcel 2. This future phase of construction will likely be funded with 1311 money and awarded in FY27 and completed by the end of 2028. But all of this is pending the release of BIL funding and the issuance of the notice to proceed with work on Phases 2, 3, and 5. Also, given the excessive delays in getting a hatchery tank house building constructed, the hatchery is also investigating the option to purchase a pre-engineered metal building kit and install it in house, or through a maintenance action team with wage grade employees from the Service, to expedite having permanent and functional culture space before we lose further infrastructure when BIL funded construction activities ensue with normally planned demolition work.

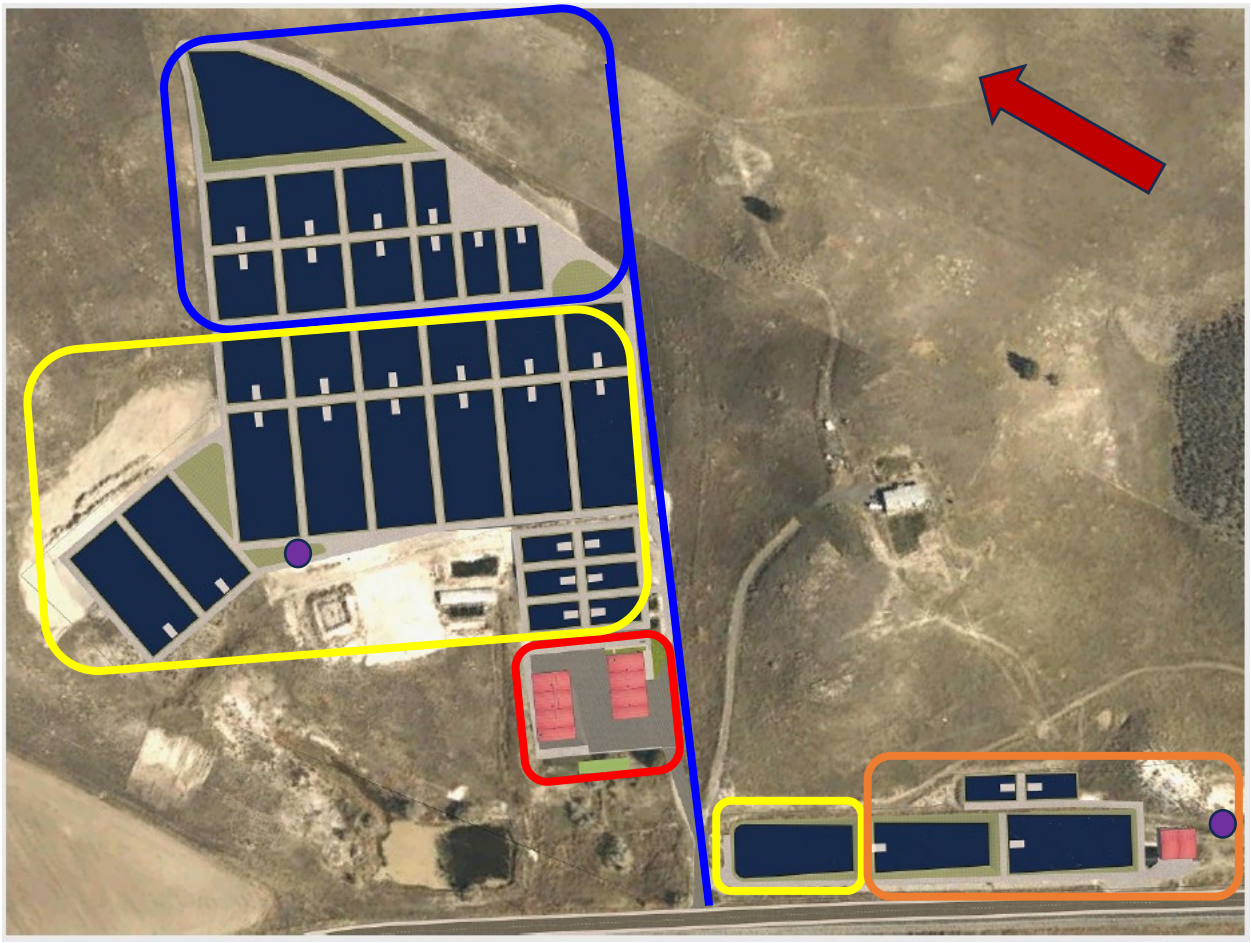

Figure 20 – Conceptual aerial overlay of the full buildout of the Klamath Falls National Fish Hatchery. Phase 1 construction (blue) will include a 1.0-acre influent retention pond at the top, followed by six 0.25-acre and four 0.12-acre fry and fingerling ponds in the Tier-1 series of juvenile ponds below. Phase 2 construction (red) will include the maintenance building, the hatchery building and office, and a feed and chemical storage building. Phase 3 construction (yellow) will include six 0.25-acre and eight 0.5-acre juvenile production ponds in the Tier-2 series below the Tier-1 series, followed by six 0.08-acre broodstock ponds below that, and lastly the 0.5-acre effluent retention pond. Phase 4 construction (purple dots) will include a new primary geothermal well by the Tier-2 series of ponds and possibly a new backup geothermal well on the lower lease parcel by the road. Phase 5 construction (orange) will include two 0.08-acre and one or two 0.5-acre quarantine ponds on the lower lease parcel by the road, as well as a new influent retention pond.

Future Work

The following is a list of various priority activities that will be carried out by the Hatchery staff during FY26. While not all things may be fully achieved, this outlines our normal planned hatchery operations, special projects, and potential outcomes for each project.

Hatchery Construction Management (All Year) – Lead: Mark Yost

- Monitor progress of the existing six phases of construction currently awarded and underway.
    - Phase 1 – Completion of the new head ponds, ten tier-1 production ponds, utility and plumbing installations, and bird predation cabling and netting on ponds, tentative by January 2026. The construction contract will likely remain open until summer of 2026 to complete pending electrical modifications to aid in getting the new well operational; this is proving challenging with the BIL fund freeze on Phase 4.
    - Phases 2-5 – Currently on hold since January 2025 and awaiting word on whether BIL funds will be released.
    - Phase 6 – Completion of a fencing contract to provide site security to the hatchery grounds on both lease areas. Tentative completion is in the winter or spring of 2026.
- Assisting with the development of a new design package and solicitation contract to award for Phase 7 construction, which may include a mop up effort to finish unfunded work, or incomplete work needed in earlier phases, with a tentative award in the spring of 2026.

PIT Tag Array Construction and Deployment in UKL (All Year) – Lead: Josh Gondek

- Monitor fixed and portable arrays for detections of suckers into late Fall 2025 and then pull gear for the winter.
- Assemble and set up four additional litz array systems purchased in FY25 and deploy in the spring of 2026 and monitor through the fall of 2026 prior to removal for winter. In addition, utilize the new towable antenna developed by Abernathy to do active population monitoring in UKL.
- Prepare a proposal for upscaling fixed array construction/assembly, installation, and monitoring for FY26 and FY27.

Fall 2025 Telemetry Transfer/Stocking of KFNFH Fish / Telemetry Monitoring for USGS/USFWS Acoustic Study (All Year) – Lead: Josh Gondek

- Provide USGS and Klamath Tribes Hatchery ~400 30-month old suckers to install acoustic telemetry tags in November and to release in December 2025.
- Assist USGS with monitoring efforts and acoustic receiver installation and maintenance as needed.
- Receive regular updates on preliminary results and management lessons, at least quarterly and get access to raw data as needed, as well as any summary reports to better inform management decisions.

Harvest, Process, Stock Production Lots for Repatriation and Winter Storage (October/November 2025) – Lead: Michelle Jackson

- Perform normal fall pond harvest and inventory work to include redistributing broodstock and production lots to new ponds for the winter season.
- Stock the normal 18-month production lots that are to the average target size for repatriation and hold onto ~2,500 fish from wild and ESS lots for a 30-month production into FY26.
- Select CY2023 and CY2024 broodstock fish and combine for winter holding into production ponds.
- PIT tag all fingerlings that are taggable and will be carried over for the CY2025 production lots and stock into new ponds for the winter season.
- Prepare pond space, buildings, tanks, and incubation systems for winter.
- Assist the Klamath Tribes Hatchery with a transfer of ~30-35,000 "excess" fingerlings as needed.

Draft, Edit, and Finalize the FY25 Annual Report (Fall/Winter 2025) – Lead: Mark Yost

- Compile hatchery data and staff accomplishments from the previous fiscal year and rearing cycles and draft a report that documents our practices, activities, and production results. The first edited draft will be completed by January 2026, and the final, with partner edits, by the end of February/March 2026.

Prepare Presentations for the Sucker Science Symposium (Winter 2026) – Lead: Michelle Jackson / Josh Gondek / Javier Linares / Mark Yost

- Compile hatchery data and draft a hatchery presentation for conferences. (Michelle Jackson)
- Compile data from the spring spawning activities and draft a report and presentation for conferences. (Javier Linares)
- Compile data from the USGS PIT tag detection release from August 2025 and draft a presentation for conferences. (Javier Linares/Mark Yost)
    - SARP Recruitment
    - Survival by location, release type, size, body condition, etc.
    - Differential survival between sexes.

Process PIT Tag Array SARP Detections (Winter 2026) – Lead: Javier Linares / Josh Gondek

- Develop code to extract SARP PIT tag detections from the USGS data file released in August 2025.
- Correlate SARP PIT tag detections to the Klamath Falls NFH master release file to determine the total number of unique detections and a length frequency plot for all unique detections to date of August 2024.
- Correlate SARP PIT tag detections to the Klamath Falls NFH master release file to compile a table of unique detections by year for each release year of fish to look at persistence in the system over time.

- Correlate SARP PIT tag detections to the Klamath Falls NFH master release file to compile a table or matrix of unique detections by stocking season (spring versus fall releases), stocking locations, release type (soft vs. hard), and by release year.

Draft, Edit, and Finalize a Captive Broodstock Management Plan (Winter 2026) – Michelle Jackson / Javier Linares

- Compile results from past year's spawning efforts and drafting standard procedures for future years.

Draft, Edit, and Finalize a Multi-Year Program Summary Report for KFNFH, FY2020-2025 (Winter 2026) – Lead: Mark Yost

- Compiling a multi-year program summary report draft for review internally before finalizing.

Review and Edit the Comprehensive SOPs for the Klamath Falls NFH (Winter 2026) – Lead: Mark Yost

- Review, edit, and refine the various SOPs currently drafted for the Klamath Falls NFH, so that the documents represent the operation of the current and/or new infrastructure used to date.
- Include in these SOP documents ample amounts of pictures of infrastructure and staff performing various tasks to allow continuity of operations and to help prevent loss of institutional knowledge about methodologies used in the event of staff turnover.
- Finalize a copy of the comprehensive SOPs by individual chapters and print for a binder so sections can be updated, created, and/or included as needed. Include redundant copies of this document to be kept at the hatchery, the office, and by local and RO leadership.

Harvest, Process, Stock Production Lots for Grow Out (February/March 2026) – Lead: Michelle Jackson

- Perform normal spring harvest and inventory work to include redistributing broodstock and production lots to new ponds for the growing season.
- Prepare pond space, tanks, and incubation systems for larval rearing season.
- Assist the Klamath Tribes with pond harvest and fish processing work for the spring season as needed.

Upper Klamath Lake/Gerber Reservoir Net Pen Operations (March/April-September/October 2026) – Josh Gondek

- Pending adequate water levels and conditions for net pen rearing at the sites selected, stocking excess CY2025 year class holdover fish into the Upper Klamath Lake and Gerber Reservoir net pens in April for grow out through September or October.
- Possibly disassemble and move the net pen from Agency Lake to Gerber Reservoir for additional grow out space in the future.

- Harvest net pens perhaps one of each lakes net in July or August and one in September or October, process fish, and repatriate fish to Upper Klamath Lake and/or its tributaries. In addition, possibly restock the empty net pen in the late summer and/or early fall for some limited acclimation prior to a fall release.

Spawning East Side Springs LRS / Incubation (April 2026) – Lead: Michelle Jackson

- Collect and fertilize wild LRS gametes from the East Side Springs of Upper Klamath Lake, utilizing at least four males to every female, to produce a minimum of 48 family groups across at least two weeks of the spawn.
- Transport, disinfect, enumerate, and incubate eggs from each family group in a separate hatching jar, which will be pooled by each weekly collection for hatching into a common trough, circular tank, and/or production pond after individual jars are inventoried upon hatch.
- Excess eggs may be used for research.
- Viable larval fish will be ponded from each family group cross for broodstock and possibly limited production rearing. Excess fish may be used for research during the spring or stocked to off channel growout locations.

Spawning Captive Broodstock SNS / Incubation (April 2026) – Lead: Javier Linares / Mark Yost

- Artificially warm the broodstock ponds in late March or early April to expedite maturation and to attempt to better control water temperature fluctuations common in the spring.
- Collect and fertilize wild SNS gametes from the captive broodstock at Klamath Fall NFH, utilizing at least four males to every female, to produce as many family groups as possible across at least two weeks of the spawn.
    - This will likely require the use of HGC hormone injections to help ovulation but ripeness checks will occur at least once each week or every other week if handling needs to be limited.
    - Most gravid fish from the 2017, 2018, and 2019 year classes will be pooled into clean ponds in several large floating net pen structures for ease of handling for hormone injections and ripeness checks.
    - Some gravid fish from the 2017, 2018, and 2019 year classes may be put in tanks to administer hormone injections and ripeness checks.
- Transport, disinfect, enumerate, and incubate eggs from each family group in a separate hatching jar, which will be pooled by each weekly collection for hatching into a common trough after individual jars are inventoried upon hatch.
- Excess eggs may be used for research.
- Viable larval fish may be used for larval feed trials during the spring but no fish are planned for production or release unless decided appropriate by Abernathy geneticists and management.

Spawning Williamson/Sprague Rivers LRS / Incubation (April/May 2026) – Lead: Josh Gondek

- Collect and fertilize wild LRS gametes from the Williamson/Sprague Rivers, utilizing at least four males to every female, to produce as many family groups as possible across at least two weeks of the spawn.
- Transport, disinfect, enumerate, and incubate eggs from each family group in a separate hatching jar, which will be pooled by each weekly collection for hatching into a common trough after individual jars are inventoried upon hatch.
- Excess eggs may be used for research.
- Viable larval fish may be ponded from each family group cross for limited production rearing. Excess fish may be used for research during the spring or stocked to off channel growout locations.

Extending and Cryopreservation of Milt (April/May 2026) – Lead: Javier Linares

- Collect milt from the various spawning efforts from both LRS and SNS.
- Perform motility checks over time using milt extender solutions for both LRS and SNS.
- Cryopreserve milt from both LRS and SNS for post thaw motility checks.
- Fertilize SNS eggs with thawed SNS milt and incubate to assess fertilization, development, and hatch rates.
- Cryopreserve milt from both LRS and SNS for banking.

Thiamine Treatments and Sample Analysis (April/May 2026) – Lead: Javier Linares

- Utilizing gametes collected in other efforts, performing egg treatments, incubation, and early rearing, and sample collection and analysis to compare baseline thiamine levels and dose responses.

Wild Larval Collection of LRS and SNS in Williamson River (May/June 2026) – Lead: Josh Gondek

- Collect at least 25-30,000 wild caught larval suckers from the Williamson River in May/June for production purposes using active methods, such as dip netting, as well as passive methods, such as drift netting.
- Transporting and enumerating larval suckers to the Klamath Falls NFH isolation building for disease treatment and monitoring, as well as early rearing.

Early Rearing LRS and SNS to Ponding (April-June 2026) – Lead: Michelle Jackson

- Perform initial prophylactic treatments and early rearing techniques to pond at least 25-30,000 wild caught larval suckers from the Williamson River in May/June for production purposes.
- Possibly stock excess larval suckers directly into backwater locations upon collection.

Appendices

1. Thiamine Egg Results

2. Sprague River Laboratory Results

3. Neilson Laboratory Results

4. Matrix Science Results

5. Bend Genetics LLC Results

6. California Nevada Fish Health Center Inspection Results

Appendix 1

Thiamine Egg Results

Klamath Falls National Fish Hatchery

	nmol/g				Average (nmol/g)			
	TPP	TMP	TH	Total	TPP	TMP	TH	Total
F1	13.86	1.32	0.55	15.73	12.20	1.22	0.54	13.96
	10.54	1.11	0.54	12.19				
F2	11.63	1.07	0.18	12.88	11.59	1.06	0.16	12.81
	11.56	1.05	0.14	12.75				
F3	2.01	0.19	0.05	2.25	2.01	0.19	0.06	2.26
	2.01	0.19	0.06	2.26				
F4	0.80	0.05	0.03	0.88	0.78	0.04	0.03	0.85
	0.76	0.02	0.02	0.81				
F5	1.00	0.08	0.02	1.10	1.04	0.07	0.02	1.14
	1.09	0.07	0.01	1.17				
F6	0.71	0.03	0.01	0.75	0.71	0.03	0.01	0.76
	0.72	0.03	0.01	0.76				
F7	1.07	0.08	0.01	1.16	1.02	0.06	0.01	1.09
	0.97	0.04	0.01	1.02				
F8	4.83	0.53	0.32	5.68	4.82	0.53	0.31	5.66
	4.82	0.53	0.29	5.64				
F9	0.48	0.03	0.01	0.52	0.48	0.02	0.01	0.51
	0.47	0.02	0.01	0.50				
F10	8.91	0.65	0.11	9.67	8.99	0.36	0.10	9.45
	9.07	0.07	0.10	9.23				
F11	1.68	0.13	0.02	1.84	1.70	0.13	0.02	1.85
	1.72	0.13	0.02	1.87				
F13	6.14	0.53	0.06	6.73	6.14	0.53	0.08	6.74
	6.13	0.53	0.09	6.75				
F14	2.55	0.28	0.09	2.92	2.54	0.28	0.09	2.92
	2.53	0.29	0.10	2.92				
F15	7.14	0.77	0.07	7.97	7.12	0.76	0.07	7.95
	7.10	0.76	0.07	7.93				
F16	1.18	0.10	0.01	1.29	1.18	0.10	0.01	1.29
	1.18	0.10	0.01	1.29				
F17	0.96	0.07	0.01	1.04	0.94	0.07	0.01	1.02
	0.92	0.06	0.01	0.99				
F18	0.76	0.06	0.02	0.84	0.75	0.06	0.02	0.83
	0.74	0.06	0.02	0.83				
F20	2.04	0.17	0.17	2.39	2.03	0.17	0.18	2.39
	2.02	0.17	0.19	2.38				

TPP = Thiamine pyrophosphate
TMP = Thiamine monophosphate
TH = Free thiamine
Total = TPP + TMP + TH

Method (See Futia et al. 2017)

Free thiamine (TH), thiamine monophosphate (TMP), and thiamine pyrophosphate (T
by Brown et al. (1998a), with modifications. Concentrations from the three vitamers w
For thiamine extraction, approximately one gram of unfertilized egg was used and e
thiamine levels were determined using a high performance liquid chromatograph (HPL
included a delivery pump, automatic sample injector, Hamilton PRP-HI column (150
(25 × 2.3 mm; 12 to 20-mm mesh size), and a fluorometric detector (375-nm excit
thiochrome detection). The column thermostat was set to 30 °C. Flow rate was 0.87 or
phase was comprised of 25mM potassium phosphate buffer (pH 8.4) and N,N-dimeth
was comprised of 25 mM potassium phosphate buffer (pH 8.4). From 1 to 5 min, tl
phosphate buffer. From 5 to 9 min, the mobile phase was 10% DFM and 25 mM pot
phase was 30% DFM and 25 mM potassium phosphate buffer. From 11.0 to
25mMpotassiumphosphate. Afterwards, the mobile phase was returned to the initial c
A six-point standard curve with known concentrations of thiamine (0, 1, 2.5, 5, 10 an
samples run.

'PP) were extracted and quantified (nmol/g) as described
ere combined to determine total thiamine concentrations.
:ach sample was run in duplicate. Following extraction,
C) system (Agilent Technologies 1200 series). TheHPLC
× 4.1 mm; 5-µm mesh size) with attached guard column
.ation wavelength and 433-nm emission wavelength for
r1 ml/min and the total run time was 25 min. The mobile
ylformamide (DMF). From 0 to 1 min, the mobile phase
he mobile phase was 0.5% DFM and 25mM potassium
tassium phosphate buffer. From 9 to 11 min, the mobile
o 11.1 min, the mobile phase was 35% DFM and
:onditions, equilibrating the column for the next sample.
d 30 nmol/g) was generated at the start of each group of

Appendix 2

Sprague River Laboratory Results

5671 Sprague River Road
PO Box 436
Chiloquin, OR 97624
541.783.2149 Phone
541.783.2148 Fax

Sprague River Water Quality Lab

Data Report: WO# 5090905

Michelle Jackson
US Fish & Wildlife Service
1936 California Avenue
Klamath Falls, OR 97601

Project: Pond Water Quality
Project Number: 2025

Sampled:09-Sep-25
Received:09-Sep-25
Reported:17-Sep-25 15:26

SRWQL ID	Client Sample ID	Analyte	TP as P	PO4 as P	NH3 as N	NO2+3 as N	TN as N	-	-	-
		Specific Method	EPA 365.2	EPA 365.1	EPA 350.1	EPA 353.2	USGS I-2650-03	-	-	-
		MRL	0.036	0.006	0.030	0.030	0.060	-	-	-
		MDL	0.018	0.003	0.015	0.015	0.030	-	-	-
		Date	mg P/L	mg P/L	mg N/L	mg N/L	mg N/L	-	-	-
5090905-01	C1	9/09/25	0.243	0.008	0.015	< 0.015	2.29	-	-	-
5090905-02	C2	9/09/25	0.152	0.008	< 0.015	< 0.015	1.18	-	-	-
5090905-03	Control (C6)	9/09/25	0.168	0.012	0.018	< 0.015	1.09	-	-	-

The SRWQL is Accredited through ORELAP for the following analyses, per the methods referenced in the report: TP as P, PO4 as P, NH3 as N, TN as N, NO2+NO3 as N, NO3 as N, SiO2, TSS, and Turbidity. The Lab is not accredited for any other analyses, per the referenced methods.

The results in this report relate only to the parameters tested or to the sample as received by the laboratory.

All samples were received as required unless noted otherwise.

Notes:

1. Shaded values indicate results that are detectable, but are below the Minimum Reporting Limit.

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Quality Control

Analyte	Result	MDL	MRL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes /Qual
Batch 2509013 Analyzed 09/15/25											
Blank (2509013-BLK1)	-										
Total Phosphorus	<0.018	0.018	0.036	mg P/L	-	-	-	-		-	
Total Nitrogen	<0.030	0.030	0.060	mg N/L	-	-	-	-		-	
Blank (2509013-BLK2)	-										
Total Phosphorus	<0.018	0.018	0.036	mg P/L	-	-	-	-		-	
Total Nitrogen	<0.030	0.030	0.060	mg N/L	-	-	-	-		-	
LCS (2509013-BS1)	-										
Total Phosphorus	0.050	0.018	0.036	mg P/L	0.05000		101	80-120		-	
Total Nitrogen	0.088	0.030	0.060	mg N/L	0.1000		88.4	80-120		-	
LCS (2509013-BS2)	-										
Total Phosphorus	0.053	0.018	0.036	mg P/L	0.05000		106	80-120		-	
Total Nitrogen	0.092	0.030	0.060	mg N/L	0.1000		92.1	80-120		-	
LCS (2509013-BS3)	-										
Total Phosphorus	2.91	0.018	0.036	mg P/L	3.000		97.0	80-120		-	
Total Nitrogen	1.74	0.030	0.060	mg N/L	2.000		86.8	80-120		-	
LCS (2509013-BS4)	-										
Total Phosphorus	2.93	0.018	0.036	mg P/L	3.000		97.8	80-120		-	
Total Nitrogen	1.78	0.030	0.060	mg N/L	2.000		89.1	80-120		-	
Duplicate (2509013-DUP1)	**Source:5090204-01**										
Total Phosphorus	0.050	0.018	0.036	mg P/L	-	0.051	-	-	2.84	20	
Total Nitrogen	0.077	0.030	0.060	mg N/L	-	0.073	-	-	5.02	20	
Duplicate (2509013-DUP2)	**Source:5090302-01**										
Total Phosphorus	0.060	0.018	0.036	mg P/L	-	0.060	-	-	0.635	20	
Total Nitrogen	0.128	0.030	0.060	mg N/L	-	0.128	-	-	0.0938	20	
Matrix Spike (2509013-MS1)	**Source:5090204-01**										
Total Phosphorus	0.249	0.018	0.036	mg P/L	0.1895	0.051	105	80-120		-	
Total Nitrogen	1.02	0.030	0.060	mg N/L	1.053	0.073	89.6	80-120		-	
Matrix Spike (2509013-MS2)	**Source:5090302-01**										
Total Phosphorus	0.247	0.018	0.036	mg P/L	0.1895	0.060	98.8	80-120		-	
Total Nitrogen	1.13	0.030	0.060	mg N/L	1.053	0.128	95.3	80-120		-	
Matrix Spike (2509013-MS3)	**Source:5090402-01**										
Total Phosphorus	0.237	0.018	0.036	mg P/L	0.1895	0.064	91.1	80-120		-	
Total Nitrogen	1.05	0.030	0.060	mg N/L	1.053	0.180	83.1	80-120		-	
Matrix Spike Dup (2509013-MSD1)	**Source:5090204-01**										
Total Phosphorus	0.238	0.018	0.036	mg P/L	0.1895	0.051	98.6	80-120	4.66	20	
Total Nitrogen	0.983	0.030	0.060	mg N/L	1.053	0.073	86.4	80-120	3.39	20	
Reference (2509013-SRM1)	-										

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Quality Control

Analyte	Result	MDL	MRL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes /Qual
Batch 2509013 Analyzed 09/15/25											
Reference (2509013-SRM1)	-										
Total Phosphorus	0.818	0.018	0.036	mg P/L	0.8000		102	0-200		-	
Total Nitrogen	0.704	0.030	0.060	mg N/L	0.7500		93.9	0-200		-	
Reference (2509013-SRM2)	-										
Total Phosphorus	7.88	0.018	0.036	mg P/L	8.000		98.5	0-200		-	
Total Nitrogen	5.55	0.030	0.060	mg N/L	6.000		92.4	0-200		-	
Batch 2509018 Analyzed 09/16/25											
Blank (2509018-BLK1)	-										
SRP	<0.003	0.003	0.006	mg P/L	-	-	-	-		-	
Ammonia	<0.015	0.015	0.030	mg N/L	-	-	-	-		-	
Nitrate+Nitrite	<0.015	0.015	0.030	mg N/L	-	-	-	-		-	
Blank (2509018-BLK2)	-										
SRP	<0.003	0.003	0.006	mg P/L	-	-	-	-		-	
Ammonia	<0.015	0.015	0.030	mg N/L	-	-	-	-		-	
Nitrate+Nitrite	<0.015	0.015	0.030	mg N/L	-	-	-	-		-	
LCS (2509018-BS1)	-										
SRP	0.022	0.003	0.006	mg P/L	0.02000		109	80-120		-	
Ammonia	0.050	0.015	0.030	mg N/L	0.05000		101	80-120		-	
Nitrate+Nitrite	0.099	0.015	0.030	mg N/L	0.1000		99.3	80-120		-	
LCS (2509018-BS2)	-										
SRP	0.022	0.003	0.006	mg P/L	0.02000		109	80-120		-	
Ammonia	0.050	0.015	0.030	mg N/L	0.05000		101	80-120		-	
Nitrate+Nitrite	0.097	0.015	0.030	mg N/L	0.1000		96.7	80-120		-	
Duplicate (2509018-DUP1)	**Source:5090803-01**										
SRP	0.017	0.003	0.006	mg P/L	-	0.017	-	-	1.73	20	
Ammonia	<0.015	0.015	0.030	mg N/L	-	<0.030	-	-		20	
Nitrate+Nitrite	<0.015	0.015	0.030	mg N/L	-	<0.030	-	-		20	
Duplicate (2509018-DUP2)	**Source:5090903-01**										
SRP	0.004	0.003	0.006	mg P/L	-	0.006	-	-	29.7	20	1
Ammonia	0.026	0.015	0.030	mg N/L	-	0.026	-	-	0.308	20	
Nitrate+Nitrite	0.041	0.015	0.030	mg N/L	-	0.041	-	-	0.755	20	
Matrix Spike (2509018-MS1)	**Source:5090803-01**										
SRP	0.064	0.003	0.006	mg P/L	0.04737	0.017	99.6	80-120		-	
Ammonia	0.106	0.015	0.030	mg N/L	0.1053	<0.030	101	80-120		-	
Nitrate+Nitrite	0.207	0.015	0.030	mg N/L	0.2105	<0.030	98.3	80-120		-	
Matrix Spike (2509018-MS2)	**Source:5090903-01**										
SRP	0.052	0.003	0.006	mg P/L	0.04737	0.006	97.3	80-120		-	

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Quality Control

Analyte	Result	MDL	MRL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes /Qual
Batch 2509018 Analyzed 09/16/25											
Matrix Spike (2509018-MS2)	**Source:5090903-01**										
Ammonia	0.118	0.015	0.030	mg N/L	0.1053	0.026	87.6	80-120		-	
Nitrate+Nitrite	0.242	0.015	0.030	mg N/L	0.2105	0.041	95.7	80-120		-	
Matrix Spike (2509018-MS3)	**Source:5091002-01**										
SRP	0.076	0.003	0.006	mg P/L	0.04737	0.028	100	80-120		-	
Ammonia	0.136	0.015	0.030	mg N/L	0.1053	0.037	94.1	80-120		-	
Nitrate+Nitrite	0.328	0.015	0.030	mg N/L	0.2105	0.127	95.2	80-120		-	
Matrix Spike Dup (2509018-MSD1)	**Source:5090803-01**										
SRP	0.065	0.003	0.006	mg P/L	0.04737	0.017	101	80-120	0.900	20	
Ammonia	0.110	0.015	0.030	mg N/L	0.1053	<0.030	104	80-120	3.22	20	
Nitrate+Nitrite	0.212	0.015	0.030	mg N/L	0.2105	<0.030	101	80-120	2.37	20	
Batch 5I15003 Analyzed 09/15/25											
Cal Standard (5I15003-CAL1)	-										
Total Phosphorus	0.020			mg/L	0.01800		114	-		-	
Total Nitrogen	0.029			mg/L	0.03000		95.4	-		-	
Cal Standard (5I15003-CAL2)	-										
Total Phosphorus	0.050			mg/L	0.05000		100	-		-	
Total Nitrogen	0.055			mg/L	0.05000		111	-		-	
Cal Standard (5I15003-CAL3)	-										
Total Phosphorus	0.191			mg/L	0.2000		95.4	-		-	
Total Nitrogen	0.099			mg/L	0.1000		99.0	-		-	
Cal Standard (5I15003-CAL4)	-										
Total Phosphorus	0.411			mg/L	0.4000		103	-		-	
Total Nitrogen	0.244			mg/L	0.2500		97.7	-		-	
Cal Standard (5I15003-CAL5)	-										
Total Phosphorus	0.596			mg/L	0.6000		99.3	-		-	
Total Nitrogen	0.503			mg/L	0.5000		101	-		-	
Cal Standard (5I15003-CAL6)	-										
Total Phosphorus	0.807			mg/L	0.8000		101	-		-	
Total Nitrogen	0.737			mg/L	0.7500		98.3	-		-	
Cal Standard (5I15003-CAL7)	-										
Total Phosphorus	1.01			mg/L	1.000		101	-		-	
Total Nitrogen	1.02			mg/L	1.000		102	-		-	
Initial Cal Blank (5I15003-ICB1)	-										
Total Phosphorus	0.007			mg/L	-		-	-		-	
Total Nitrogen	0.008			mg/L	-		-	-		-	
Initial Cal Check (5I15003-ICV1)	-										

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Quality Control

Analyte		Result	MDL	MRL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes /Qual
Batch 5I15003 Analyzed 09/15/25												
Initial Cal Check (5I15003-ICV1)	-											
Total Phosphorus		0.053			mg/L	0.05000		105	80-120		-	
Total Nitrogen		0.094			mg/L	0.1000		93.6	80-120		-	
Initial Cal Check (5I15003-ICV2)	-											
Total Phosphorus		0.402			mg/L	0.4000		100	80-120		-	
Total Nitrogen		0.216			mg/L	0.2500		86.5	80-120		-	4
Initial Cal Check (5I15003-ICV3)	-											
Total Phosphorus		0.822			mg/L	0.8000		103	80-120		-	
Total Nitrogen		0.692			mg/L	0.7500		92.2	80-120		-	
Batch 5I15004 Analyzed 09/15/25												
Cal Standard (5I15004-CAL1)	-											
Total Phosphorus		0.955			mg/L	1.000		95.5	-		-	
Total Nitrogen		1.00			mg/L	1.000		100	-		-	
Cal Standard (5I15004-CAL2)	-											
Total Phosphorus		3.04			mg/L	3.000		101	-		-	
Total Nitrogen		2.03			mg/L	2.000		102	-		-	
Cal Standard (5I15004-CAL3)	-											
Total Phosphorus		3.94			mg/L	4.000		98.6	-		-	
Total Nitrogen		2.97			mg/L	3.000		98.9	-		-	
Cal Standard (5I15004-CAL4)	-											
Total Phosphorus		5.03			mg/L	5.000		101	-		-	
Total Nitrogen		3.96			mg/L	4.000		99.0	-		-	
Cal Standard (5I15004-CAL5)	-											
Total Phosphorus		6.04			mg/L	6.000		101	-		-	
Total Nitrogen		5.04			mg/L	5.000		101	-		-	
Cal Standard (5I15004-CAL6)	-											
Total Phosphorus		8.07			mg/L	8.000		101	-		-	
Total Nitrogen		6.13			mg/L	6.000		102	-		-	
Cal Standard (5I15004-CAL7)	-											
Total Phosphorus		9.92			mg/L	10.00		99.2	-		-	
Total Nitrogen		6.95			mg/L	7.000		99.3	-		-	
Initial Cal Check (5I15004-ICV1)	-											
Total Phosphorus		3.04			mg/L	3.000		101	80-120		-	
Total Nitrogen		1.90			mg/L	2.000		95.0	80-120		-	
Initial Cal Check (5I15004-ICV2)	-											
Total Phosphorus		5.11			mg/L	5.000		102	80-120		-	
Total Nitrogen		3.84			mg/L	4.000		95.9	80-120		-	

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Quality Control

Analyte		Result	MDL	MRL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes /Qual
Batch 5I15004 Analyzed 09/15/25												
Initial Cal Check (5I15004-ICV3)	-											
Total Phosphorus		8.21			mg/L	8.000		103	80-120		-	
Total Nitrogen		5.96			mg/L	6.000		99.4	80-120		-	
Batch 5I16003 Analyzed 09/16/25												
Cal Standard (5I16003-CAL1)	-											
SRP		0.006			mg/L	0.005000		117	-		-	
Ammonia		0.018			mg/L	0.01500		117	-		-	
Nitrate+Nitrite		0.021			mg/L	0.02000		103	-		-	
Cal Standard (5I16003-CAL2)	-											
SRP		0.023			mg/L	0.02400		96.2	-		-	
Ammonia		0.049			mg/L	0.05000		97.1	-		-	
Nitrate+Nitrite		0.101			mg/L	0.1000		101	-		-	
Cal Standard (5I16003-CAL3)	-											
SRP		0.050			mg/L	0.05000		99.4	-		-	
Ammonia		0.099			mg/L	0.1000		99.3	-		-	
Nitrate+Nitrite		0.198			mg/L	0.2000		99.1	-		-	
Cal Standard (5I16003-CAL4)	-											
SRP		0.080			mg/L	0.08000		100	-		-	
Ammonia		0.198			mg/L	0.2000		99.2	-		-	
Nitrate+Nitrite		0.399			mg/L	0.4000		99.8	-		-	
Cal Standard (5I16003-CAL5)	-											
SRP		0.100			mg/L	0.1000		100	-		-	
Ammonia		0.301			mg/L	0.3000		100	-		-	
Nitrate+Nitrite		0.572			mg/L	0.5714		100	-		-	
Cal Standard (5I16003-CAL6)	-											
SRP		0.115			mg/L	0.1143		101	-		-	
Ammonia		0.400			mg/L	0.4000		100	-		-	
Nitrate+Nitrite		0.801			mg/L	0.8000		100	-		-	
Cal Standard (5I16003-CAL7)	-											
SRP		0.160			mg/L	0.1600		100	-		-	
Ammonia		0.604			mg/L	0.6000		101	-		-	
Nitrate+Nitrite		0.999			mg/L	1.000		99.9	-		-	
Initial Cal Blank (5I16003-ICB1)	-											
SRP		0.001			mg/L	-		-	-		-	
Ammonia		0.005			mg/L	-		-	-		-	
Nitrate+Nitrite		0.0007			mg/L	-		-	-		-	
Initial Cal Check (5I16003-ICV1)	-											

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Quality Control

Analyte		Result	MDL	MRL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes /Qual
Batch 5I16003 Analyzed 09/16/25												
Initial Cal Check (5I16003-ICV1)	-											
SRP		0.021			mg/L	0.02000		105	90-110		-	
Ammonia		0.052			mg/L	0.05000		104	90-110		-	
Nitrate+Nitrite		0.099			mg/L	0.1000		98.9	90-110		-	
Initial Cal Check (5I16003-ICV2)	-											
SRP		0.082			mg/L	0.08000		102	90-110		-	
Ammonia		0.197			mg/L	0.2000		98.4	90-110		-	
Nitrate+Nitrite		0.401			mg/L	0.4000		100	90-110		-	
Initial Cal Check (5I16003-ICV3)	-											
SRP		0.126			mg/L	0.1230		102	90-110		-	
Ammonia		0.698			mg/L	0.4000		174	90-110		-	4
Nitrate+Nitrite		0.804			mg/L	0.8000		101	90-110		-	

Notes/Comments

1 = IF data values are <5x the DL, then values are expected to be within ± the DL.

4 = QC Sample % Recovery is outside laboratory established control Limits

FYI,
Below are the details of the new project/station for FWS.
Sample from and data to FWS
Billing Michelle Jackson FWS
Teresa Coley (9/9/25)
Samples filterd by TC
Samples acidified by TC
PH checked by TC
PH strip lot #80A3131

Sprague River Water Quality Lab

Teresa Coley, Lead Chemist

Appendix 3

Neilson Laboratory Results

Neilson Research Corporation
245 S Grape St
Medford, OR 97501
TEL: (541) 770-5678 FAX: (541) 770-2901
Website: www.nrclabs.com

September 25, 2025

Mark Yost
US Fish & Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603
TEL: (458) 232-5417
FAX

RE: Klamath Falls NFH Order No.: 25090593

Dear Mark Yost:

Neilson Research Corporation received 1 sample(s) on 9/11/2025 for the analyses presented in the following report.

The results relate only to the parameters tested or to the sample as received by the laboratory. This report shall not be reproduced except in full, without the written approval of Neilson Research Corporation. If you have any questions regarding these test results, please feel free to call.

Sincerely,
Neilson Research Corporation

Tamra Schmedemann

Tamra Schmedemann
Director of Project Management
245 S Grape St
Medford, OR 97501

Original

Page 1 of 10

NEILSON RESEARCH CORPORATION

Neilson Research Corporation
245 S Grape St
Medford, OR 97501
TEL: (541) 770-5678 FAX: (541) 770-2901
Website: www.nrclabs.com

Case Narrative

WO#: **25090593**
Date: **9/25/2025**

CLIENT: US Fish & Wildlife Services
Project: Klamath Falls NFH

The analyses were performed according to the guidelines in the Neilson Research Corporation Quality Assurance Program. This report contains analytical results for the sample(s) as received by the laboratory.

Neilson Research Corporation certifies that this report is in compliance with the requirements of NELAP. No unusual difficulties were experienced during analysis of this batch except as noted below or qualified with data flags on the reports.

Original

NEILSON RESEARCH CORPORATION

Neilson Research Corporation
245 S Grape St
Medford, OR 97501
TEL: (541) 770-5618 FAX: (541) 770-2901
Website: www.nrclabs.com

Analytical Report

WO#: **25090593**
Date Reported: **9/25/2025**

CLIENT: US Fish & Wildlife Services
Lab ID: 25090593-01
Client Sample ID Pond C1
Project: Klamath Falls NFH
Sample Location:
Sample Address:

Collection Date: 9/11/2025 11:20:00 AM
Received Date: 9/11/2025 2:20:00 PM
Matrix: AQUEOUS
PWS#:
Source ID:
Sample Collector: MARK YOST
Residual Chlorine:

Analyses	Method	NELAP Status	Result	Qual	Units	Date Analyzed/Analyst
BACTERIA BY QUANTI-TRAY						
Total Coliform Bacteria	A9223B	A	>2420		MPN/100mL	09/11/25 17:33 EBC
E. Coli Bacteria	A9223B	A	ND		MPN/100mL	09/11/25 17:33 EBC

QUALIFIERS

*	Value exceeds Maximum or Minimum Contaminant Level	C1	Sample contains temperature is out of limits as specified in this code
DF	Dilution Factor	E	Value above quantitation range
H	Holding times for preparation or analysis exceeded	M1	Recovery outside control limits due to Matrix Interference
ND	Not Detected at the Reporting Limit	RL	Reporting Limit

Original

NELAP

NELAP A Accredited in accordance with NELAP ORELAP 100016, OR-028

Page 3 of 10

NEILSON RESEARCH CORPORATION

Neilson Research Corporation
245 S Grape St
Medford, OR 97501
TEL: (541) 770-5618 FAX: (541) 770-2901
Website: www.nrclabs.com

Analytical Report

WO#: **25090593**
Date Reported: **9/25/2025**

US Fish & Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls , OR 97603

Lab ID: 25090593-01
Collection Date: 9/11/2025 11:20:00 AM
Received Date: 9/11/2025 2:20:00 PM

Sample Information:

Client Sample ID: Pond C1
Sample Collector: Mark Yost
Matrix: AQUEOUS
Source:
Sample Location:

Top 35 Analysis™

Analyses	Method	NELAP Status	Result	Qual	DF	RL	Units	EPA Limit	Date Analyzed	Analyst
Aluminum	E200.7	A	0.622	*	1	0.0200	mg/L	0.200	09/16/25 12:17	KN
Antimony	E200.8	A	ND		1	0.000500	mg/L	0.00600	09/15/25 18:14	KN
Arsenic	E200.8	A	0.0177	*	1	0.00100	mg/L	0.0100	09/15/25 18:14	KN
Barium	E200.8	A	0.00484		1	0.00200	mg/L	2.00	09/15/25 18:14	KN
Beryllium	E200.7	A	ND		1	0.00200	mg/L	0.00400	09/16/25 12:17	KN
Boron	E200.7	A	0.929		1	0.0500	mg/L		09/16/25 12:17	KN
Cadmium	E200.8	A	ND		1	0.000500	mg/L	0.00500	09/15/25 18:14	KN
Calcium	E200.7	A	21.8		1	1.00	mg/L		09/16/25 12:17	KN
Chloride	E300.0	A	65.6		10	5.00	mg/L	250	09/12/25 18:08	EGR
Chromium	E200.8	A	ND		1	0.00200	mg/L	0.100	09/15/25 18:14	KN
Copper	E200.8	A	0.00278		1	0.00200	mg/L	1.30	09/15/25 18:14	KN
Fluoride	E300.0	A	1.72		1	0.200	mg/L	4.00	09/11/25 18:40	EGR
Hardness, Total (As CaCO3)	A2340B	A	54.5		1	6.62	mg/L	250	09/16/25 12:17	KN
Iron	E200.7	A	0.377	*	1	0.0150	mg/L	0.300	09/16/25 12:17	KN
Lead	E200.8	A	ND		1	0.000500	mg/L	0.0100	09/15/25 18:14	KN
Lithium	E200.7	A	0.133		1	0.100	mg/L		09/16/25 12:17	KN
Magnesium	E200.7	A	ND		1	1.00	mg/L		09/16/25 12:17	KN
Manganese	E200.8	A	0.0181		1	0.00500	mg/L	0.0500	09/15/25 18:14	KN
Molybdenum	E200.8	A	0.0196		1	0.00100	mg/L		09/15/25 18:14	KN
Nickel	E200.8	A	ND		1	0.00100	mg/L	0.100	09/15/25 18:14	KN

QUALIFIERS

*	Value exceeds Maximum or Minimum Contaminant Level.	C1	Sample container temperature is out of limits as specified at the tools
DF	Dilution Factor	E	Value above quantitation range
H	Holding times for preparation or analysis exceeded	J	Analyte detected below quantitation limits
MI	Recovery outside control limits due to Matrix Interference	ND	Not Detected at the Reporting Limit
R	RPD outside accepted recovery limits	RL	Reporting Limit

NELAP: NELAP A Accredited in accordance with NELAP ORELAP 100016, OR-028

Original

Results are out of the EPA limits

NEILSON RESEARCH CORPORATION

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Website: www.nrclabs.com

Analytical Report

WO#: **25090593**
Date Reported: **9/25/2025**

US Fish & Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls , OR 97603

Lab ID: 25090593-01
Collection Date: 9/11/2025 11:20:00 AM
Received Date: 9/11/2025 2:20:00 PM

Sample Information:

Client Sample ID: Pond C1
Sample Collector: Mark Yost
Matrix: AQUEOUS
Source:
Sample Location:

Top 35 Analysis™

Analyses	Method	NELAP Status	Result	Qual	DF	RL	Units	EPA Limit	Date Analyzed	Analyst
Nitrogen, Nitrate	E300.0	A	ND		1	0.200	mg/L	10.0	09/11/25 18:40	EGR
Nitrogen, Nitrite	E300.0	A	ND		1	0.0500	mg/L	1.00	09/11/25 18:40	EGR
pH	A4500-H+B	A	9.50	*	1	0.10	pH Units	6.50-8.50	09/11/25 17:05	VES
Potassium	E200.7	A	5.74		1	1.00	mg/L		09/16/25 12:17	KN
Selenium	E200.8	A	ND		1	0.00100	mg/L	0.0500	09/15/25 18:14	KN
Silica	E200.7	A	96.9		1	2.14	mg/L		09/16/25 12:17	KN
Silver	E200.8	A	ND		1	0.00100	mg/L	0.100	09/15/25 18:14	KN
Sodium	E200.7	A	240	*	10	10.0	mg/L	200	09/18/25 17:52	KN
Specific Conductance	A2510B	A	1290		1	1.00	µmhos/cm		09/12/25 15:54	CSB
Sulfate	E300.0	A	409	*	10	5.00	mg/L	250	09/12/25 18:08	EGR
Thallium	E200.8	A	ND		1	0.000500	mg/L	0.00200	09/15/25 18:14	KN
Turbidity	A2130	A	33.1	*MI	1	0.100	NTU	5.00	09/11/25 17:13	VES
Uranium	E200.8	A	0.000145		1	0.000100	mg/L	0.0300	09/15/25 18:14	KN
Vanadium	E200.8	A	ND		1	0.00500	mg/L		09/15/25 18:14	KN
Zinc	E200.7	A	ND		1	0.0500	mg/L	5.00	09/16/25 12:17	KN

QUALIFIERS

*	Value exceeds Maximum or Minimum Contaminant Level.
DF	Dilution Factor
H	Holding times for preparation or analysis exceeded
MI	Recovery outside control limits due to Matrix Interference
R	RPD outside accepted recovery limits
C1	Sample container temperature is out of limits as specified at the time
E	Value above quantitation range
J	Analyte detected below quantitation limits
ND	Not Detected at the Reporting Limit
RL	Reporting Limit

NELAP

NELAP A Accredited in accordance with NELAP ORELAP 100016, OR-028

Original

Results are out of the EPA limits

NEILSON RESEARCH CORPORATION

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245 S Grape St
Medford, OR 97501
TEL: (541) 770-5678 FAX: (541) 770-2901
Website: www.nrclabs.com

Analytical Report

WO#: **25090593**
Date Reported: **9/25/2025**

CLIENT: US Fish & Wildlife Services
Lab ID: 25090593-01
Client Sample ID Pond C1
Project: Klamath Falls NFH
Sample Address:

Collection Date: 9/11/2025 11:20:00 AM
Received Date: 9/11/2025 2:20:00 PM
Matrix: AQUEOUS

Sample Location:

Analyses	Method	NELAP Status	Result	Qual	DF	RL	Units	MCL	Date Analyzed/Analyst
ORGANOCHLORINE PESTICIDES/PCBS BY EPA 608.3									
4,4´-DDD	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
4,4´-DDE	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
4,4´-DDT	EPA 608.3	A	ND	ER	20	0.198	µg/L		09/19/25 15:45 TJW
Aldrin	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
alpha-BHC	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Aroclor 1016	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Aroclor 1221	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Aroclor 1232	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Aroclor 1242	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Aroclor 1248	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Aroclor 1254	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Aroclor 1260	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
beta-BHC	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Chlordane	EPA 608.3	A	ND		1	0.0990	µg/L		09/19/25 18:17 TJW
delta-BHC	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Dieldrin	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Endosulfan I	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Endosulfan II	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Endosulfan sulfate	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Endrin	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Endrin aldehyde	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Endrin ketone	EPA 608.3	A	ND		1	0.0198	µg/L		09/19/25 18:17 TJW
gamma-BHC (Lindane)	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Heptachlor	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Heptachlor epoxide	EPA 608.3	A	ND		1	0.00990	µg/L		09/19/25 18:17 TJW
Methoxychlor	EPA 608.3	A	ND	ER	20	0.198	µg/L		09/19/25 15:45 TJW
Total Polychlorinated Biphenyls	EPA 608.3	A	ND		1	0.248	µg/L		09/19/25 18:17 TJW
Toxaphene	EPA 608.3	A	ND		1	0.495	µg/L		09/19/25 18:17 TJW
Surr: Decachlorobiphenyl	EPA 608.3		83.9		1	20 - 117	%Rec		09/19/25 18:17 TJW

QUALIFIERS

*	Value exceeds Maximum or Minimum Contaminant Level.	C1	Sample container temperature is out of limit as specified at test code
DF	Dilution Factor	E	Value above quantitation range
H	Holding times for preparation or analysis exceeded	MI	Recovery outside control limits due to Matrix Interference
ND	Not Detected at the Reporting Limit	RL	Reporting Limit

Original

NELAP NELAP A Accredited in accordance with NELAP ORELAP 100016, OR-028

NEILSON RESEARCH CORPORATION

Neilson Research Corporation
245 S Grape St
Medford, OR 97501
TEL: (541) 770-5678 FAX: (541) 770-2901
Website: www.nrclabs.com

Analytical Report

WO#: **25090593**
Date Reported: **9/25/2025**

CLIENT: US Fish & Wildlife Services
Lab ID: 25090593-01
Client Sample ID Pond C1
Project: Klamath Falls NFH
Sample Address:

Collection Date: 9/11/2025 11:20:00 AM
Received Date: 9/11/2025 2:20:00 PM
Matrix: AQUEOUS

Sample Location:

Analyses	Method	NELAP Status	Result	Qual	DF	RL	Units	MCL	Date Analyzed/Analyst
ORGANOCHLORINE PESTICIDES/PCBS BY EPA 608.3									
Surr: TCMX	EPA 608.3		65.5		1	40 - 112	%Rec		09/19/25 18:17 TJW
GLYPHOSATE BY EPA 547									
Glyphosate	EPA 547	A	ND		1	0.0500	mg/L	0.700	09/12/25 19:09 TJW

QUALIFIERS

*	Value exceeds Maximum or Minimum Contaminant Level.	C1	Sample container temperature is out of limits as specified at the time of collection.
DF	Dilution Factor	E	Value above quantitation range
H	Holding times for preparation or analysis exceeded	MI	Recovery outside control limits due to Matrix Interference
ND	Not Detected at the Reporting Limit	RL	Reporting Limit

Original

NELAP NELAP A Accredited in accordance with NELAP ORELAP 100016, OR-028

NEILSON RESEARCH CORPORATION

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245 S Grape St
Medford, OR 97501
TEL: (541) 770-5678 FAX: (541) 770-2901
Website: www.nrclabs.com

Sample Log-In Check List

Client Name: **USFISH&WILDLIFE_Kla** Work Order Number: **25090593** RcptNo: **1**

Logged by:	**Ashley Spiegelberg**	**9/11/2025 2:20:00 PM**
Completed By:	**Coni Boyko**	**9/17/2025 10:26:48 AM**
Reviewed By:	**Ashley Spiegelberg**	**9/25/2025 11:13:37 AM**

Chain of Custody

1. Is Chain of Custody complete? Yes ☑ No ☐ Not Present ☐
2. How was the sample delivered? Client

Log In

3. Coolers are present? Yes ☑ No ☐ NA ☐
4. Shipping container/cooler in good condition? Yes ☑ No ☐
   Custody seals intact on shipping container/cooler? Yes ☐ No ☐ Not Present ☑ NA ☐
   No. Seal Date: Signed By:
5. Was an attempt made to cool the samples? Yes ☑ No ☐ NA ☐
6. Were all samples received at a temperature of >0° C to 6.0°C Yes ☑ No ☐ NA ☐
7. Sample(s) in proper container(s)? Yes ☑ No ☐
8. Sufficient sample volume for indicated test(s)? Yes ☑ No ☐
9. Are samples (except VOA and ONG) properly preserved? Yes ☑ No ☐
10. Was preservative added to bottles? Yes ☑ No ☐ NA ☐ HNO3 pH<2
11. Is the headspace in the VOA vials less than 1/4 inch or 6 mm? Yes ☐ No ☐ No VOA Vials ☑
12. Were any sample containers received broken? Yes ☐ No ☑
13. Does paperwork match bottle labels? (Note discrepancies on chain of custody) Yes ☑ No ☐
14. Are matrices correctly identified on Chain of Custody? Yes ☑ No ☐
15. Is it clear what analyses were requested? Yes ☑ No ☐
16. Were all holding times able to be met? (If no, notify customer for authorization.) Yes ☑ No ☐

Special Handling (if applicable)

17. Was client notified of all discrepancies with this order? Yes ☐ No ☐ NA ☑

Person Notified: Date:
By Whom: Via: ☐ eMail ☐ Phone ☐ Fax ☐ In Person
Regarding:
Client Instructions:

18. Additional remarks:

Cooler Information

Cooler No	Temp °C	Condition	Seal Intact	Seal No	Seal Date	Signed By
1	5.5	Good				AS

NEILSON RESEARCH CORPORATION

Chain of Custody Record

This Chain of Custody is a LEGAL DOCUMENT and must be filled out accurately.

Page ___ of ___

Section A Required Client Information

Company:	U.S. Fish and Wildlife Ser.
Address:	3745 Lower Klamath Lake Rd.
Email:	mark_yost@fws.gov
Phone:	458-232-5417 Fax:
Collected By (Print):	Mark Yost
Collected By (Sign):	
Email Report ☒	Mail Report ☐ Fax Report ☐

Section B Required Project Information

Project Name:	Klamath Falls NFH
Project Number:	
Report To:	Mark Yost
Copy To:	"

Section C Invoice Information

Attention:	Same
Company Name:	"
Address:	
P.O. #	

Section D Rush Status (Subject to Scheduling)

- ☐ Standard: 10 Business Days
- ☐ Priority: 5 Business Days (List x 1.50)
- ☐ Express: 3 Business Days (List x 1.75)
- ☐ Rush: 2 Business Days (List x 2.00)
- ☐ Rush: 1 Business Day (List x 2.50)
- ☐ Rush: Same Day (List x 3.00)

Authorized ☐ Yes ☐ No

Section E Sample Information

NRC Workorder # (Lab Use Only): 25090593

Sample ID	Comp/Grab	Matrix*	Date Collected	Time Collected	No. of Containers	6008	547	Top 35	BT	Remarks / Field Data	NRC Sample # (Lab Use Only)
Pond C1			9/11	11:20	10	X	X	X			01

*Matrix: DW - Drinking Water WW - Wastewater W - Water S - Soil/Solid SL - Sludge O - Oil WP - Wipe OT - Other

Section F Relinquish/Receive

	Sign	Print	Date	Time
Relinquished By:	Michael L. Green	Michael L. Green	9/11/25	1420
Received By:				
Relinquished By:				
Received By:				
Relinquished By:				
Received By Laboratory:		Ashley Spiegelberg	9/11/25	14:20

Section G Lab Use Only

Temp:	5.5 IR-Q
≤6°C:	(Yes) No
Received on Ice:	(Yes) No
Number of Bottles Received:	
pH Checked:	
COC Seals Intact:	___ Yes ___ No (NA)
Field Blank Included:	___ Yes ___ No

Received Via ___ UPS ___ FedEX ___ Other ___ Hand

Payment: (Invoice) ___ Cash ___ VISA, M/C ___ Check # ___ Amount 862.00

Effective 6/19/2020

PAID

NEILSON RESEARCH CORPORATION

Neilson Research Corporation
245 S Grape St
Medford, OR 97501
TEL: (541) 770-5678 FAX (541) 770-2901
Website: www.nrclabs.com

Data Flags

WO#: **25090593**
Date: **9/25/2025**

A Total Alkalinity and Bicarbonate Alkalinity results are to a pH endpoint of 4.5. Carbonate Alkalinity result is to a pH endpoint of 8.3.
A-LL The total low level alkalinity results are to a pH endpoint of 4.3-4.7 pH units per SM 2320 B.
B Analyte detected in the associated method blank.
C Sample(s) does not meet NELAP/ORELAP sample acceptance criteria. See Case Narrative.
C1 Sample(s) does not meet NELAP/ORELAP sample acceptance criteria for temperature.
CF Results confirmed by re-analysis.
CU Cleanup performed as specified by method.
E Estimated value.
ER Elevated reporting limit due to matrix. Report limits (MDLs, MRLs & PQLs) are adjusted based on variations in sample preparation amounts, analytical dilutions, and percent solids, where applicable.
FC Fecal Coliforms: Sample(s) received past 40 CFR Part 136 specified holding time. Results reported as estimated values.
HP Sample re-analysis performed outside of method specified holding time.
HR Sample received outside of method specified holding time.
HS Sample analyzed for volatile organics contained headspace.
HT At the client's request, the sample was analyzed outside of method specified holding time.
H Analysis performed outside of method specified holding time.
J Analyte detected below the Minimum Reporting Limit (MRL) and above the Method Detection Limit (MDL). The J flag result is an estimated value and the user should be aware that this data is of limited reliability.
L Dissolved metals were not filtered within 15 minutes of collection per 40 CFR Part 136.
MI Surrogate, Duplicate Sample (DUP) or Matrix Spikes recoveries are out of control limits due to matrix interference. Sample results may be biased.
N See Case Narrative on page 2 of report.
Q Initial calibration verification (ICV), continuing calibration verification (CCV) or laboratory control sample (LCS), and/or matrix spikes exceeded high recovery limits, but associated samples are non-detect and the sample results are not affected. Data meets EPA/NELAP requirements.
R Relative percent difference (RPD) is outside of the accepted recovery limits.
R1 The numerical difference between the parent sample and the duplicate (DUP) is outside of the accepted recovery limits. Greater than 5 degrees for Flashpoint, or greater than 0.1 pH units for pH.
R3 The relative percent difference (RPD) and/or percent recovery for the duplicate (DUP) or matrix spike (MS)/matrix spike duplicate (MSD) cannot be accurately calculated due to the concentration of analyte already present in the sample.
R4 The Relative percent difference (RPD) is not within control limits because the concentration of the sample result is too low to represent proper statistical error.
R5 The difference between the BOD/CBOD results for the highest and lowest dilution used for the calculation is >30% because the results are too low to represent proper statistical error. The BOD/CBOD sample result is an average of all qualified bottles for each dilution series. The sample results are not affected.
R6 The difference between the BOD/CBOD results for the highest and lowest dilution used for the calculation is >30%. This may indicate a possible matrix interference. The BOD/CBOD sample result is an average of all qualified bottles for each dilution series.
S Surrogate and/or matrix spike recovery is outside of the accepted recovery limits. Sample results may be biased.
S1 Surrogate or matrix spike recovery is outside of control limits due to dilution necessary for analysis.
SC Sub-contracted to another laboratory for analysis.
SP Sample(s) were not collected per EPA Method 5035A protocols. The results are considered minimum values.

* Value exceeds Maximum Contaminant Level or is outside the acceptable range.<<>>
1 Value exceeds one half of the Maximum Contaminant Level.

Appendix 4

Matrix Sciences Results

Matrix SCIENCES | PAL PACIFIC AGRICULTURAL LABORATORY

PACAGLAB.COM
503.626.7943
21830 S.W. Alexander Ln
Sherwood, OR 97140

U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: IRP
PAL Sample ID: P251811-01
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes	Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8270D (GC-MS/MS)									
09/23/2025	a-BHC	ND	0.060		09/23/2025	Acetochlor	ND	0.060	
09/23/2025	Alachlor	ND	0.060		09/23/2025	Aldrin	ND	0.060	
09/23/2025	Ametryn	ND	0.060		09/23/2025	Aspon	ND	0.060	
09/23/2025	b-BHC	ND	0.060		09/23/2025	Benfluralin	ND	0.060	
09/23/2025	Bifenthrin	ND	0.060		09/23/2025	Bolstar	ND	0.060	
09/23/2025	Bromopropylate	ND	0.060		09/23/2025	Buprofezin	ND	0.060	
09/23/2025	Captan	ND	0.60		09/23/2025	Chlordane	ND	0.060	
09/23/2025	Chlorfenapyr	ND	0.060		09/23/2025	Chlorfenvinphos	ND	0.060	
09/23/2025	Chlorobenzilate	ND	0.060		09/23/2025	Chloroneb	ND	0.060	
09/23/2025	Chlorpropham	ND	0.060		09/23/2025	Chlorpyrifos	ND	0.060	
09/23/2025	Chlorpyrifos-methyl	ND	0.060		09/23/2025	cis-Nonachlor	ND	0.060	
09/23/2025	Cyfluthrin	ND	0.30		09/23/2025	Cypermethrin	ND	0.30	
09/23/2025	Dacthal	ND	0.060		09/23/2025	d-BHC	ND	0.060	
09/23/2025	Deltamethrin	ND	0.30		09/23/2025	Demeton	ND	0.060	
09/23/2025	Diazinon	ND	0.060		09/23/2025	Dichlobenil	ND	0.060	
09/23/2025	Dichlorofenthion	ND	0.060		09/23/2025	Dichlorvos	ND	0.060	
09/23/2025	Diclofop-methyl	ND	0.060		09/23/2025	Dicloran	ND	0.30	
09/23/2025	Dicofol	ND	0.060		09/23/2025	Dieldrin	ND	0.060	
09/23/2025	Dimethenamid	ND	0.060		09/23/2025	Diphenamid	ND	0.060	
09/23/2025	Diphenylamine	ND	0.060		09/23/2025	Disulfoton	ND	0.060	
09/23/2025	Dithiopyr	ND	0.060		09/23/2025	Endosulfan I	ND	0.12	
09/23/2025	Endosulfan II	ND	0.12		09/23/2025	Endosulfan sulfate	ND	0.12	
09/23/2025	Endrin	ND	0.060		09/23/2025	Endrin ketone	ND	0.060	
09/23/2025	EPN	ND	0.060		09/23/2025	EPTC	ND	0.060	
09/23/2025	Esfenvalerate	ND	0.060		09/23/2025	Ethalfluralin	ND	0.060	
09/23/2025	Ethofumesate	ND	0.060		09/23/2025	Ethoprop	ND	0.060	
09/23/2025	Etoxazole	ND	0.060		09/23/2025	Etridiazole	ND	0.060	
09/23/2025	Fenarimol	ND	0.060		09/23/2025	Fenitrothion	ND	0.060	
09/23/2025	Fenoxaprop-ethyl	ND	0.060		09/23/2025	Fenthion	ND	0.060	
09/23/2025	Fenvalerate	ND	0.060		09/23/2025	Fipronil	ND	0.060	
09/23/2025	Fluazifop-p-butyl	ND	0.060		09/23/2025	Fludioxonil	ND	0.060	
09/23/2025	Fluroxypyr-meptyl	ND	0.060		09/23/2025	Flutolanil	ND	0.060	
09/23/2025	g-BHC	ND	0.060		09/23/2025	Heptachlor	ND	0.060	
09/23/2025	Heptachlor epoxide	ND	0.060		09/23/2025	Hexachlorobenzene	ND	0.060	
09/23/2025	Kresoxim-methyl	ND	0.060		09/23/2025	lambda-Cyhalothrin	ND	0.060	
09/23/2025	Leptophos	ND	0.060		09/23/2025	Malathion	ND	0.060	
09/23/2025	Mefenoxam	ND	0.060		09/23/2025	Methoxychlor	ND	0.060	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: IRP
PAL Sample ID: P251811-01
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis
(Continued)

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes	Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8270D (GC-MS/MS) (Continued)									
09/23/2025	Metolachlor	ND	0.060		09/23/2025	MGK-264	ND	0.060	
09/23/2025	Myclobutanil	ND	0.060		09/23/2025	Napropamide	ND	0.060	
09/23/2025	o-Phenylphenol	ND	0.060		09/23/2025	Ovex	ND	0.060	
09/23/2025	Oxadiazon	ND	0.060		09/23/2025	Oxyfluorfen	ND	0.060	
09/23/2025	p,p'-DDD	ND	0.060		09/23/2025	p,p'-DDE	ND	0.060	
09/23/2025	p,p'-DDT	ND	0.060		09/23/2025	Paclobutrazol	ND	0.060	
09/23/2025	Parathion	ND	0.060		09/23/2025	Parathion-methyl	ND	0.060	
09/23/2025	PCA	ND	0.060		09/23/2025	PCB	ND	0.060	
09/23/2025	PCNB	ND	0.060		09/23/2025	Pendimethalin	ND	0.060	
09/23/2025	Pentachlorothioanisole	ND	0.060		09/23/2025	Permethrin	ND	0.12	
09/23/2025	Phorate	ND	0.060		09/23/2025	Procymidone	ND	0.060	
09/23/2025	Prodiamine	ND	0.060		09/23/2025	Pronamide	ND	0.060	
09/23/2025	Propachlor	ND	0.060		09/23/2025	Pyriproxyfen	ND	0.060	
09/23/2025	Quinoxyfen	ND	0.060		09/23/2025	Ronnel	ND	0.060	
09/23/2025	Spirodiclofen	ND	0.060		09/23/2025	Sulfotep	ND	0.060	
09/23/2025	Tefluthrin	ND	0.060		09/23/2025	Terbufos	ND	0.060	
09/23/2025	Tetraconazole	ND	0.060		09/23/2025	Tetradifon	ND	0.060	
09/23/2025	Thionazin	ND	0.060		09/23/2025	Tokuthion	ND	0.060	
09/23/2025	trans-Nonachlor	ND	0.060		09/23/2025	Trichloronate	ND	0.060	
09/23/2025	Trifluralin	ND	0.060		09/23/2025	Vinclozalin	ND	0.060	
Modified EPA 8321B (LC-MS/MS)									
09/24/2025	Abamectin	ND	0.060		09/24/2025	Acetamiprid	ND	0.060	
09/24/2025	Acibenzolar-S-methyl	ND	0.12		09/24/2025	Afidopyropen	ND	0.060	
09/24/2025	Aldicarb	ND	0.060		09/24/2025	Allethrin	ND	0.060	
09/24/2025	Ametoctradin	ND	0.060		09/24/2025	Atrazine	ND	0.060	
09/24/2025	Azinphos-ethyl	ND	0.060		09/24/2025	Azinphos-methyl	ND	0.12	
09/24/2025	Azoxystrobin	ND	0.060		09/24/2025	Bendiocarb	ND	0.060	
09/24/2025	Bensulide	ND	0.060		09/24/2025	Bicyclopyrone	ND	0.060	
09/24/2025	Bitertanol	ND	0.060		09/24/2025	Boscalid	ND	0.060	
09/24/2025	Bromacil	ND	0.060		09/24/2025	Carbaryl	ND	0.060	
09/24/2025	Carbendazim	ND	0.060		09/24/2025	Carbofuran	ND	0.060	
09/24/2025	Carfentrazone-ethyl	ND	0.060		09/24/2025	Chlorantraniliprole	ND	0.060	
09/24/2025	Clethodim	ND	0.12		09/24/2025	Clofentezine	ND	0.060	
09/24/2025	Clomazone	ND	0.060		09/24/2025	Cyanazine	ND	0.060	
09/24/2025	Cyantraniliprole	ND	0.060		09/24/2025	Cyazofamid	ND	0.060	
09/24/2025	Cyclaniliprole	ND	0.060		09/24/2025	Cycloate	ND	0.12	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: IRP
PAL Sample ID: P251811-01
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis
(Continued)

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes	Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8321B (LC-MS/MS) (Continued)									
09/24/2025	Cyflufenamid	ND	0.060		09/24/2025	Cyflumetofen	ND	0.060	
09/24/2025	Cyhalofop-butyl	ND	0.12		09/24/2025	Cymoxanil	ND	0.060	
09/24/2025	Cyprodinil	ND	0.060		09/24/2025	Cyprosulfamide	ND	0.060	
09/24/2025	DCPMU	ND	0.060		09/24/2025	Diazoxon	ND	0.060	
09/24/2025	Difenoconazole	ND	0.060		09/24/2025	Diflubenzuron	ND	0.060	
09/24/2025	Diflufenican	ND	0.060		09/24/2025	Dimethoate	ND	0.060	
09/24/2025	Dimethomorph	ND	0.060		09/24/2025	Dioxathion	ND	0.060	
09/24/2025	Disulfoton sulfone	ND	0.060		09/24/2025	Diuron	ND	0.060	
09/24/2025	d-Phenothrin	ND	0.060		09/24/2025	Ethion	ND	0.060	
09/24/2025	Etofenprox	ND	0.060		09/24/2025	Famoxadone	ND	0.060	
09/24/2025	Famphur	ND	0.060		09/24/2025	Fenamidone	ND	0.060	
09/24/2025	Fenamiphos sulfone	ND	0.060		09/24/2025	Fenamiphos sulfoxide	ND	0.060	
09/24/2025	Fenazaquin	ND	0.060		09/24/2025	Fenbuconazole	ND	0.060	
09/24/2025	Fenbutatin oxide	ND	0.060		09/24/2025	Fenhexamid	ND	0.060	
09/24/2025	Fenobucarb	ND	0.060		09/24/2025	Fenoxycarb	ND	0.060	
09/24/2025	Fenpropathrin	ND	0.060		09/24/2025	Fenpyroximate	ND	0.060	
09/24/2025	Fenuron	ND	0.060		09/24/2025	Fluazinam	ND	0.060	
09/24/2025	Flubendiamide	ND	0.12		09/24/2025	Flufenacet	ND	0.060	
09/24/2025	Flumioxazin	ND	0.060		09/24/2025	Fluometuron	ND	0.060	
09/24/2025	Fluopicolide	ND	0.060		09/24/2025	Fluopyram	ND	0.060	
09/24/2025	Fluoxastrobin	ND	0.060		09/24/2025	Flupyradifurone	ND	0.060	
09/24/2025	Fluridone	ND	0.060		09/24/2025	Flutianil	ND	0.060	
09/24/2025	Flutriafol	ND	0.060		09/24/2025	Fluvalinate	ND	0.060	
09/24/2025	Fluxapyroxad	ND	0.060		09/24/2025	Fonofos	ND	0.12	
09/24/2025	Hexaconazole	ND	0.060		09/24/2025	Hexazinone	ND	0.060	
09/24/2025	Hexythiazox	ND	0.060		09/24/2025	Imazalil	ND	0.060	
09/24/2025	Imidacloprid	ND	0.060		09/24/2025	Indaziflam	ND	0.060	
09/24/2025	Indoxacarb	ND	0.060		09/24/2025	Iodosulfuron-methyl	ND	0.060	
09/24/2025	Ipconazole	ND	0.060		09/24/2025	Iprodione	ND	0.30	
09/24/2025	Isofetamid	ND	0.060		09/24/2025	Isoxaben	ND	0.060	
09/24/2025	Isoxadifen-ethyl	ND	0.060		09/24/2025	Lactofen	ND	0.060	
09/24/2025	Linuron	ND	0.060		09/24/2025	Malaoxon	ND	0.060	
09/24/2025	Mandipropamid	ND	0.060		09/24/2025	Metconazole	ND	0.060	
09/24/2025	Methidathion	ND	0.060		09/24/2025	Methiocarb	ND	0.060	
09/24/2025	Methomyl	ND	0.060		09/24/2025	Methoxyfenozide	ND	0.060	
09/24/2025	Metrafenone	ND	0.060		09/24/2025	Metribuzin	ND	0.060	
09/24/2025	Mevinphos	ND	0.060		09/24/2025	Monuron	ND	0.060	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: IRP
PAL Sample ID: P251811-01
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis
(Continued)

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8321B (LC-MS/MS) (Continued)				
09/24/2025	Neburon	ND	0.060	
09/24/2025	Novaluron	ND	0.060	
09/24/2025	Oxadixyl	ND	0.060	
09/24/2025	Penoxsulam	ND	0.060	
09/24/2025	Phorate Sulfone	ND	0.060	
09/24/2025	Phosalone	ND	0.060	
09/24/2025	Phosphamidon	ND	0.060	
09/24/2025	Piperonyl Butoxide	ND	0.060	
09/24/2025	Pirimiphos-methyl	ND	0.060	
09/24/2025	Prometon	ND	0.060	
09/24/2025	Propanil	ND	0.060	
09/24/2025	Propazine	ND	0.060	
09/24/2025	Propoxur	ND	0.060	
09/24/2025	Pyraflufen-ethyl	ND	0.060	
09/24/2025	Pyridaben	ND	0.060	
09/24/2025	Pyrimethanil	ND	0.060	
09/24/2025	Quizalofop-p-ethyl	ND	0.060	
09/24/2025	Saflufenacil	ND	0.060	
09/24/2025	Siduron	ND	0.060	
09/24/2025	Simetryn	ND	0.060	
09/24/2025	Spinosad	ND	0.060	
09/24/2025	Spirotetramat	ND	0.060	
09/24/2025	Sulfentrazone	ND	0.060	
09/24/2025	Tebuconazole	ND	0.060	
09/24/2025	Tebuthiuron	ND	0.060	
09/24/2025	Terbuthylazine	ND	0.060	
09/24/2025	Thiabendazole	ND	0.060	
09/24/2025	Thiamethoxam	ND	0.060	
09/24/2025	Thiobencarb	ND	0.060	
09/24/2025	Tolfenpyrad	ND	0.060	
09/24/2025	Triadimenol	ND	0.12	
09/24/2025	Trifloxystrobin	ND	0.060	
09/24/2025	Triflumizole	ND	0.060	
09/24/2025	Triticonazole	ND	0.060	

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
09/24/2025	Norflurazon	ND	0.060	
09/24/2025	Oryzalin	ND	0.060	
09/24/2025	Oxamyl	ND	0.060	
09/24/2025	Penthiopyrad	ND	0.060	
09/24/2025	Phorate Sulfoxide	ND	0.060	
09/24/2025	Phosmet	ND	0.060	
09/24/2025	Picoxystrobin	ND	0.060	
09/24/2025	Pirimicarb	ND	0.060	
09/24/2025	Prallethrin	ND	0.060	
09/24/2025	Prometryn	ND	0.060	
09/24/2025	Propargite	ND	0.060	
09/24/2025	Propiconazole	ND	0.12	
09/24/2025	Pyraclostrobin	ND	0.060	
09/24/2025	Pyrethrin	ND	0.30	
09/24/2025	Pyridalyl	ND	0.060	
09/24/2025	Pyroxasulfone	ND	0.060	
09/24/2025	Rotenone	ND	0.060	
09/24/2025	Sethoxydim	ND	0.12	
09/24/2025	Simazine	ND	0.060	
09/24/2025	Spinetoram	ND	0.060	
09/24/2025	Spiromesifen	ND	0.12	
09/24/2025	Spiroxamine	ND	0.060	
09/24/2025	Sulfoxaflor	ND	0.060	
09/24/2025	Tebufenozide	ND	0.060	
09/24/2025	Terbacil	ND	0.060	
09/24/2025	Terbutryn	ND	0.060	
09/24/2025	Thiacloprid	ND	0.060	
09/24/2025	Thiencarbazone-methyl	ND	0.060	
09/24/2025	Thiodicarb	ND	0.060	
09/24/2025	Triadimefon	ND	0.060	
09/24/2025	Triallate	ND	0.060	
09/24/2025	Trifloxysulfuron-sodium	ND	0.060	
09/24/2025	Trinexapac-ethyl	ND	0.060	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number:	P251811		
Report Date:	September 30, 2025		
Client Project ID:			
Client Sample ID:	IRP	**Sample Date:**	09/17/2025
PAL Sample ID:	P251811-01	**Received Date:**	09/18/2025
Matrix:	water	**Extraction Date:**	09/22/2025

Notes and Definitions

Notes	Definition
LOQ	Limit of Quantitation
ND	Not Detected
*	Not included under current scope of accreditation

The results contained in this report relate only to the items tested.
The results reflect the condition of the samples as received by PAL.
Samples will be stored for a minimum of 60 days after the final report is issued, as described in our Quality Manual.
Reports should not be reproduced, except in full, without written approval from PAL.
PAL is accredited to ISO/IEC 17025:2017 Standard, by ANAB, Accreditation #AT-2875, Testing.

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: Pond C6
PAL Sample ID: P251811-02
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes	Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8270D (GC-MS/MS)									
09/23/2025	a-BHC	ND	0.060		09/23/2025	Acetochlor	ND	0.060	
09/23/2025	Alachlor	ND	0.060		09/23/2025	Aldrin	ND	0.060	
09/23/2025	Ametryn	ND	0.060		09/23/2025	Aspon	ND	0.060	
09/23/2025	b-BHC	ND	0.060		09/23/2025	Benfluralin	ND	0.060	
09/23/2025	Bifenthrin	ND	0.060		09/23/2025	Bolstar	ND	0.060	
09/23/2025	Bromopropylate	ND	0.060		09/23/2025	Buprofezin	ND	0.060	
09/23/2025	Captan	ND	0.60		09/23/2025	Chlordane	ND	0.060	
09/23/2025	Chlorfenapyr	ND	0.060		09/23/2025	Chlorfenvinphos	ND	0.060	
09/23/2025	Chlorobenzilate	ND	0.060		09/23/2025	Chloroneb	ND	0.060	
09/23/2025	Chlorpropham	ND	0.060		09/23/2025	Chlorpyrifos	ND	0.060	
09/23/2025	Chlorpyrifos-methyl	ND	0.060		09/23/2025	cis-Nonachlor	ND	0.060	
09/23/2025	Cyfluthrin	ND	0.30		09/23/2025	Cypermethrin	ND	0.30	
09/23/2025	Dacthal	ND	0.060		09/23/2025	d-BHC	ND	0.060	
09/23/2025	Deltamethrin	ND	0.30		09/23/2025	Demeton	ND	0.060	
09/23/2025	Diazinon	ND	0.060		09/23/2025	Dichlobenil	ND	0.060	
09/23/2025	Dichlorofenthion	ND	0.060		09/23/2025	Dichlorvos	ND	0.060	
09/23/2025	Diclofop-methyl	ND	0.060		09/23/2025	Dicloran	ND	0.30	
09/23/2025	Dicofol	ND	0.060		09/23/2025	Dieldrin	ND	0.060	
09/23/2025	Dimethenamid	ND	0.060		09/23/2025	Diphenamid	ND	0.060	
09/23/2025	Diphenylamine	ND	0.060		09/23/2025	Disulfoton	ND	0.060	
09/23/2025	Dithiopyr	ND	0.060		09/23/2025	Endosulfan I	ND	0.12	
09/23/2025	Endosulfan II	ND	0.12		09/23/2025	Endosulfan sulfate	ND	0.12	
09/23/2025	Endrin	ND	0.060		09/23/2025	Endrin ketone	ND	0.060	
09/23/2025	EPN	ND	0.060		09/23/2025	EPTC	ND	0.060	
09/23/2025	Esfenvalerate	ND	0.060		09/23/2025	Ethalfluralin	ND	0.060	
09/23/2025	Ethofumesate	ND	0.060		09/23/2025	Ethoprop	ND	0.060	
09/23/2025	Etoxazole	ND	0.060		09/23/2025	Etridiazole	ND	0.060	
09/23/2025	Fenarimol	ND	0.060		09/23/2025	Fenitrothion	ND	0.060	
09/23/2025	Fenoxaprop-ethyl	ND	0.060		09/23/2025	Fenthion	ND	0.060	
09/23/2025	Fenvalerate	ND	0.060		09/23/2025	Fipronil	ND	0.060	
09/23/2025	Fluazifop-p-butyl	ND	0.060		09/23/2025	Fludioxonil	ND	0.060	
09/23/2025	Fluroxypyr-meptyl	ND	0.060		09/23/2025	Flutolanil	ND	0.060	
09/23/2025	g-BHC	ND	0.060		09/23/2025	Heptachlor	ND	0.060	
09/23/2025	Heptachlor epoxide	ND	0.060		09/23/2025	Hexachlorobenzene	ND	0.060	
09/23/2025	Kresoxim-methyl	ND	0.060		09/23/2025	lambda-Cyhalothrin	ND	0.060	
09/23/2025	Leptophos	ND	0.060		09/23/2025	Malathion	ND	0.060	
09/23/2025	Mefenoxam	ND	0.060		09/23/2025	Methoxychlor	ND	0.060	

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Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: Pond C6
PAL Sample ID: P251811-02
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis
(Continued)

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes	Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8270D (GC-MS/MS) (Continued)									
09/23/2025	Metolachlor	ND	0.060		09/23/2025	MGK-264	ND	0.060	
09/23/2025	Myclobutanil	ND	0.060		09/23/2025	Napropamide	ND	0.060	
09/23/2025	o-Phenylphenol	ND	0.060		09/23/2025	Ovex	ND	0.060	
09/23/2025	Oxadiazon	ND	0.060		09/23/2025	Oxyfluorfen	ND	0.060	
09/23/2025	p,p'-DDD	ND	0.060		09/23/2025	p,p'-DDE	ND	0.060	
09/23/2025	p,p'-DDT	ND	0.060		09/23/2025	Paclobutrazol	ND	0.060	
09/23/2025	Parathion	ND	0.060		09/23/2025	Parathion-methyl	ND	0.060	
09/23/2025	PCA	ND	0.060		09/23/2025	PCB	ND	0.060	
09/23/2025	PCNB	ND	0.060		09/23/2025	Pendimethalin	ND	0.060	
09/23/2025	Pentachlorothioanisole	ND	0.060		09/23/2025	Permethrin	ND	0.12	
09/23/2025	Phorate	ND	0.060		09/23/2025	Procymidone	ND	0.060	
09/23/2025	Prodiamine	ND	0.060		09/23/2025	Pronamide	ND	0.060	
09/23/2025	Propachlor	ND	0.060		09/23/2025	Pyriproxyfen	ND	0.060	
09/23/2025	Quinoxyfen	ND	0.060		09/23/2025	Ronnel	ND	0.060	
09/23/2025	Spirodiclofen	ND	0.060		09/23/2025	Sulfotep	ND	0.060	
09/23/2025	Tefluthrin	ND	0.060		09/23/2025	Terbufos	ND	0.060	
09/23/2025	Tetraconazole	ND	0.060		09/23/2025	Tetradifon	ND	0.060	
09/23/2025	Thionazin	ND	0.060		09/23/2025	Tokuthion	ND	0.060	
09/23/2025	trans-Nonachlor	ND	0.060		09/23/2025	Trichloronate	ND	0.060	
09/23/2025	Trifluralin	ND	0.060		09/23/2025	Vinclozalin	ND	0.060	
Modified EPA 8321B (LC-MS/MS)									
09/24/2025	Abamectin	ND	0.060		09/24/2025	Acetamiprid	ND	0.060	
09/24/2025	Acibenzolar-S-methyl	ND	0.12		09/24/2025	Afidopyropen	ND	0.060	
09/24/2025	Aldicarb	ND	0.060		09/24/2025	Allethrin	ND	0.060	
09/24/2025	Ametoctradin	ND	0.060		09/24/2025	Atrazine	ND	0.060	
09/24/2025	Azinphos-ethyl	ND	0.060		09/24/2025	Azinphos-methyl	ND	0.12	
09/24/2025	Azoxystrobin	ND	0.060		09/24/2025	Bendiocarb	ND	0.060	
09/24/2025	Bensulide	ND	0.060		09/24/2025	Bicyclopyrone	ND	0.060	
09/24/2025	Bitertanol	ND	0.060		09/24/2025	Boscalid	ND	0.060	
09/24/2025	Bromacil	ND	0.060		09/24/2025	Carbaryl	ND	0.060	
09/24/2025	Carbendazim	ND	0.060		09/24/2025	Carbofuran	ND	0.060	
09/24/2025	Carfentrazone-ethyl	ND	0.060		09/24/2025	Chlorantraniliprole	ND	0.060	
09/24/2025	Clethodim	ND	0.12		09/24/2025	Clofentezine	ND	0.060	
09/24/2025	Clomazone	ND	0.060		09/24/2025	Cyanazine	ND	0.060	
09/24/2025	Cyantraniliprole	ND	0.060		09/24/2025	Cyazofamid	ND	0.060	
09/24/2025	Cyclaniliprole	ND	0.060		09/24/2025	Cycloate	ND	0.12	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: Pond C6
PAL Sample ID: P251811-02
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis
(Continued)

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8321B (LC-MS/MS) (Continued)				
09/24/2025	Cyflufenamid	ND	0.060	
09/24/2025	Cyhalofop-butyl	ND	0.12	
09/24/2025	Cyprodinil	ND	0.060	
09/24/2025	DCPMU	ND	0.060	
09/24/2025	Difenoconazole	ND	0.060	
09/24/2025	Diflufenican	ND	0.060	
09/24/2025	Dimethomorph	ND	0.060	
09/24/2025	Disulfoton sulfone	ND	0.060	
09/24/2025	d-Phenothrin	ND	0.060	
09/24/2025	Etofenprox	ND	0.060	
09/24/2025	Famphur	ND	0.060	
09/24/2025	Fenamiphos sulfone	ND	0.060	
09/24/2025	Fenazaquin	ND	0.060	
09/24/2025	Fenbutatin oxide	ND	0.060	
09/24/2025	Fenobucarb	ND	0.060	
09/24/2025	Fenpropathrin	ND	0.060	
09/24/2025	Fenuron	ND	0.060	
09/24/2025	Flubendiamide	ND	0.12	
09/24/2025	Flumioxazin	ND	0.060	
09/24/2025	Fluopicolide	ND	0.060	
09/24/2025	Fluoxastrobin	ND	0.060	
09/24/2025	Fluridone	ND	0.060	
09/24/2025	Flutriafol	ND	0.060	
09/24/2025	Fluxapyroxad	ND	0.060	
09/24/2025	Hexaconazole	ND	0.060	
09/24/2025	Hexythiazox	ND	0.060	
09/24/2025	Imidacloprid	ND	0.060	
09/24/2025	Indoxacarb	ND	0.060	
09/24/2025	Ipconazole	ND	0.060	
09/24/2025	Isofetamid	ND	0.060	
09/24/2025	Isoxadifen-ethyl	ND	0.060	
09/24/2025	Linuron	ND	0.060	
09/24/2025	Mandipropamid	ND	0.060	
09/24/2025	Methidathion	ND	0.060	
09/24/2025	Methomyl	ND	0.060	
09/24/2025	Metrafenone	ND	0.060	
09/24/2025	Mevinphos	ND	0.060	
09/24/2025	Cyflumetofen	ND	0.060	
09/24/2025	Cymoxanil	ND	0.060	
09/24/2025	Cyprosulfamide	ND	0.060	
09/24/2025	Diazoxon	ND	0.060	
09/24/2025	Diflubenzuron	ND	0.060	
09/24/2025	Dimethoate	ND	0.060	
09/24/2025	Dioxathion	ND	0.060	
09/24/2025	Diuron	ND	0.060	
09/24/2025	Ethion	ND	0.060	
09/24/2025	Famoxadone	ND	0.060	
09/24/2025	Fenamidone	ND	0.060	
09/24/2025	Fenamiphos sulfoxide	ND	0.060	
09/24/2025	Fenbuconazole	ND	0.060	
09/24/2025	Fenhexamid	ND	0.060	
09/24/2025	Fenoxycarb	ND	0.060	
09/24/2025	Fenpyroximate	ND	0.060	
09/24/2025	Fluazinam	ND	0.060	
09/24/2025	Flufenacet	ND	0.060	
09/24/2025	Fluometuron	ND	0.060	
09/24/2025	Fluopyram	ND	0.060	
09/24/2025	Flupyradifurone	ND	0.060	
09/24/2025	Flutianil	ND	0.060	
09/24/2025	Fluvalinate	ND	0.060	
09/24/2025	Fonofos	ND	0.12	
09/24/2025	Hexazinone	ND	0.060	
09/24/2025	Imazalil	ND	0.060	
09/24/2025	Indaziflam	ND	0.060	
09/24/2025	Iodosulfuron-methyl	ND	0.060	
09/24/2025	Iprodione	ND	0.30	
09/24/2025	Isoxaben	ND	0.060	
09/24/2025	Lactofen	ND	0.060	
09/24/2025	Malaoxon	ND	0.060	
09/24/2025	Metconazole	ND	0.060	
09/24/2025	Methiocarb	ND	0.060	
09/24/2025	Methoxyfenozide	ND	0.060	
09/24/2025	Metribuzin	ND	0.060	
09/24/2025	Monuron	ND	0.060	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number: P251811
Report Date: September 30, 2025
Client Project ID:

Client Sample ID: Pond C6
PAL Sample ID: P251811-02
Matrix: water

Sample Date: 09/17/2025
Received Date: 09/18/2025
Extraction Date: 09/22/2025

Certificate of Analysis
(Continued)

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
Modified EPA 8321B (LC-MS/MS) (Continued)				
09/24/2025	Neburon	ND	0.060	
09/24/2025	Novaluron	ND	0.060	
09/24/2025	Oxadixyl	ND	0.060	
09/24/2025	Penoxsulam	ND	0.060	
09/24/2025	Phorate Sulfone	ND	0.060	
09/24/2025	Phosalone	ND	0.060	
09/24/2025	Phosphamidon	ND	0.060	
09/24/2025	Piperonyl Butoxide	ND	0.060	
09/24/2025	Pirimiphos-methyl	ND	0.060	
09/24/2025	Prometon	ND	0.060	
09/24/2025	Propanil	ND	0.060	
09/24/2025	Propazine	ND	0.060	
09/24/2025	Propoxur	ND	0.060	
09/24/2025	Pyraflufen-ethyl	ND	0.060	
09/24/2025	Pyridaben	ND	0.060	
09/24/2025	Pyrimethanil	ND	0.060	
09/24/2025	Quizalofop-p-ethyl	ND	0.060	
09/24/2025	Saflufenacil	ND	0.060	
09/24/2025	Siduron	ND	0.060	
09/24/2025	Simetryn	ND	0.060	
09/24/2025	Spinosad	ND	0.060	
09/24/2025	Spirotetramat	ND	0.060	
09/24/2025	Sulfentrazone	ND	0.060	
09/24/2025	Tebuconazole	ND	0.060	
09/24/2025	Tebuthiuron	ND	0.060	
09/24/2025	Terbuthylazine	ND	0.060	
09/24/2025	Thiabendazole	ND	0.060	
09/24/2025	Thiamethoxam	ND	0.060	
09/24/2025	Thiobencarb	ND	0.060	
09/24/2025	Tolfenpyrad	ND	0.060	
09/24/2025	Triadimenol	ND	0.12	
09/24/2025	Trifloxystrobin	ND	0.060	
09/24/2025	Triflumizole	ND	0.060	
09/24/2025	Triticonazole	ND	0.060	

Analysis Date	Analyte	Amount Detected	LOQ (ug/L)	Notes
09/24/2025	Norflurazon	ND	0.060	
09/24/2025	Oryzalin	ND	0.060	
09/24/2025	Oxamyl	ND	0.060	
09/24/2025	Penthiopyrad	ND	0.060	
09/24/2025	Phorate Sulfoxide	ND	0.060	
09/24/2025	Phosmet	ND	0.060	
09/24/2025	Picoxystrobin	ND	0.060	
09/24/2025	Pirimicarb	ND	0.060	
09/24/2025	Prallethrin	ND	0.060	
09/24/2025	Prometryn	ND	0.060	
09/24/2025	Propargite	ND	0.060	
09/24/2025	Propiconazole	ND	0.12	
09/24/2025	Pyraclostrobin	ND	0.060	
09/24/2025	Pyrethrin	ND	0.30	
09/24/2025	Pyridalyl	ND	0.060	
09/24/2025	Pyroxasulfone	ND	0.060	
09/24/2025	Rotenone	ND	0.060	
09/24/2025	Sethoxydim	ND	0.12	
09/24/2025	Simazine	ND	0.060	
09/24/2025	Spinetoram	ND	0.060	
09/24/2025	Spiromesifen	ND	0.12	
09/24/2025	Spiroxamine	ND	0.060	
09/24/2025	Sulfoxaflor	ND	0.060	
09/24/2025	Tebufenozide	ND	0.060	
09/24/2025	Terbacil	ND	0.060	
09/24/2025	Terbutryn	ND	0.060	
09/24/2025	Thiacloprid	ND	0.060	
09/24/2025	Thiencarbazone-methyl	ND	0.060	
09/24/2025	Thiodicarb	ND	0.060	
09/24/2025	Triadimefon	ND	0.060	
09/24/2025	Triallate	ND	0.060	
09/24/2025	Trifloxysulfuron-sodium	ND	0.060	
09/24/2025	Trinexapac-ethyl	ND	0.060	

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U.S. Fish and Wildlife Services
3745 Lower Klamath Lake Rd
Klamath Falls, OR 97603

Report Number:	P251811		
Report Date:	September 30, 2025		
Client Project ID:			
Client Sample ID:	Pond C6	**Sample Date:**	09/17/2025
PAL Sample ID:	P251811-02	**Received Date:**	09/18/2025
Matrix:	water	**Extraction Date:**	09/22/2025

Notes and Definitions

Notes	Definition
LOQ	Limit of Quantitation
ND	Not Detected
*	Not included under current scope of accreditation

The results contained in this report relate only to the items tested.
The results reflect the condition of the samples as received by PAL.
Samples will be stored for a minimum of 60 days after the final report is issued, as described in our Quality Manual.
Reports should not be reproduced, except in full, without written approval from PAL.
PAL is accredited to ISO/IEC 17025:2017 Standard, by ANAB, Accreditation #AT-2875, Testing.

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Appendix 5

Bend Genetics LLC Results

Bend Genetics

Date: 9/16/2025

Subject: Cyanobacteria testing results

From: Tim Otten, Laboratory Director

To: Michelle Jackson
U.S. Fish and Wildlife Service

Testing results are enclosed for microscopy and ELISA analyses conducted on one sample collected from C2 Pond on 9/9/2025. All data have been reviewed and are considered final.

Analyses included in this report:

- Quantification of total cyanobacterial toxins (anatoxin-a, cylindrospermopsin, microcystin/nodularin (EPA Method 546), and saxitoxin) by enzyme linked immunosorbent assay (ELISA) methods.
- Microscope photos and identification of potentially toxigenic (PTOX) cyanobacteria.

Submitted by: _____
Timothy G. Otten, MPH, PhD

107 Scripps Drive, Ste. 210
Sacramento, CA 95825

(916) 550-1048
www.bendgenetics.com

Bend Genetics

Received: 9/11/2025
Time: 9:28 AM
Matrix: Water Grab
BG ID: TL149
Reported: 9/16/2025 13:31

Customer: USFWS
Project:
Location: C2 Pond
Collected: 9/9/2025 7:05

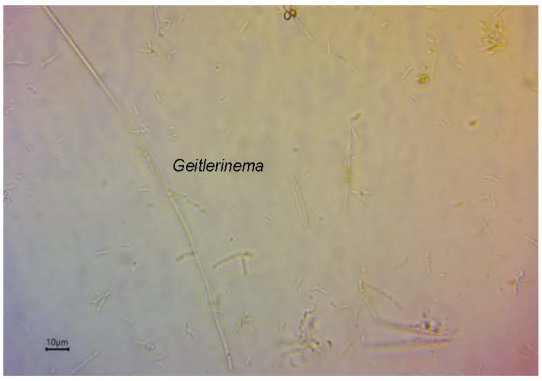
Sample ID: C2

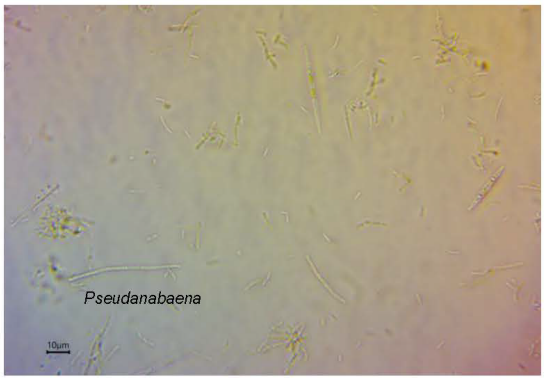
SAMPLE RESULTS

Method	Analyte	Result	Reporting Limit	Units	Qualifiers	Batch
ELISA	Anatoxin-a	ND	0.15	µg/L	U	
ELISA	Cylindrospermopsin	ND	0.05	µg/L	U	
ELISA	Microcystin/Nod.	ND	0.20	µg/L	U	
ELISA	Saxitoxin	ND	0.02	µg/L	U	

MICROSCOPY RESULTS - Identification of CyanoHABs

Dominant	Sub-dominant	Also present	Notes
Geitlerinema	*Pseudanabaena*	*Anabaenopsis*	This sample contained a low amount of *Geitlerinema* sp. and *Pseudanabaena* sp., and a very low amount of *Anabaenopsis* sp.; the photomicrographs were taken at 400X magnification.





Bend Genetics

Customer: USFWS
Project:

Reported: 9/16/2025 13:31

QUALITY CONTROL

Method	Analyte	Result	Units	Qualifiers	Spike	%Rec	%Rec Limit	Batch
ELISA	ATX - Blank	ND	µg/L	U	0			
ELISA	ATX - Positive	0.63	µg/L		0.75	83.7	70-130	
ELISA	ATX - Spike	1.23	µg/L		1.25	98.2	70-130	
ELISA	CYN - Blank	ND	µg/L	U	0			
ELISA	CYN - Positive	0.73	µg/L		0.75	97.5	70-130	
ELISA	CYN - Spike	1.03	µg/L		1.00	102.6	70-130	
ELISA	MC - Blank	ND	µg/L	U	0			
ELISA	MC - Positive	0.81	µg/L		0.75	107.8	70-130	
ELISA	MC - Spike	1.10	µg/L		1.00	110.3	70-130	
ELISA	STX - Blank	ND	µg/L	U	0			
ELISA	STX - Positive	0.069	µg/L		0.075	92.0	70-130	
ELISA	STX - Spike	0.193	µg/L		0.200	96.5	70-130	

QUALIFIERS:

C1	The reported result for this analyte is below the quantification limit.
C2	The reported result for this analyte is above the calibration range of the assay.
D	Sample diluted - reported result accounts for dilution factor.
J	The reported result for this analyte should be considered as an estimated value.
U	Undetected

Appendix 6

California Nevada Fish Health Center Inspection Results

Diagnostic Summary: Lost River / Short-nose Sucker Mortality Investigation

Location: Klamath Falls National Fish Hatchery (KFNFH)

Timeframe: June – September 2025

Prepared by: Ron Stone, CA-NV Fish Health Center (FHC)

Executive Summary

Between June and September 2025, elevated mortality was observed in multiple ponds at KFNFH, primarily affecting Lost River Suckers. Diagnostic investigations identified *Aeromonas* spp. and *Vibrio* spp. as the primary bacterial agents, with *Ichthyobodo* (Costia) detected in later samples. Environmental stressors—particularly elevated water temperatures—likely contributed to immunosuppression, leading to opportunistic infections. A novel intervention involving indoor holding and treatment during peak mortality appeared to reduce losses and improve fish condition.

Timeline of Events

June – Early July

- Slight increase in mortality in ponds C2 and C4.
- On July 2, FHC sampled fish from C2 and C4.
    - *Aeromonas / Pseudomonas* isolated from kidney.
    - Histology (C4): inflammation of visceral fat, abnormal kidney tubules with hyaline deposits and necrosis.
    - No parasites observed.
    - No immediate action taken.

Early August

- Mortality increased in B-ponds (B3 & B4).
- *Aeromonas* spp. isolated from kidneys.
- Histology (B3): abnormal gill lamellae, edema, epithelial hyperplasia, and depleted liver glycogen/lipid stores.

Late August

- Significant mortality spike in C2 and C4.
- On August 28, 59 clinical fish were transported to FHC; 28 died in transit.
- *Aeromonas* spp. and *Vibrio* spp. isolated and sent to WADDL for identification.
- Antibiotic sensitivity testing initiated.

September 10

- Ponds C1 and C2 harvested; fish moved indoors for treatment and observation.
- *Aeromonas* spp., *Vibrio* spp., and *Ichthyobodo* detected.
- One adult and four juvenile Tilapia were removed from C2.

Diagnostic Findings

Virology (n = 60)

- No cytopathic effect (CPE) observed on EPC or CHSE-214 cell lines after 28 days.

Microscopy (Wet Mount, n = 18)

- No external parasites observed on skin or gills in early samples.
- *Ichthyobodo* (Costia) detected in C2 fish in September.

Bacteriology (n = 109)

- Kidney samples streaked on TSA; isolates identified via MALDI-TOF MS at WADDL.
- Most common isolates:
  - *Aeromonas veronii* (32%)
  - *Vibrio cholerae* (8%) – low human risk
  - Other *Aeromonas* spp., *Pseudomonas anguilliseptica*, and miscellaneous bacteria
- *Aeromonas* spp. are endemic and often isolated from asymptomatic fish at KFNFH.

Antibiotic Sensitivity (Case 25-086)

- Ciprofloxacin showed the highest efficacy across isolates.
- Use of Ciprofloxacin not approved due to tribal consumption of suckers as food fish.
- Coordination ongoing with Dr. Katie Royer and FDA.

Kidney Imprints (Case 25-088)

- 69% of imprints showed gram-negative rods.
- Some showed two morphotypes, suggesting mixed infections.
- Blood smears: low WBC counts, predominantly macrophages; bacterial rods in 3 of 14 samples.

Histopathology Summary

Case 25-068

- Wild-caught suckers: inflammation of visceral fat, abnormal kidney tubules with hyaline deposits and necrosis.
- No parasites; gill, liver, and intestine normal.

Case 25-083

- B3 fish: gill edema, epithelial hyperplasia, depleted liver energy stores.
- No parasites; intestines and kidneys normal.

Case 25-088

- C1 & C2 fish: hyperplastic gills, signs of hypoxia, presence of *Ichthyobodo*.
- Low feed intake; no evidence of bacteremia or necrosis.

Interpretation & Conclusions

- Elevated water temperatures and associated water quality stressors (e.g., low DO, pH fluctuations) likely initiated the mortality event.
- Fish went off feed and failed to recover, becoming susceptible to opportunistic pathogens (*Aeromonas*, *Vibrio*, *Ichthyobodo*).
- The presence of Tilapia in C2 may have contributed to pathogen introduction or persistence.
- The decision to bring fish indoors for treatment was a critical and commendable step that likely reduced mortality and improved fish welfare.

Recommendations

1. **Continue Monitoring:** Regular water quality checks and early detection of feeding behavior changes.
2. **Biosecurity:** Prevent introduction of non-native species (e.g., Tilapia).
3. **Treatment Planning:** Explore alternative therapeutics or supportive care options given antibiotic restrictions.
4. **Data Integration:** Use mortality trends and diagnostic data to refine future response protocols.
5. **Documentation:** Maintain detailed records of interventions and outcomes for adaptive management.

Appendix

Appendix A. Bacterial isolated from fish sampled July – September. Bacterial identification from WADDL MALDI-TOF MS assay.

	25-068	25-083	25-086	25-088	**Totals**	
Total Number of Bacte Samples	**13**	**29**	**46**	**21**	**109**	**%**
Aeromonas veronii	6	5	6	18	**35**	**32 %**
Aeromonas hydrophila	0	0	3	1	**4**	**4 %**
Aeromonas sobria	0	0	0	1	**1**	**1 %**
Psuedomonas anguilliseptica	1	0	0	1	**2**	**2 %**
Vibrio sp.	0	0	6	9	**9**	**8 %**
Other	2	8	2	6	**18**	**16 %**
No Growth	4	13	23	0	**40**	**37 %**

Appendix B. Antibiotic Sensitivity Testing results, mean and range (min-max). T30 – Oxytetracycline 30µg, TE30 – Tetracycline 30µg, FFC30 – Florfenicol 30µg, Romet, SXT – Sulfamethox + trimethroprim 23.75µg / 1.25µg, CIP5 – Ciprofloxacin 5µg

	T30	TE30	FFC30	*Romet	SXT	CIP5
A. veronii	19 (9-27)	22 (10-29)	36 (32-40)	15 (14-16)	25 (14-33)	28 (18-39)
A. hydrophila	18 (9-35)	17 (11-30)	34 (28-39)	29	31 (27-33)	39 (34-42)
Vibrio sp.	29 (28-31)	29 (28-31)	30 (27-34)	19 (18-20)	20 (18-22)	38 (34-41)
Enterobacter	22 (19-24)	16 (12-20)	30 (26-33)	20	30	40 (39-41)

*Romet acquired from Aquatactic expired 9/15/2023 .

Appendix C. Kidney Imprints Results from CANV case 25-088 observationally quantifying levels of gram-negative rods (GNR) in kidney imprints as sell types of blood cells observed. CA-NV Fish Health Center case no. 25-088

	GNR			Cell Types				
Moribund	Limited	Diffuse	None	Macrophage	Neutrophil	Thrombocytes	Lymphocytes	Bacterial Rods
Yes		x		6	1	9		yes
Yes	x			6	4		1	yes
Yes		x		9				yes
Yes			x	5		2		no
Yes	x			11		11		no
Yes	x			4	1		1	no
Yes			x	1			1	no
Yes	x			2		1		no
No			x			4	2	no
No	x					3		no
No	x							no
No		x		3				no
No	x			1	1			no
No			x			1		no

Appendix D. Fish transported to the Ca-NV Fis Health Center 25, August 2025, if they were clinical, alive or dead, fish identification number, fork length (mm), weight (g), condition factor, observed external hemorrhaging, observed visceral fat &, food observed in the gut.

Klamath Sucker Diagnostic 8-28-2025 CANV Case no. 25-088								
Group	**Live**	**ID**	**FL**	**Wt**	**KFL**	**Ex_Hem**	**VFAT**	**Food**
Clin	N	1	85	5.5	0.90	2	0	0
Clin	N	2	73	4	1.03	2	0	0
Clin	N	3	73	3.6	0.93	2	1	0
Clin	N	4	62	3	1.26	1	0	0
Clin	N	5	74	3	0.74	3	0	0
Clin	N	6	54	2	1.27	3	0	0
Clin	N	7	73	3.6	0.93	2	0	0
Clin	N	8	72	4.4	1.18	2	0	0
Clin	N	9	65	2.8	1.02	3	0	0
Clin	N	10	79	4.2	0.85	0	1	0
Clin	N	11	62	2.4	1.01	3	1	0
Clin	N	12	78	4.4	0.93	3	0	0
Clin	N	13	65	2.5	0.91	2	0	0
Clin	N	14	52	1.6	1.14	3	0	0
Clin	N	15	64	2.3	0.88	3	0	0
Clin	N	16	65	2.3	0.84	3	0	0
Clin	N	17	63	2.8	1.12	4	0	0
Clin	N	18	80	4.8	0.94	2	1	0
Clin	N	19	80	3.9	0.76	4	0	0
Clin	N	20	66	2.5	0.87	3	0	0
Clin	N	21	72	2.8	0.75	2	0	0
Clin	N	22	66	2.6	0.90	3	0	0
Clin	N	23	77	3.3	0.72	3	0	0
Clin	N	24	72	3.2	0.86	2	0	0
Clin	N	25	77	4	0.88	3	0	0
Clin	N	26	75	3.7	0.88	3	0	0
Clin	N	27	62	2.1	0.88	2	0	0
Clin	N	28	83	4.8	0.84	3	0	0
Clin	Y	1	95	7.2	0.84	0	1	0
Clin	Y	2	78	4.6	0.97	1	0	0
Clin	Y	3	70	2.9	0.85	3	0	0
Clin	Y	4	62	2.1	0.88	2	0	0
Clin	Y	5	85	5.6	0.91	2	1	0
Clin	Y	6	80	5.1	1.00	0	1	1
Clin	Y	7	84	5.2	0.88	0	1	1

Clin	Y	8	76	4.3	0.98	0	1	0
Clin	Y	9	84	5.6	0.94	2	1	0
Clin	Y	10	78	4.3	0.91	0	1	0

Appendix E.

PATHOLOGY REPORT

US Fish & Wildlife Service
CA-NV Fish Health Center
24411 Coleman Hatchery Rd
Anderson, CA 96007

phone 530-365-4271
fax 530-365-7150

FHC Case No. : **25-068 UKL suckers (SNS&LRS)** Submittal date: **7/2/2025**
Sample Collector: **R. Stone** Sample Site(s): **KF NFH**

juvenile from Pond c2 eastside springs spawn (6) and c4 wild caught larvae captured in netpens *not necessarily moribund* 45mm+

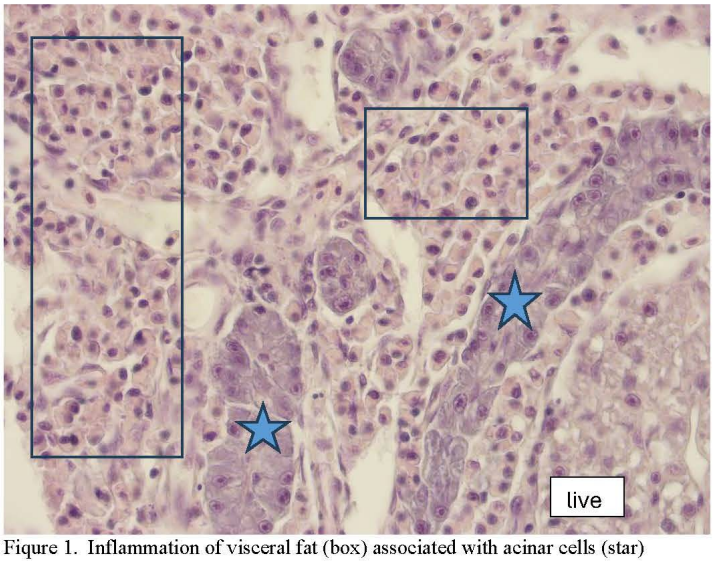
Note: high prevalence of Aeromonas sp. Isolated from fish.

Histological specimen examiner: **JS Foott**
Species **immature LRS and/or SNS** Age: **Juvenile, fry, larvae**
Tissues: **gill/head viscera mass**
Fixative: **Davidson (xx), PREFER-ETOH (), 10%BF ( ), ZFIX ( ), Bouins ( )**
Stains: **Hematoxylin & eosin (xx), PAS ( ), Iron ( ) ACID FAST () Gram ()**
Block No. **14797 - 14807** Block / slide deposition: **FHC**
Blood Smear (Number): **ND** Blood smear Stain: **Leishman-Giemsa ( ), DiffQuick ( )**
Clinical chemistry: **ND**

Summary:
Wild caught juveniles had inflammation (Fig. 1) of the visceral fat associated with acinar cells (not necrotic). No microbiological stimuli was associated with this inflammation. It is possible that rapid metabolism of lipid invoked the inflammation (arachidonic acid release from adipose cells). These fish also had abnormal kidney tubules (likely distal) with extensive protein deposits (hyaline) and necrosis of the tubule epithelium (Fig 2). Gill, liver (low glycogen vacuolization but no necrosis), and intestine was normal. No parasites observed.

C2 fry had normal tissues.

NO DIGANOSIS OF REPORTED MORTALITY



Fiqure 1. Inflammation of visceral fat (box) associated with acinar cells (star)

Figure 2. Hyaline deposits (arrow) in necrotic tubule epithelium.

Appendix F.

PATHOLOGY REPORT

US Fish & Wildlife Service
CA-NV Fish Health Center
24411 Coleman Hatchery Rd
Anderson, CA 96007

phone 530-365-4271
fax 530-365-7150

FHC Case No. : **25-083 UKL suckers (SNS&LRS)** Submittal date: **8/6/2025**
Sample Collector: **R. Stone** Sample Site(s): **KF NFH**

3 CY25 juveniles from Pond B3 captured in baited trap – normal appearance
2 CY25 juveniles from Pond B4 captured in baited trap – normal appearance

Mortality was elevated the week prior to capture – 0 on 8/6/2025

Histological specimen examiner: **JS Foott**
Species **immature LRS and/or SNS** Age: **Juvenile CY25**
Tissues: **gill, intestine, acinar cell, kidney, liver**

Fixative: **Davidson (xx), PREFER-ETOH (), 10%BF ( ), ZFIX ( ), Bouins ( )**
Stains: **Hematoxylin & eosin (xx), PAS ( ), Iron ( ) ACID FAST () Gram ()**
Block No. **14814 - 14819** Block / slide deposition: **FHC**
Blood Smear (Number): **ND** Blood smear Stain: **Leishman-Giemsa ( ), DiffQuick ( )**
Clinical chemistry: **ND**

Summary:
Fish from B3 had abnormal gill lamellae – edema and epithelial hyperplasia. No parasites observed. Liver hepatocytes without glycogen/lipid vacuoles. Intestine and kidney normal.

Water quality irritant is a possibility for B3

Gills of fish from B4 – lamellar structure normal, moderate level of interlamellar cell mass hyperplasia (can occur without irritation). Other tissues normal

The two B4 fish were normal.

Appendix G.

PATHOLOGY REPORT

US Fish & Wildlife Service
CA-NV Fish Health Center
24411 Coleman Hatchery Rd
Anderson, CA 96007

phone 530-365-4271
fax 530-365-7150

FHC Case No. : **25-088 UKL suckers (LRS)** Submittal date: **9/10/2025**
Sample Collector**: JS Foott** Sample Site(s): **KF NFH**

C2 = 8 moribund LRS (CY25) and 4 "normal" swimming
1 adult tilapia 1 fry fish collected as pond was harvested
C1 = 1 normal, 1 moribund LRS (CY25)

Histological specimen examiner: **JS Foott**
Species **immature LRS Tilapia** Age**: Juvenile CY25**
Tissues**: gill, intestine, acinar cell, kidney, liver**
Kidney imprints (gramstain), bloodsmear (diff quick stain)
Fixative: **Davidson (xx), PREFER-ETOH (), 10%BF ( ), ZFIX ( ), Bouins ( )**
Stains: **Hematoxylin & eosin (xx), PAS ( ), Iron ( ) ACID FAST () Gram (kidney imprint)**
Block No**. 14820 - 14835** Block / slide deposition: **FHC**
Blood Smear (Number): **ND** Blood smear Stain**: Leishman-Giemsa ( ), DiffQuick (x)**
Clinical chemistry: **ND**

Summary: Suckers in C2 and C1 ponds had hyperplastic gills and showed signs of hypoxia. *Ichthyobodo (*costia) was present in C2 fish. Low feed intake apparent. Bacterial infection was not associated with clinical signs of bacteremia or necrosis.

Likely scenario was that fish had been off feed for weeks and succumbed to endemic bacteria and external parasites.

Pond C2:
All "moribund" **and** "normal" LRS had gill lamellar hyperplasia and intralamellar cell mass (Fig 1a). Varying numbers of the ciliate *Ichthyobodo* sp (Fig.1b). Observed in the majority of gill sections. No clinical signs of bacterial septicemia or necrosis seen in either group. Potential lesions associated with hypoxia (gill hyperplasia) include liver hepatocyte coagulative necrosis / protein deposits and kidney tubule epithelium protein deposits (Fig 2 a&b). Intestine was normal. Signs of low food intake included no hepatocyte glycogen vacuoles, full gall bladders, and acinar cells lacking zymogen granules. During necrosis, the gall bladder of all fish was extended with bile.

Kidney imprints of both "moribund" **and** "normal" LRS contained varying degrees of gram-negative rod (9 of 13 (69%) supporting the bacterial isolations. Two different sized rods were seen in 3 of 9 positive imprints (multiple isolates).

Bloodsmears had low numbers of white blood cells with the majority being macrophages (particularly in "clinical" fish). A low frequency of bacterial rods observed in 3 of 14 smears. Erythrocytes were normal.

Tilapia gill sections with normal lamellae and no parasites. Liver with glycogen deposits reflective of normal energy reserves (feeding).

Pond C1
Moribund fish similar to C2 fish (hyperplastic gill, hepatocyte eosinophilic vacuoles), however no cilates observed.

Normal fish gill and liver was normal.

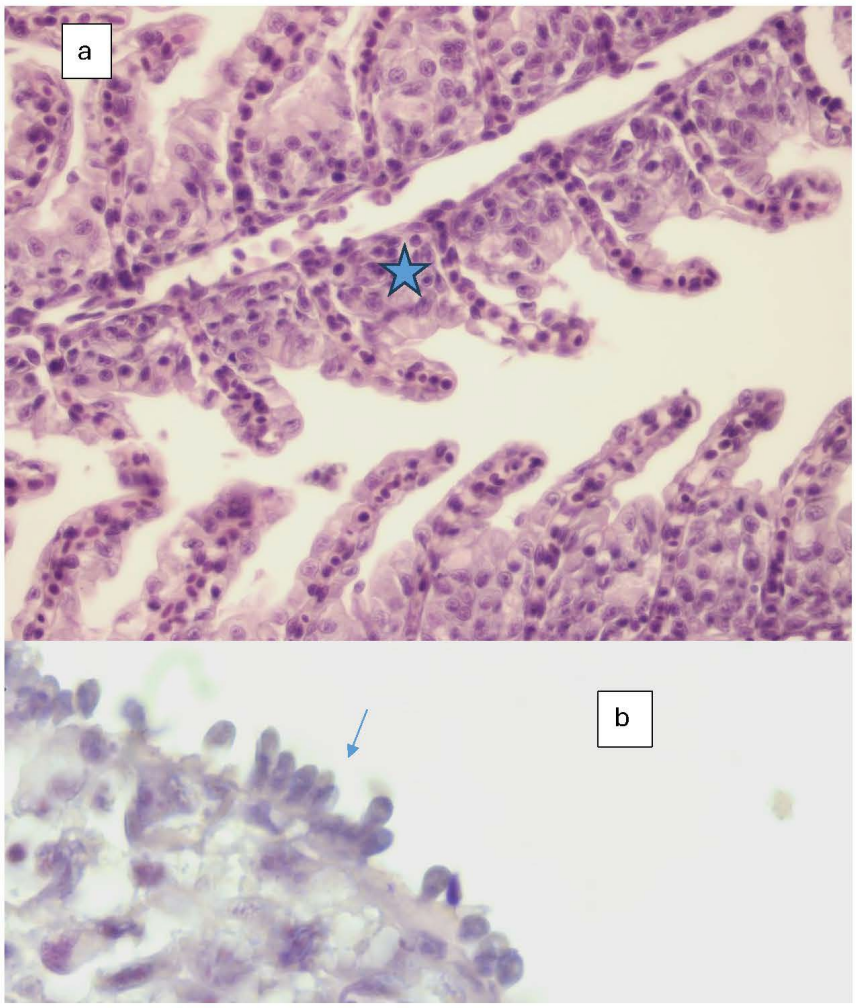


Figure 1. a. Hyperplastic gill from C2 pond LRS, intralamellar cell mass noted by star icon., b. *Ichthyoboda sp.* ("Costia") on C2 LRS gill.

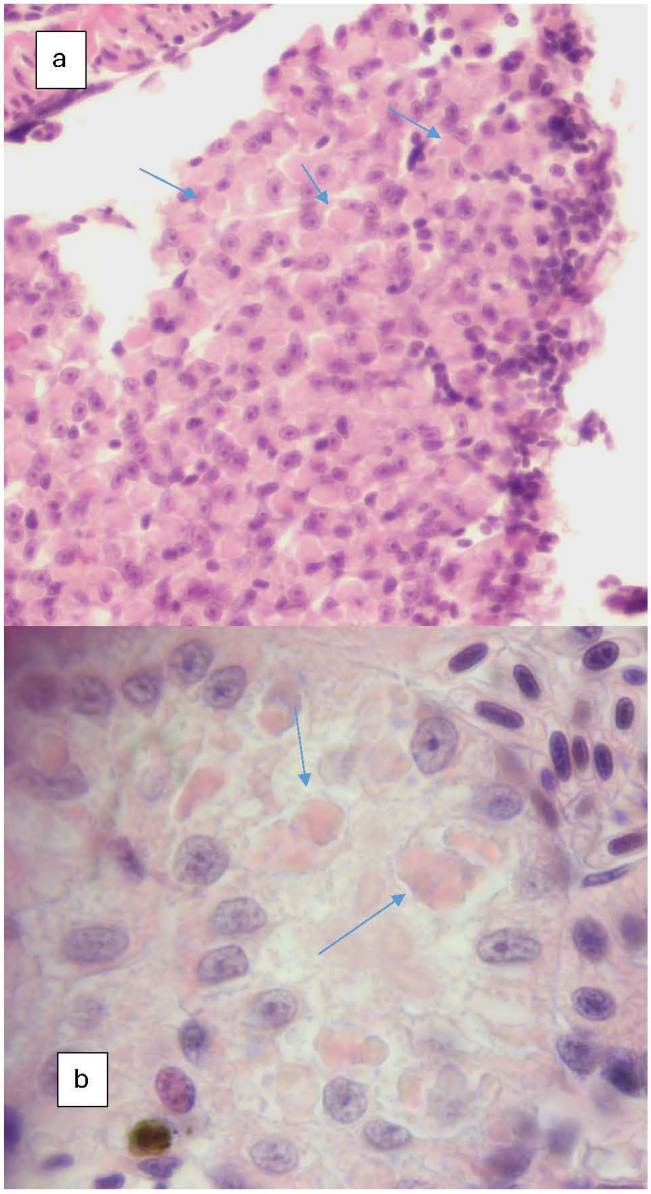


Figure 2. a. C2 LRS liver with eosinophilic cytoplasmic deposits. b. kidney tubule with eosinophilic deposits (protein).

Appendix H. WADDL results, Ca-NV FHC case 25-068

WASHINGTON STATE UNIVERSITY
Washington Animal Disease Diagnostic Laboratory

Accession Number: W251920063
Account Number: 11723

Testing Lab:	WADDL - Pullman	**Client:**	CA-NV Fish Health Center (USFWS)
Case Coordinator:	Claire Burbick, DVM, PhD, DACVM		24411 Coleman Fish Hatchery Road,
Date Received:	07/11/2025		Fish and Wildlife Service
Report Date:	07/14/2025		Anderson, CA 96007
Medical Records #:	CA-NV FHC 25-068	**Veterinarian:**	Ron Stone (ron_stone@fws.gov)

FINAL REPORT

Animal ID: CY25 Pond C4 Lost River Sucker :: NFS / Unknown / Fry

BACTERIOLOGY

Test: Bacterial Identification MALDI-TOF MS

Animals::Specimens	Isolate #	Organism	Result	Level
CY25 Pond C4 :: Isolate :: Isolate 2[1, 2]	1	Aeromonas veronii	Identified	2.22
CY25 Pond C4 :: Isolate :: Isolate 3[1, 2]	2	Aeromonas veronii	Identified	2.07
CY25 Pond C4 :: Isolate :: Isolate 6[2]	3	Pseudomonas anguilliseptica	Identified	1.92
CY25 Pond C4 :: Isolate :: Isolate 7[1, 2]	4	Aeromonas veronii	Identified	2.32
CY25 Pond C4 :: Isolate :: Isolate 8[1, 2]	5	Aeromonas veronii	Identified	2.19
CY25 Pond C4 :: Isolate :: Isolate 9[1, 2]	6	Aeromonas veronii	Identified	2.25
CY25 Pond C4 :: Isolate :: Isolate 11[1, 2]	7	Aeromonas veronii	Identified	2.17

[2]

Footnotes

1: **Species of this genus are difficult to differentiate with MALDI-TOF MS. To confirm species-level identification, molecular methods are recommended.**

2: Isolate name provided by MALDI-TOF MS. Please refer to the following score interpretation for identification confidence. MALDI-TOF MS score interpretation:

- 0-1.699 No reliable identification
- 1.7-1.999 Probable genus identification
- 2.0-2.299 Secure genus identification, probable species identification
- 2.3-3.0 Highly probable species identification

Authorized by: Claire Burbick, DVM, PhD, DACVM

Section Head

WADDL - Pullman
1940 SE Olympia Ave

PO Box 647034

Pullman, WA 99164

Phone: (509) 335-9696

Website: https://waddl.vetmed.wsu.edu/

WASHINGTON STATE UNIVERSITY
Washington Animal Disease Diagnostic Laboratory

Accession Number: W2519
Account Number: 11723

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Previous versions of this report

We have previously issued interim and/or final reports for this accession with the following names: preliminary_report_W251920063_2025-07-14_14-24-16.pdf

Appendix I. WADDL results CANV FHC 25-083

WASHINGTON STATE UNIVERSITY
Washington Animal Disease Diagnostic Laboratory

Accession Number: W252260043
Account Number: 11723

Testing Lab:	WADDL - Pullman	**Client:**	CA-NV Fish Health Center (USFWS) 24411 Coleman Fish Hatchery Road Fish and Wildlife Service Anderson, CA 96007
Case Coordinator:	Claire Burbick, DVM, PhD, DACVM		
Date Received:	08/14/2025		
Report Date:	08/15/2025	**Veterinarian:**	Ron Stone

FINAL REPORT

Animal ID: 25-083

BACTERIOLOGY

Test: Bacterial Identification MALDI-TOF MS

Animals::Specimens	Isolate #	Organism	Result	Level
25-083 :: 9 :: Isolate-Bacteria :: Isolate 1[1, 2]	1	Aeromonas veronii	Identified	2.22
25-083 :: 10 :: Isolate-Bacteria :: Isolate 2[1, 2]	2	Aeromonas veronii	Identified	2.25
25-083 :: 11 :: Isolate-Bacteria :: Isolate 3[1, 2]	3	Aeromonas veronii	Identified	2.28
25-083 :: 12 :: Isolate-Bacteria :: Isolate 4[1, 2]	4	Aeromonas veronii	Identified	2.24
25-083 :: 16 :: Isolate-Bacteria :: Isolate 5[1]	5	Planococcus okeanokoites	Identified	2.49
25-083 :: 17 :: Isolate-Bacteria :: Isolate 6[1]	6	Planococcus okeanokoites	Identified	1.72
25-083 :: B3-3 :: Isolate-Bacteria :: Isolate 7[1, 2]	7	Shewanella oneidensis	Identified	2.04
25-083 :: B4-2 :: Isolate-Bacteria :: Isolate 8[1, 2]	8	Aeromonas veronii	Identified	2.17

Footnotes

1: Isolate name provided by MALDI-TOF MS. Please refer to the following score interpretation for identification confidence. MALDI-TOF MS score interpretation:

- 0-1.699 No reliable identification
- 1.7-1.999 Probable genus identification
- 2.0-2.299 Secure genus identification, probable species identification
- 2.3-3.0 Highly probable species identification

2: **Species of this genus are difficult to differentiate with MALDI-TOF MS. To confirm species-level identification, molecular methods are recommended.**

WADDL - Pullman
1940 SE Olympia Ave
PO Box 647034
Pullman, WA 99164
Phone: (509) 335-9696
Website: https://waddl.vetmed.wsu.edu/

WASHINGTON STATE UNIVERSITY
Washington Animal Disease Diagnostic Laboratory

Accession Number: W252260
Account Number: 11723

Authorized by: Claire Burbick, DVM, PhD, DACVM
Section Head

Notice: *This report contains information that is confidential and is intended for the use of the individual or entity named on page 1. If you have received this report in error, please notify WADDL staff immediately.*

Previous versions of this report
We have previously issued interim and/or final reports for this accession with the following names:
preliminary_report_W252260043_2025-08-15_15-40-04.pdf

1940 SE Olympia Ave
PO Box 647034
Pullman, WA 99164
Phone: (509) 335-9696
Website: https://waddl.vetmed.wsu.edu/

Appendix J WADDL results CANV FHC 25-086

Accession Number: W252470129
Account Number: 11723

Testing Lab:	WADDL - Pullman	**Client:**	CA-NV Fish Health Center (USFWS)
Case Coordinators:	Claire Burbick, DVM, PhD, DACVM, Kevin Snekvik, DVM, PhD, DACVP		24411 Coleman Fish Hatchery Road,
Date Received:	09/04/2025		
Report Date:	09/12/2025	**Owner:**	Fish and Wildlife Service Anderson, CA 96007 Ron Stone
		Veterinarian:	Ron Stone

FINAL REPORT

Animal ID: 086 Lost River Sucker :: NFS

BACTERIOLOGY

Test: Bacterial Identification MALDI-TOF MS

Animals::Specimens	Isolate #	Organism	Result	Level
086 :: 8A :: Isolate :: Bacteria[1,2]	1	Aeromonas hydrophila	Identified	2.16
086 :: 4A :: Isolate :: Bacteria[2,3]	2	Enterobacter kobei	Identified	2.48
086 :: 4B :: Isolate :: Bacteria[1,2]	3	Aeromonas veronii	Identified	2.22
086 :: 10A :: Isolate :: Bacteria[1,2]	4	Vibrio albensis	Identified	1.74
086 :: 10B :: Isolate :: Bacteria[1,2]	5	Aeromonas veronii	Identified	2.19
086 :: 11A :: Isolate :: Bacteria[1,2]	6	Vibrio albensis	Identified	2.02
086 :: 12A :: Isolate :: Bacteria[1,2]	7	Aeromonas veronii	Identified	2.23

086 :: 15A :: Isolate :: Bacteria[1,2]	8	Vibrio albensis	Identified	2.02
086 :: 16A :: Isolate :: Bacteria[1,2]	9	Vibrio albensis	Identified	1.92
086 :: 16B :: Isolate :: Bacteria[1,2]	10	Aeromonas veronii	Identified	2.21
086 :: 24A :: Isolate :: Bacteria[1,2]	11	Aeromonas hydrophila	Identified	2.22
086 :: 24B :: Isolate :: Bacteria[1,2]	12	Aeromonas veronii	Identified	2.11
086 :: 25A :: Isolate :: Bacteria[1,2]	13	Aeromonas hydrophila	Identified	2.21
086 :: 25C :: Isolate :: Bacteria[2,3]	14	Enterobacter bugandensis	Identified	2.36
086 :: 29A :: Isolate :: Bacteria[1,2]	15	Vibrio mimicus	Identified	2.00
086 :: 29B :: Isolate :: Bacteria[1,2]	16	Aeromonas veronii	Identified	2.14
086 :: 31A :: Isolate :: Bacteria[1,2]	17	Vibrio albensis	Identified	1.89

Footnotes

1: **Species of this genus are difficult to differentiate with MALDI-TOF MS. To confirm species-level identification, molecular methods are recommended.**

2: Isolate name provided by MALDI-TOF MS. Please refer to the following score interpretation for identification confidence. MALDI-TOF MS score interpretation:

- 0-1.699 No reliable identification

- 1.7-1.999 Probable genus identification
- 2.0-2.299 Secure genus identification, probable species identification
- 2.3-3.0 Highly probable species identification

3: **This organism is a member of the *Enterobacter cloacae* complex.**

Authorized by:

Claire Burbick, DVM, PhD,
DACVM Section Head

MOLECULAR DIAGNOSTICS

Test: Direct PCR/Sequencing, Bacteria **Specimen:** 10A :: Isolate :: Bacteria **Collection date:** 09/03/2025

Organism	Vibrio cholerae
Result	***Detected***

Comments

After DNA extraction from the sample, a portion of the rpoB gene was amplified by PCR using universal eubacterial primers. When this PCR amplicon was directly sequenced the sequence most closely matched that of *Vibrio cholerae* (99.61% sequence identity to GenBank acc # CP104356 and others) when compared with sequences in GenBank.

Test: Direct PCR/Sequencing, Bacteria **Specimen:** 11A :: Isolate :: Bacteria **Collection date:** 09/03/2025

Organism	Vibrio cholerae
Result	***Detected***

Comments

After DNA extraction from the sample, a portion of the rpoB gene was amplified by PCR using universal eubacterial primers. When this PCR amplicon was directly sequenced the sequence most closely matched that of *Vibrio cholerae* (100% sequence identity to GenBank acc # CP189019and others) when compared with sequences in GenBank.

Test: Direct PCR/Sequencing, Bacteria **Specimen:** 15A :: Isolate :: Bacteria **Collection date:** 09/03/2025

Organism	Vibrio cholerae
Result	***Detected***

Comments

After DNA extraction from the sample, a portion of the rpoB gene was amplified by PCR using universal eubacterial primers. When this PCR amplicon was directly sequenced the sequence most closely matched that of *Vibrio cholerae* (100% sequence identity to GenBank acc # CP104356 and others) when compared with sequences in GenBank.

Test: Direct PCR/Sequencing, Bacteria **Specimen:** 16A :: Isolate :: Bacteria **Collection date:** 09/03/2025

Organism	Vibrio cholerae
Result	***Detected***

Comments

After DNA extraction from the sample, a portion of the rpoB gene was amplified by PCR using universal eubacterial primers. When this PCR amplicon was directly sequenced the sequence most closely matched that of *Vibrio cholerae* (100% sequence identity to GenBank acc # CP104356 and others) when compared with sequences in GenBank.

Accession Number: W252470129
Account Number: 11723

Test: Direct PCR/Sequencing, Bacteria **Specimen:** 29A :: Isolate :: Bacteria **Collection date:** 09/03/2025

Organism	Vibrio cholerae
Result	***Detected***

Comments

After DNA extraction from the sample, a portion of the rpoB gene was amplified by PCR using universal eubacterial primers. When this PCR amplicon was directly sequenced the sequence most closely matched that of *Vibrio cholerae* (100% sequence identity to GenBank acc # CP184808and others) when compared with sequences in GenBank.

Test: Direct PCR/Sequencing, Bacteria **Specimen:** 31A :: Isolate :: Bacteria **Collection date:** 09/03/2025

Organism	Vibrio cholerae
Result	***Detected***

Comments

After DNA extraction from the sample, a portion of the rpoB gene was amplified by PCR using universal eubacterial primers. When this PCR amplicon was directly sequenced the sequence most closely matched that of *Vibrio cholerae* (100% sequence identity to GenBank acc # CP189019and others) when compared with sequences in GenBank.

Authorized by: Dr. Kevin Snekvik, DVM, PhD, DACVP

Section Head

Notice: *This report contains information that is confidential and is intended for the use of the individual or entity named on page 1. If you have received this report in error, please notify WADDL staff immediately.*

Previous versions of this report
We have previously issued interim and/or final reports for this accession with the following names:

preliminary_report_W252470129_2025-09-05_16-37-51.pdf,
preliminary_report_W252470129_2025-09-12_11-49-07.pdf

Appendix K. WADDL results CANV FHC 25-088

Testing Lab:	WADDL - Pullman	**Client:**	CA-NV Fish Health Center (USFWS)
Case Coordinator:	Claire Burbick, DVM, PhD, DACVM		24411 Coleman Fish Hatchery Road
Date Received:	09/17/2025		Fish and Wildlife Service
Report Date:	09/19/2025		Anderson, CA 96007
		Veterinarian:	Ron Stone

Previous Accession #: W252470129

FINAL REPORT

Animal IDs:

088 Lost River Sucker :: NFS
089 tilapia :: NFS

BACTERIOLOGY

Test: Bacterial Identification MALDI-TOF MS

Animals::SpecimensIsolate # Level		OrganismResult		
088 :: 088-1A :: Isolate :: Bacteria[1,2] 1		Aeromonas veronii	Identified	2.21
088 :: 088-1B :: Isolate :: Bacteria[1]	2	Plesiomonas shigelloides	Identified	2.33
088 :: 088-1C :: Isolate :: Bacteria[1]	3	Pseudomonas anguilliseptica	Identified	1.88
088 :: 088-2A :: Isolate :: Bacteria[1,2]	4	Aeromonas veronii	Identified	2.19
088 :: 088-2B :: Isolate :: Bacteria[1,2]	5	Vibrio mimicus	Identified	1.89
088 :: 088-3A :: Isolate :: Bacteria[1,2]	6	Aeromonas veronii	Identified	2.12
088 :: 088-3C :: Isolate :: Bacteria[1,2]	7	Vibrio mimicus	Identified	2.02
088 :: 088-4 :: Isolate :: Bacteria[1]	8	Chryseobacterium shigense	Identified	1.93
088 :: 088-5A :: Isolate :: Bacteria[1,2]	9	Aeromonas veronii	Identified	2.31
088 :: 088-5B :: Isolate :: Bacteria[1]	10	Chryseobacterium shigense	Identified	2.13
088 :: 088-5C :: Isolate :: Bacteria[1,2]	11	Aeromonas veronii	Identified	2.22
088 :: 088-6A :: Isolate :: Bacteria[1,2]	12	Aeromonas veronii	Identified	2.32
088 :: 088-6B :: Isolate :: Bacteria[1,2]	13	Aeromonas veronii	Identified	2.24
088 :: 088-7A :: Isolate :: Bacteria[1,2]	14	Aeromonas veronii	Identified	2.38
088 :: 088-7B :: Isolate :: Bacteria[1,2]	15	Aeromonas veronii	Identified	2.30
088 :: 088-8A :: Isolate :: Bacteria[1,2]	16	Vibrio mimicus	Identified	1.92

Animals::Specimens		Organism	Result	Level
088 :: 088-14A :: Isolate :: Bacteria[1,2]	21	Aeromonas veronii	Identified	2.13

Isolate #				
088 :: 088-8B :: Isolate :: Bacteria[1,2]	17	Aeromonas veronii	Identified	2.37
088 :: 088-13A :: Isolate :: Bacteria[1,2]	18	Aeromonas veronii	Identified	2.31
088 :: 088-13B :: Isolate :: Bacteria[1,2]	19	Aeromonas veronii	Identified	2.30
088 :: 088-13C :: Isolate :: Bacteria[1,2]	20	Aeromonas veronii	Identified	2.23
2				
088 :: 088-14B :: Isolate :: Bacteria[1,2]	22	Vibrio mimicus	Identified	2.04
088 :: 088-15A :: Isolate :: Bacteria[1,2]	23	Vibrio albensis	Identified	1.91
088 :: 088-15B :: Isolate :: Bacteria[1,2]	24	Aeromonas veronii	Identified	2.32
088 :: 088-15D :: Isolate :: Bacteria[1,2]	25	Vibrio mimicus	Identified	1.97
088 :: 088-16A :: Isolate :: Bacteria[1,2]	26	Vibrio albensis	Identified	1.95
088 :: 088-16B :: Isolate :: Bacteria[1]	27	Shewanella xiamenensis	Identified	2.36
088 :: 088-16D :: Isolate :: Bacteria[1,2]	28	Aeromonas veronii	Identified	2.31
088 :: 088-17A :: Isolate :: Bacteria[1,2]	29	Aeromonas sobria	Identified	1.87
088 :: 088-17B :: Isolate :: Bacteria[1,2]	30	Vibrio mimicus	Identified	2.07
088 :: 088-17C :: Isolate :: Bacteria[1]	31	Plesiomonas shigelloides	Identified	2.37
088 :: 088-18A :: Isolate :: Bacteria[1,2]	32	Aeromonas hydrophila	Identified	2.14
088 :: 088-18B :: Isolate :: Bacteria[1,2]	33	Aeromonas ichthiosmia	Identified	2.27

088 :: 088-19A :: Isolate :: Bacteria[1, 2]	34	Vibrio mimicus	Identified	2.18
088 :: 088-19B :: Isolate :: Bacteria[1, 2]	35	Aeromonas veronii	Identified	2.27
089 :: 089-9 :: Isolate :: Bacteria[1, 2]	36	Massilia tieshanensis	Identified	2.05
089 :: 089-11 :: Isolate :: Bacteria[1]	37		No Reliable ID	1.38
089 :: 089-12 :: Isolate :: Bacteria[1, 2]	38	Aeromonas veronii	Identified	2.27

Footnotes

1: Isolate name provided by MALDI-TOF MS. Please refer to the following score interpretation for identification confidence. MALDI-TOF MS score interpretation:

- 0-1.699 No reliable identification
- 1.7-1.999 Probable genus identification
- 2.0-2.299 Secure genus identification, probable species identification
- 2.3-3.0 Highly probable species identification

2: **Species of this genus are difficult to differentiate with MALDI-TOF MS. To confirm species-level identification, molecular methods are recommended.**

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Section Head

Notice: *This report contains information that is confidential and is intended for the use of the individual or entity named on page 1. If you have received this report in error, please notify WADDL staff immediately.*

Previous versions of this report

We have previously issued interim and/or final reports for this accession with the following names:
preliminary_report_W252600070_2025-09-18_15-11-30.pdf

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